An Efficient Minimax-Optimal Algorithm for Adversarial m-Set Bandits*

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November 2, 2026

Abstract

We study adversarial combinatorial bandits with m-set actions, where at each round the learner selects m out of d items and observes only the aggregate loss of the selected items. The resulting action set contains $K = \binom{d}{m}$ elements and can therefore be exponentially large. We propose a computationally efficient algorithm that exploits this structure without explicitly enumerating the action set. Against adaptive non-anticipating adversaries, it guarantees, with probability at least $1-\delta$, regret against the best action in hindsight of $R_T = O(\sqrt{dT \log(K/\delta)})$. This matches the high-probability regret bound of the finite-action EXP3 with Kiefer-Wolfowitz exploration algorithm of Zimmert and Lattimore [33], whose direct implementation may require exponential space. Our algorithm instead represents each sampling distribution with d parameters and runs in polynomial time without enumerating the action set. This resolves an open problem posed by Maiti et al. [30]. We complement this upper bound with a matching high-probability lower bound. For all sufficiently small δ and all sufficiently large horizons, every randomized algorithm admits an adaptive non-anticipating adversary for which, with probability at least δ, $R_T = \Omega(\sqrt{dT \log(K/\delta)})$. Thus, the rate is minimax optimal up to universal constants in this regime. In particular, setting $m = 1$ shows that the $\log K$ factor in the high-probability regret bound for ordinary K-armed bandits against adaptive non-anticipating adversaries is unavoidable. This strengthens the lower bound of Gerchinovitz and Lattimore [18], closing the remaining $\sqrt{\log K}$ gap between high-probability upper and lower regret bounds.

*A Lean formalization accompanying this paper, together with instructions for independent verification, is available in the companion repository.

1 Introduction

In this paper, we study adversarial m-set bandits. At each round, the learner selects m out of d items and incurs the sum of their losses, but observes only the aggregate loss, not the loss of each selected item. Since any m-set may be chosen, the learner has $K = \binom{d}{m}$ possible actions, and its goal is to minimize regret with respect to the best fixed m-set in hindsight. In this work, we assume that, at every round, the loss of each action, namely each m-set, lies in $[-1, 1]$. The loss vectors are chosen by an adaptive non-anticipating adversary and may depend on the past interaction history but not on the learner's random choice.

Zimmert and Lattimore [33, Theorem 6 and Algorithm 3] show that, in this setting, EXP3 with Kiefer-Wolfowitz exploration, abbreviated as EXP3-KW, achieves $O(\sqrt{dT \log(K/\delta)})$ regret with probability at least $1 - \delta$, where T represents the number of rounds and δ the failure probability. For m-sets with $m \le d/2$, this gives $\widetilde{O}(\sqrt{dmT})$.[1] However, the direct implementation maintains one weight for each of the $K = \binom{d}{m}$ actions and may therefore require exponential space.

Maiti et al. [30, Appendix E.6] obtain a polynomial-time high-probability algorithm with regret $\widetilde{O}(d\sqrt{mT})$ and ask whether the extra $\sqrt{d}$ factor can be removed efficiently. Furthermore, applying their general directed acyclic graph (DAG) guarantee (Theorem 7) to the m-set DAG constructed in their Appendix E.6, which has $O(md)$ edges and $\binom{d}{m}$ paths, yields the sharper bound $\widetilde{O}(m\sqrt{dT})$.

Our algorithm attains the finite-action EXP3-KW regret rate in polynomial time, representing each sampling distribution with only d parameters. Writing $s := \min\{m, d - m\}$, its regret is $\widetilde{O}(\sqrt{dsT})$. For $m \le d/2$, this becomes $\widetilde{O}(\sqrt{dmT})$, improving the two guarantees of Maiti et al. [30] by factors of $\sqrt{d}$ and $\sqrt{m}$, respectively. Thus, this resolves their open question. Table 1 compares our regret bound and computational complexity with those of state-of-the-art algorithms.

Work	Regret bound	Computational complexity
Zimmert and Lattimore [33] (Theorem 6 and Algorithm 3)	$\widetilde{O}(\sqrt{dmT})$	Exponential
Maiti et al. [30] (Theorem 7 applied to the DAG of Appendix E.6)	$\widetilde{O}(m\sqrt{dT})$	Polynomial
This work	$\boldsymbol{\widetilde{O}(\sqrt{dmT})}$	**Polynomial**

Table 1: Statistical and computational comparison for the regime $m \le d/2$. All upper bounds hold with high probability against adaptive non-anticipating adversaries.

We also provide a matching lower bound. For every d and m, all sufficiently small δ, and all horizons above a polynomial threshold, every randomized algorithm admits an adaptive non-anticipating adversary for which $R_T = \Omega(\sqrt{dT \log(K/\delta)})$ with probability at least δ. Together, these results identify the minimax high-probability rate in this regime.

The special case $m = 1$, and hence $d = K$, gives a consequence for ordinary adversarial K-armed bandits. Here it is essential to distinguish oblivious adversaries, which fix the entire loss sequence in advance, from adaptive non-anticipating adversaries. For oblivious adversaries, the algorithm of Audibert and Bubeck [3, Theorem 18] gives an upper bound of $O(\sqrt{KT \log(1/\delta)})$, matched by the lower bound of Gerchinovitz and Lattimore [18, Theorem 1]. For adaptive non-anticipating adversaries, the upper bound presented in Cesa-Bianchi and Lugosi [11, Theorem 6.10] is instead $O(\sqrt{KT \log(K/\delta)})$. Our lower bound shows that the additional $\log K$ is necessary in this model, closing the remaining $\sqrt{\log K}$ gap. We summarize these results in Table 2.

[1] The notation $\widetilde{O}$ suppresses logarithmic factors.

	Oblivious	Adaptive non-anticipating
Upper bound	$O\Big(\sqrt{KT\log(1/\delta)}\Big)$ Audibert and Bubeck [3, Theorem 18] (2010)	$O\Big(\sqrt{KT\log(K/\delta)}\Big)$ Cesa-Bianchi and Lugosi [11, Theorem 6.10] (2006)
Lower bound	$\Omega\Big(\sqrt{KT\log(1/\delta)}\Big)$ Gerchinovitz and Lattimore [18, Theorem 1] (2016)	$\boldsymbol{\Omega\Big(\sqrt{KT\log(K/\delta)}\Big)}$ **This work** (Corollary 1)

Table 2: High-probability regret bounds for ordinary K-armed bandits ($m = 1$, $d = K$), with losses in $[0, 1]$. Upper bounds hold with probability at least $1 - \delta$, and lower bounds with probability at least δ. The comparison is for sufficiently small δ and sufficiently large T.

2 Setting and problem formulation

In this section, we introduce the notation used throughout the paper and formulate the adversarial m-set bandit problem.

Preliminary definitions. For a positive integer n, we write $[n] := \{1, \dots, n\}$. In the following, we fix two integers $d \geq 2$ and $1 \leq m \leq d - 1$. Then, we define the action set and its size as

$$\mathcal{X} := \mathcal{X}_{d,m} := \left\{x \in \{0,1\}^d : \|x\|_1 = m\right\}, \qquad K := |\mathcal{X}| = \binom{d}{m}. \tag{1}$$

Each action $x \in \mathcal{X}$ selects exactly m of the d items, where $x_i = 1$ if item i is selected and $x_i = 0$ otherwise. The set of items selected by each action is an m-element subset of $[d]$, which we call an *m-set*. Conversely, each m-set $S \subseteq [d]$ corresponds to the indicator vector $\mathbf{1}_S \in \mathcal{X}$. We therefore use actions in $\mathcal{X}$ and m-sets interchangeably. Finally, to express the regret bounds in terms of d and m, we also introduce $s := \min\{m, d - m\}$.

Linear-algebra notation. We write $\mathbf{0} := (0, \dots, 0)^\top$ for the zero vector. For a symmetric matrix $A \in \mathbb{R}^{d\times d}$, we denote its kernel by $\ker(A) := \{z \in \mathbb{R}^d : Az = \mathbf{0}\}$, its range (or image) by $\mathrm{range}(A) := \{Az : z \in \mathbb{R}^d\}$, and its Moore-Penrose pseudoinverse by A^+. A symmetric matrix A is positive semidefinite if $z^\top Az \geq 0$ for every $z \in \mathbb{R}^d$. We write $A \succ 0$ if A is positive definite, that is, $z^\top Az > 0$ for every nonzero $z \in \mathbb{R}^d$. Let $\mathbf{1} := (1, \dots, 1)^\top$, and define

$$\mathrm{span}\{\mathbf{1}\} := \{a\mathbf{1} : a \in \mathbb{R}\}, \qquad H_0 := \{y \in \mathbb{R}^d : y^\top \mathbf{1} = 0\}.$$

Thus, H_0 is the subspace orthogonal to $\mathrm{span}\{\mathbf{1}\}$. Formally, $H_0^\perp = \mathrm{span}\{\mathbf{1}\}$.

Probability notation. We denote the probability simplex over $\mathcal{X}$ by

$$\Delta(\mathcal{X}) := \left\{p : \mathcal{X} \to [0,1] : \sum_{x\in\mathcal{X}} p(x) = 1\right\}.$$

For probability laws P and Q on a common measurable space, the Kullback-Leibler (KL) divergence is $\mathrm{KL}(P\|Q) := \int \log(dP/dQ)\, dP$ if $P \ll Q$, and $+\infty$ otherwise. Here $P \ll Q$ means that P is absolutely continuous with respect to Q, and dP/dQ is the Radon-Nikodym derivative of P with

respect to Q. For $p, q \in \Delta(\mathcal{X})$, this specializes to $\mathrm{KL}(p \,\|\, q) = \sum_{x \in \mathcal{X}} p(x) \log(p(x)/q(x))$.[2] To describe the moments of a distribution, let X be a random vector drawn from $p \in \Delta(\mathcal{X})$. We define the marginal vector, second-moment matrix, and covariance matrix of p as

$$\mu(p) := \mathbb{E}_{X \sim p}[X], \qquad M(p) := \mathbb{E}_{X \sim p}[XX^\top], \qquad \Sigma(p) := \mathrm{Cov}_{X \sim p}(X) = M(p) - \mu(p)\mu(p)^\top.$$

The i-th marginal probability $\mu_i(p)$ satisfies $\mu_i(p) = \mathbb{P}_{X \sim p}[X_i = 1]$ and represents the probability that action X selects item i. In addition, if $M(p)$ is invertible, we refer to the quantity

$$x^\top M(p)^{-1} x \tag{2}$$

as the *leverage score* of x under p.

We also use the total variation distance $\mathrm{TV}(P, Q) := \sup_E |P(E) - Q(E)|$, where the supremum is over measurable events. We write $\mathrm{kl}(a, b) := \mathrm{KL}(\mathrm{Ber}(a) \| \mathrm{Ber}(b))$ for binary KL, with the same zero-probability conventions as above.

Weighted m-set distributions. Given $\theta \in \mathbb{R}^d$, define $w_i := e^{\theta_i} > 0$ for every $i \in [d]$. For every m-set $S \subseteq [d]$, the weighted m-set distribution p_θ is

$$p_\theta(S) := \frac{\exp\left(\sum_{i \in S} \theta_i\right)}{\sum_{A \subseteq [d]:\, |A| = m} \exp\left(\sum_{i \in A} \theta_i\right)} = \frac{\prod_{i \in S} w_i}{\sum_{A \subseteq [d]:\, |A| = m} \prod_{i \in A} w_i}.$$

Thus, $p_\theta(S) \propto e^{\theta^\top \mathbf{1}_S}$. Notice that, although p_θ may assign positive probability to exponentially many actions, only the d parameters $\theta_1, \ldots, \theta_d$ represent it. Finally, we define a set of distributions whose marginal probabilities stay away from 0 and 1. Let $r := m/d$ and fix $\lambda \in (0, 1)$. Define

$$\mathcal{P}_\lambda := \{p \in \Delta(\mathcal{X}) : \lambda r \le \mu_i(p) \le 1 - \lambda(1 - r) \text{ for every } i \in [d]\}. \tag{3}$$

Notice that the uniform distribution U over $\mathcal{X}$ satisfies $\mu_i(U) = r$ for every $i \in [d]$ and therefore belongs to $\mathcal{P}_\lambda$ for every $\lambda \in (0, 1)$.

Adversarial m-set bandit protocol. Fix a time horizon $T \ge 1$. Let $(\mathcal{F}_t)_{t=0}^T$ denote the filtration generated by the interaction history, where $\mathcal{F}_{t-1}$ contains the information available before round t. At each round $t \in [T]$, the learner selects an $\mathcal{F}_{t-1}$-measurable distribution $p_t \in \Delta(\mathcal{X})$, where $\mathcal{X}$, defined in Equation (1), represents the set of actions available to the learner. The adversary selects a loss vector $\ell_t \in \mathbb{R}^d$ that is also $\mathcal{F}_{t-1}$-measurable, where $\ell_{t,i}$ represents the loss associated with item $i \in [d]$. Thus, the adversary is *adaptive non-anticipating*. The loss vector ℓ_t may depend on the entire interaction history up to round $t - 1$, but not on the learner's random draw at round t. Therefore, conditionally on $\mathcal{F}_{t-1}$, both p_t and ℓ_t are fixed. The learner then draws an action $X_t \sim p_t$ and observes only the (action) loss $Y_t := X_t^\top \ell_t$. We use the standard action-level normalization

$$|x^\top \ell_t| \le 1 \qquad \forall x \in \mathcal{X},\ t \in [T],$$

also adopted by Maiti et al. [30, Assumption 1] and Zimmert and Lattimore [33, Section 1]. Under this protocol, the learner's realized regret after T rounds, measured against the best fixed action in hindsight, is

$$R_T := \sum_{t=1}^T X_t^\top \ell_t - \min_{x \in \mathcal{X}} \sum_{t=1}^T x^\top \ell_t. \tag{4}$$

[2] We use the convention $0 \log(0/q) = 0$; a term with $p(x) > 0 = q(x)$ is $+\infty$.

Under an adaptive adversary, this action may depend on the realized interaction history, while remaining fixed across rounds.[3] Throughout, we use the shorthand $\mathbb{E}_t[\,\cdot\,] := \mathbb{E}[\,\cdot\,|\mathcal{F}_{t-1}]$ for conditional expectation given the information available before round $t \in [T]$.

3 Original contributions

3.1 Upper bound

Our first contribution is an algorithm (Algorithm 1) for adversarial m-set bandits that achieves the regret guarantee summarized in the following.

Theorem (Informal version of Theorem 1)**.** *There exists an algorithm that can be implemented in polynomial time such that, against any adaptive non-anticipating adversary, with probability at least $1-\delta$, it achieves*

$$R_T = O\left(\sqrt{dT\left(\log\binom{d}{m} + \log\frac{1}{\delta}\right)}\right).$$

For example, one may take Algorithm 1.

Challenges and techniques. In their finite-action EXP3-KW algorithm, Zimmert and Lattimore [33, Algorithm 3] obtain a tight high-probability regret bound by applying exponential weights (also called multiplicative weights) to the biased surrogate loss

$$x^\top \widehat{\ell}_t - \eta x^\top M_t^{-1} x, \tag{5}$$

where $\eta > 0$ is the learning rate, $M_t = \mathbb{E}_{X\sim p_t}[XX^\top]$ is the second-moment matrix of the sampling distribution p_t, and

$$\widehat{\ell}_t := M_t^{-1} X_t (X_t^\top \ell_t) \tag{6}$$

is the loss estimate constructed after drawing $X_t \sim p_t$ and observing $X_t^\top \ell_t$.

The main obstacle to an efficient implementation of this approach is that the leverage score $x^\top M_t^{-1} x$ is quadratic in x. As a result, even if we start from a weighted m-set distribution and use the biased surrogate loss defined in (5) in a multiplicative-weights update, the updated distribution need not remain in the weighted m-set family. Thus, this update does not in general preserve the d-parameter weighted m-set representation.

Our first step is to replace the leverage score $x^\top M_t^{-1} x$ in (5) with an affine upper bound. Combining the pseudoinverse estimate of Cesari and Colomboni [13, Corollary 1.2] with an identity relating the second-moment matrix to the covariance, we derive

$$x^\top M_t^{-1} x \le \phi_{p_t}(x) := 1 + \sum_{i=1}^{d} \frac{(x_i - \mu_i(p_t))^2}{\mu_i(p_t)(1-\mu_i(p_t))}. \tag{7}$$

Although the expression defining ϕ_{p_t} appears quadratic, it is affine on $\mathcal{X}$. Indeed, every $x \in \mathcal{X}$ satisfies $x_i^2 = x_i$ and $\sum_{i=1}^d x_i = m$. For later convenience, we scale this affine majorant by four and write

$$c_{p_t}^\top x = 4\phi_{p_t}(x) \qquad \text{for every } x \in \mathcal{X}.$$

[3] More generally, for a comparator distribution $q \in \Delta(\mathcal{X})$, which is fixed across rounds but may be chosen in hindsight, one can define the regret with respect to q as $R_T(q) := \sum_{t=1}^T \left(X_t^\top \ell_t - \mathbb{E}_{X\sim q}\left[X^\top \ell_t\right]\right)$. Our notion of regret introduced in (4) then satisfies $R_T = \max_{q\in\Delta(\mathcal{X})} R_T(q)$, with the maximum attained by the Dirac mass at a best action in hindsight.

We then define the surrogate loss

$$Z_t(x) := x^\top \widehat{\ell}_t - \eta c_{p_t}^\top x. \tag{8}$$

Here, $\widehat{\ell}_t$ is the loss estimate defined in (6). The correction in Equation (8) plays the same role as the quadratic correction used by Zimmert and Lattimore [33], but remains affine in x. Consequently, the multiplicative-weights step maps a weighted m-set distribution to another weighted m-set distribution.

However, the affine leverage majorant $\phi_{p_t}(x)$ can become arbitrarily large when a marginal probability $\mu_i(p_t)$ approaches 0 or 1. The bound in (7) then no longer provides uniform control of the estimated action losses $x^\top \widehat{\ell}_t$. Therefore, we combine the exponential-weights update with an approximate KL projection toward the set $\mathcal{P}_\lambda$, introduced in Section 2. Specifically, at each round, we first apply an exponential-weights update to p_t using the surrogate loss $Z_t(x)$ defined in (8), obtaining a weighted m-set distribution $\widetilde{p}_{t+1}$. We then compute a distribution p_{t+1} that approximates the exact KL projection $p^\star_{t+1}$, defined by

$$p^\star_{t+1} \in \operatorname*{arg\,min}_{p \in \mathcal{P}_\lambda} \mathrm{KL}(p \,\|\, \widetilde{p}_{t+1})\,.^{4} \tag{9}$$

The distribution p_{t+1} has two crucial properties. It retains the weighted m-set structure and belongs to the slightly larger set $\mathcal{P}_{\lambda/2}$ for a suitably chosen λ. The former condition ensures that every iterate can be represented as p_{θ_t} using the d parameters $\theta_{t,1}, \ldots, \theta_{t,d}$, rather than one weight for each of the K actions, while the latter condition keeps the marginal probabilities away from 0 and 1, ensuring that the estimated action losses remain under control.

One of the challenges in our analysis is that our algorithm samples from distributions in $\mathcal{P}_{\lambda/2}$. However, an optimal action x in hindsight corresponds to the Dirac distribution δ_x. This distribution puts all its probability on x, has marginal probabilities in $\{0, 1\}$, and therefore does not belong to $\mathcal{P}_\lambda$. Therefore, in our analysis, we use the smoothed comparator

$$q_x := (1 - \lambda)\delta_x + \lambda U, \tag{10}$$

where U is the uniform distribution over $\mathcal{X}$. By construction, $q_x \in \mathcal{P}_\lambda$. Thus, under our action-level normalization, replacing the pure comparator δ_x with q_x increases the regret bound by at most $2\lambda T$.

Another challenge in achieving the same high-probability regret rate as Zimmert and Lattimore [33] arises from replacing their quadratic correction $x^\top M_t^{-1} x$ in (5) with the larger affine correction $c_{p_t}^\top x$ in (8). This is because the exponential-weights analysis yields a regret bound that depends on the sum over the T rounds of the second moments $\mathbb{E}_{X \sim p_t}[Z_t(X)^2]$. Since the larger correction $c_{p_t}^\top x$ may increase this quantity, we need to show that its contribution to the regret bound remains of the same order as in their analysis. To control this contribution, we show that the affine correction $c_{p_t}^\top x$ is nonnegative and satisfies

$$\mathbb{E}_{X \sim p_t}[c_{p_t}^\top X] = O(d), \qquad \max_{x \in \mathcal{X}} c_{p_t}^\top x = O(d/\lambda). \tag{11}$$

For sufficiently large T, combining these two bounds and using our choice $\lambda = \Theta(\sqrt{d \log(K/\delta)/T})$, we obtain, with probability at least $1 - \delta/2$,

$$\sum_{t=1}^{T} \mathbb{E}_{X \sim p_t}[Z_t(X)^2] \le O(dT). \tag{12}$$

[4] This two-step procedure is an approximate implementation of online mirror descent (OMD) over $\mathcal{P}_\lambda$ with the negative entropy regularizer, as discussed in Section 5.2.

Since the sum in (12) enters the regret bound multiplied by $\eta = \Theta(\sqrt{\log(K/\delta)/(dT)})$, its contribution is $O(\sqrt{dT\log(K/\delta)})$. Thus, replacing the quadratic correction $x^\top M_t^{-1}x$ with the larger affine correction $c_{p_t}^\top x$ in the surrogate loss defined in (8) does not increase the order of this term in our algorithm's regret bound.

These bounds also allow us to avoid the Kiefer-Wolfowitz exploration used by Zimmert and Lattimore [33]. Indeed, their algorithm mixes the exponential-weights distribution with a fixed exploration distribution to guarantee that $x^\top M_t^{-1}x = O(d/\lambda)$ for every action $x \in \mathcal{X}$ and every round $t \in [T]$. This bound, in turn, allows them to control the sum over $t \in [T]$ of the second moments of their surrogate losses defined in (5), obtaining a contribution to regret of the same order as the one we obtain from (12). Intuitively, Kiefer-Wolfowitz exploration is unnecessary here because our projection step keeps each sampling distribution p_t in $\mathcal{P}_{\lambda/2}$ while preserving its weighted m-set structure. Equations (7) and (11) therefore already guarantee that the leverage score $x^\top M_t^{-1}x$ is $O(d/\lambda)$ for every action $x \in \mathcal{X}$ and every round $t \in [T]$.

Finally, we discuss how to implement our algorithm in polynomial time. Since our algorithm guarantees that each p_t is a weighted m-set distribution, the methods of Chen and Liu [14, Sections 2 and 4] allow us to sample an action $X_t \sim p_t$ and compute the marginal vector $\mu(p_t)$ and second-moment matrix $M(p_t)$ in $\mathrm{poly}(d, m)$ time.[5]

The step that requires careful attention is the KL projection in (9). We need to implement this step in polynomial time while ensuring that the resulting distribution is a weighted m-set distribution. To achieve this, we allow the resulting distribution to lie in the larger set $\mathcal{P}_{\lambda/2}$ and to satisfy the optimality condition for the KL projection approximately. We compute such a distribution by solving a suitably defined convex optimization problem using the ellipsoid method, with a computational cost of $\mathrm{poly}(d, m, \log T)$ per projection.

3.2 Lower bound

Our second contribution is a matching high-probability lower bound.

Theorem (Informal version of Theorem 2)**.** *There exists a universal constant $\delta_0 > 0$ such that the following holds. For every $\delta \in (0, \delta_0]$ and every horizon T above a threshold bounded by a polynomial in d, m, and $\log(1/\delta)$, for every randomized algorithm there exists an adaptive non-anticipating adversary with action losses in $[0, 1]$ for which, with probability at least δ,*

$$R_T = \Omega\left(\sqrt{dT\left(\log\binom{d}{m} + \log(1/\delta)\right)}\right).$$

Together with Theorem 1, this identifies the minimax high-probability regret up to universal constants for every d and m.

Challenges and techniques. We give an intuition for how adaptivity creates the $\log K$ term in the lower bound. To make the argument easier to follow, we start with the case $m = 1$ and $d = K$. This coincides with the ordinary K-armed bandit problem. Thus, we use the terms item and arm interchangeably. In the following, we choose a uniformly random set W of $K' + 1$ arms, where $K' = K/32$, and select J uniformly from W. For simplicity, we assume that $K/32$ and $4T/K$ are integers.

We compare two instances, writing P and Q for the corresponding probability distributions of W, J, and all the learner's selected arms and observed losses. In P, the favorable arms are those

[5] The notation $\mathrm{poly}(\cdot)$ denotes a quantity bounded above by a polynomial in the input arguments.

in $W \setminus \{J\}$. In Q, they are all the arms in W. By a favorable arm, we mean an arm that receives a loss discount, which we call a bonus, while it is active. Here, by an active arm we mean an arm that has been played fewer than $N_0 := 4T/K$ times. Indeed, we construct the instances so that the adversary switches off the bonus after the arm's N_0th play, making it adaptive. Arms outside the favorable set receive only the baseline loss throughout. Thus, the two instances differ only on J. Specifically, the arm J receives the bonus while active in Q, and no bonus in P.

At each round t, we draw Z_t from the same fixed distribution, independently of W, J, the previous draws, and the learner's randomness. We use this random variable, shared by all arms, to define the loss of a baseline arm (an arm receiving no bonus) as $b(Z_t)$, while we let the loss of an active favorable arm be $b(Z_t) - \varepsilon g(Z_t)$, where b and g are fixed functions.

We choose this distribution and the functions b and g so that, for sufficiently small $\varepsilon > 0$, both losses lie in $[0, 1]$. Moreover, we require $0 \le g(Z_t) \le 1$ and $\mathbb{E}[g(Z_t)] = \gamma$, where γ is a sufficiently large universal constant.[6]

Writing ν_0 and ν_ε for the distributions of $b(Z_t)$ and $b(Z_t) - \varepsilon g(Z_t)$, respectively, their mean difference is $\mathbb{E}_{\nu_0}[X] - \mathbb{E}_{\nu_\varepsilon}[X] = \varepsilon\gamma = \Theta(\varepsilon)$, and we also require $\mathrm{KL}(\nu_0 \| \nu_\varepsilon) = O(\varepsilon^2)$. The bonus obtained at round t by playing an active favorable arm compared with a non-favorable one is therefore $\varepsilon g(Z_t)$, and the common baseline $b(Z_t)$ cancels when computing regret.

We notice that in instance P, the learner can receive a bonus on at most $N_0 K' = T/8$ rounds, so its total bonus is at most $\varepsilon T/8$. A favorable arm that remains active throughout all T rounds, however, receives the bonus on every round, for a total of $\varepsilon \sum_{t=1}^{T} g(Z_t)$. This sum has expectation $\varepsilon\gamma T$, since $\mathbb{E}[g(Z_t)] = \gamma$. For our sufficiently large choice of γ, this exceeds the learner's maximum total bonus by an amount of order εT. Thus, ideally, to incur small regret, the learner would like to switch off all favorable arms with high probability. We now explain why this is not the case and some favorable arm remains active with constant probability under P.

Let $n_i(T)$ be the number of times the learner plays arm i, and let A be the event that every arm in $W \setminus \{J\}$ is played at least N_0 times. Consider the event

$$E = \big\{n_i(T) \ge N_0 \text{ for every } i \in W \setminus \{J\}\big\} \cap \{n_J(T) < N_0\} = A \cap \{n_J(T) < N_0\}.$$

Thus, E is the event A together with the requirement that J remains active. We observe that, for any algorithm, at most $T/N_0 = K/4$ arms can be played at least N_0 times, so at least $3K/4$ arms remain active at the end of the T rounds, according to the definition of an active arm.

In instance P, the initial sampling is equivalent to first choosing the favorable set $W \setminus \{J\}$ uniformly among all sets of K' arms and then selecting J uniformly outside this set. Consequently, conditional on the set $W \setminus \{J\}$, J remains uniform among the arms outside $W \setminus \{J\}$ and is independent of the learner's selected arms.

Assume first that $P(A) > 0$. Under the event A, at the end of the T rounds, all the active arms are outside $W \setminus \{J\}$, so the probability that J is active is at least $3/4$. Consequently, averaging over the favorable set and using the fact that at least $3K/4$ arms remain active at the end of the T rounds, we have $P(E) = P(E \mid A)P(A) \ge 3P(A)/4$. If $P(A) = 0$, then $P(E) = 0$ since $E \subseteq A$, so the same bound holds.

Under Q, every arm in W receives the bonus up to and including its N_0th play, regardless of which arm is selected as J. Given W, J therefore remains uniform in W and is independent of which arms remain active. For E to occur, exactly one arm in W must remain active at the end of the T rounds, and it must be J. If exactly one arm in W remains active, the probability that this

[6] Throughout this argument, $\mathbb{P}$ and $\mathbb{E}$ refer to the learner's randomness, the choices of W and J, and the variables $Z_1, \ldots, Z_T$ used to generate the adversary's losses, in instance P.

arm is J is $1/(K'+1)$. If none or more than one remain active in W at the end, E cannot occur. Hence $Q(E) \le 1/(K'+1) \le 32/K$.

We now choose ε so that $N_0\varepsilon^2 = c\log K$, where $c > 0$ is a sufficiently small fixed constant. The two instances differ only when the learner plays J while it is active, which happens at most N_0 times. The KL chain rule of Lattimore and Szepesvári [28, Exercise 14.12, Eq. (14.17)] therefore gives $\mathrm{KL}(P\|Q) = O(N_0\varepsilon^2)$. By data processing applied to the indicator of E, and the binary-KL inequality[7], we obtain, for sufficiently large K,

$$P(A)\log K \le O(\mathrm{KL}(P\|Q)+1) \le O(N_0\varepsilon^2).$$

For a sufficiently small c and sufficiently large K, this gives $P(A) \le 1/4$.

Finally, we translate this into a lower bound on realized regret. On A^c, some favorable arm remains active throughout and receives total bonus $\varepsilon\sum_{t=1}^T g(Z_t)$. On A, every arm's total bonus is still nonnegative. Since the common term $b(Z_t)$ cancels, the $\varepsilon T/8$ bound on the learner's total bonus gives

$$\begin{aligned}\mathbb{E}[R_T] &\ge \varepsilon\left(\mathbb{E}\left[\sum_{t=1}^T g(Z_t)\mathbf{1}_{A^c}\right] - \frac{T}{8}\right)\\ &= \varepsilon\left(\mathbb{E}\left[\sum_{t=1}^T g(Z_t)\left(\mathbf{1}_{A^c}+\mathbf{1}_A-\mathbf{1}_A\right)\right] - \frac{T}{8}\right) \ge \varepsilon\left(\gamma T - TP(A) - \frac{T}{8}\right).\end{aligned}$$

Here we used $\mathbb{E}[\sum_{t=1}^T g(Z_t)] = \gamma T$, the choice of γ, $P(A) \le 1/4$, and $0 \le \sum_{t=1}^T g(Z_t) \le T$.

Since $R_T \le \varepsilon T$ on every realization and $\mathbb{E}[R_T] = \Omega(\varepsilon T)$, an elementary probabilistic argument (see Lemma 13) gives $\mathbb{P}(R_T \ge c_0\varepsilon T) \ge c_0$ for some universal constant $c_0 > 0$. Our choice of $\varepsilon = \Theta(\sqrt{cK\log K/T})$ gives $\varepsilon T = \Theta(\sqrt{KT\log K})$, proving the claimed order with constant probability for sufficiently large K and $T \ge CK\log K$, where C is a sufficiently large universal constant. Notice that the adversary above randomizes its choices. The probability guarantee is an average over its choices of W, J, and $Z_1,\dots,Z_T$. By an averaging argument, some fixed choice of these values preserves the same guarantee over the learner's randomness alone. This gives a deterministic adaptive non-anticipating adversary of the type introduced in Section 2.

In our lower bound construction, we can take Z_t uniform on $[-1,1]$, $b(z) = (1+z)/2$, and $g(z) = 1-z^2$. These choices satisfy the properties used in the preceding argument. The bonus $\varepsilon g(Z_t)$ always lies between 0 and ε and has mean $\Theta(\varepsilon)$, while the common baseline $b(Z_t)$ cancels in the regret. Moreover, since $g(-1) = g(1) = 0$, for sufficiently small ε the losses with and without the bonus have the same range of possible values, $[0,1]$. This choice of b and g also guarantees the KL bound used above.

The same properties are useful when the learner selects m items and observes only the sum of their losses. By the complementation reduction in Lemma 11, it suffices to consider $m \le d/2$. We divide each item's loss by m and compare two instances that differ only in whether one item is favorable. Suppose that the learner selects this item while it is still active. Writing h for the number of other selected items that are favorable and active, the observed loss has the distribution of $b(Z_t) - (\varepsilon h/m)g(Z_t)$ in the first instance and of $b(Z_t) - (\varepsilon(h+1)/m)g(Z_t)$ in the second. The common noise therefore remains of the same size, while the two bonus coefficients differ by only ε/m. The choice of b and g guarantees that the KL divergence between these distributions is $O(\varepsilon^2/m^2)$, regardless of how many other selected items are favorable and active. Thus, the smaller

[7] For $a \in [0,1]$ and $b \in (0,1)$, we have $\mathrm{KL}(\mathrm{Ber}(a)\|\,\mathrm{Ber}(b)) \ge a\log(1/b) - \log 2$.

KL allows us to choose $\varepsilon = \Theta(\sqrt{dm\log(d/m)/T})$ while keeping at least m favorable items active with sufficiently high probability.

We use a construction based on this idea, choosing each item to be favorable independently with a fixed probability, to prove a lower bound of order $\sqrt{dT\log K}$, where $K = \binom{d}{m}$, when m/d is at most a universal constant $r_\star > 0$, as shown in Appendix E.2.1. We also use a similar construction to prove the same bound in the regime considered in Appendix E.2.2, where $r_\star < m/d \le 1/2$. Here, all items remain active throughout, and the adversary is therefore oblivious. This construction gives a lower bound of order $m\sqrt{T}$, which already yields $\Omega(\sqrt{dT\log K})$ since $m > r_\star d$ and $\log K \le d\log 2$. Finally, a further construction gives the lower bound of order $\sqrt{dT\log(1/\delta)}$ in Appendix E.3.

All three constructions use item losses with a structure similar to the one above, combining a random baseline shared by all items with small changes proportional to $g(Z_t)$, with different scales and fixed additional terms. In each construction, the common random baseline cancels in the regret and the losses are guaranteed to remain in a bounded range.

Finally, we choose the construction that gives the larger bound. Since $\max\{\log(K), \log(1/\delta)\} \ge (\log(K) + \log(1/\delta))/2$, we obtain the lower bound $\Omega(\sqrt{dT(\log(K) + \log(1/\delta))})$ with probability at least δ, under the confidence and time horizon assumptions of Theorem 2.

4 Related work

Our work belongs to the literature on adversarial combinatorial bandits, where, over T rounds, a learner selects one of K feasible subsets of d items and observes only the total loss of the selected subset. In our setting, the feasible actions are all subsets of size m, so $K = \binom{d}{m}$.

Cesa-Bianchi and Lugosi [12] introduced ComBand for general adversarial combinatorial bandits. For finite action sets, Bubeck et al. [10] proposed EXP2 with John's exploration, which achieves $O(\sqrt{dT\log K})$ pseudo-regret against adaptive non-anticipating adversaries under our action-level normalization. For oblivious adversaries, this also bounds expected regret against the best action in hindsight. Later, Zimmert and Lattimore [33] obtained the high-probability bound $O(\sqrt{dT\log(K/\delta)})$ with probability at least $1-\delta$ against adaptive non-anticipating adversaries. The direct implementations of these finite-action algorithms maintain one weight per action and can therefore require exponential space in d.

In parallel, computational efficiency has been studied through structured representations of the action set. Combes et al. [17] introduced CombEXP, which admits a polynomial-time implementation for m-set actions. For $m \le d/2$, item losses in $[0, 1/m]$, and sufficiently large T, it achieves $\widetilde{O}(\sqrt{dmT})$ expected regret against oblivious adversaries. Subsequently, Maiti et al. [30] gave a polynomial-time algorithm whose guarantee yields $\widetilde{O}(m\sqrt{dT})$ high-probability regret against adaptive adversaries for $m \le d/2$. In this paper, we improve this bound to $\widetilde{O}(\sqrt{dmT})$ while retaining computational efficiency.

Under our action-level normalization, Ito et al. [22] obtained an expected-regret lower bound of $\Omega(\sqrt{dmT})$ against oblivious adversaries for the full class of m-set actions considered here, when $m \le d/2$ and $T = \Omega(dm^{3/2})$. In this work, we prove a high-probability lower bound such that for every sufficiently small δ and sufficiently large T, every randomized algorithm admits an adaptive non-anticipating adversary for which regret is at least $c\sqrt{dT\log(K/\delta)}$ with probability at least δ, where $c > 0$ is a universal constant. However, the two lower bounds are not directly comparable, as theirs concerns expected regret, whereas ours holds with a given probability.

For ordinary bandits, corresponding to $m = 1$ and hence $K = d$, Gerchinovitz and Lattimore [18] prove a high-probability lower bound of order $\sqrt{KT\log(1/\delta)}$ against oblivious adversaries. Against adaptive non-anticipating adversaries, our lower bound has order $\sqrt{KT\log(K/\delta)}$, matching the

corresponding upper bound and closing the $\sqrt{\log K}$ gap.

Finally, we combine the pseudoinverse estimate of Cesari and Colomboni [13, Corollary 1.2] with an identity for the inverse second-moment matrix to derive an affine upper bound on the leverage score for weighted m-set distributions. This yields a surrogate loss that preserves this family under exponential-weights updates. An extended discussion appears in Appendix A.

5 An efficient minimax-optimal algorithm for m-set bandits

5.1 Affine upper bound on the leverage score

For weighted m-set distributions $p \in \mathcal{P}_\lambda$, in Lemma 1 we derive an affine upper bound on the leverage score $x^\top M(p)^{-1}x$ introduced in Eq. (2) by combining an identity relating the inverse of the second-moment matrix $M(p)$ to its covariance pseudoinverse with the pseudoinverse estimate of Cesari and Colomboni [13, Corollary 1.2]. The complete proof appears in Appendix B.

Lemma 1. *Let $p \in \mathcal{P}_\lambda$ be a weighted m-set distribution, where $1 \le m \le d-1$, and let μ, M, and Σ be its marginal vector, second-moment matrix, and covariance matrix, respectively. For every $x \in \mathcal{X}$, define*

$$\phi_p(x) := 1 + \sum_{i=1}^{d} \frac{(x_i - \mu_i)^2}{\mu_i(1-\mu_i)}.$$

Then, for every $x \in \mathcal{X}$, $x^\top M^{-1}x \le \phi_p(x)$. Moreover, $4\phi_p(x) = c_p^\top x$, where

$$c_{p,i} := 4\left(\frac{1 + \sum_{j=1}^{d} \frac{\mu_j}{1-\mu_j}}{m} + \frac{1-2\mu_i}{\mu_i(1-\mu_i)}\right) \qquad \text{for every } i \in [d]. \tag{13}$$

Finally, $\mathbb{E}_{X\sim p}[\phi_p(X)] = d+1$, and, for every $x \in \mathcal{X}$, $0 \le \phi_p(x) \le 1 + 2d/\lambda$.

5.2 Algorithm and high-probability regret

In this section, we present Algorithm 1 and provide its high-probability regret guarantee. At each round t, the learner draws an action X_t from a weighted m-set distribution $p_t = p_{\theta_t} \in \mathcal{P}_{\lambda/2}$, represented by $\theta_t \in \mathbb{R}^d$, and observes only the scalar loss $Y_t = X_t^\top \ell_t$. Using this feedback, it forms the loss estimate $\widehat{\ell}_t = M_t^{-1}X_tY_t$, where $M_t = \mathbb{E}_{X\sim p_t}[XX^\top]$, and can thus estimate, for every $x \in \mathcal{X}$, the surrogate loss $Z_t(x) = x^\top\widehat{\ell}_t - \eta c_t^\top x$ defined in Equation (8), where we write $c_t := c_{p_t}$. The distribution p_t is then updated through an approximate version of online mirror descent (OMD) with the negative entropy regularizer $\sum_{x\in\mathcal{X}} p(x)\log p(x)$, with $0\log 0 = 0$. The exact update would be

$$p_{t+1}^\star \in \operatorname*{arg\,min}_{p\in\mathcal{P}_\lambda} \left\{\eta\mathbb{E}_{X\sim p}[Z_t(X)] + \mathrm{KL}(p\|p_t)\right\}. \tag{14}$$

Equivalently, this consists of the multiplicative-weights (MWU) step

$$\widetilde{p}_{t+1}(x) = \frac{p_t(x)e^{-\eta Z_t(x)}}{\sum_{y\in\mathcal{X}} p_t(y)e^{-\eta Z_t(y)}}, \qquad x \in \mathcal{X}, \tag{15}$$

followed by the exact KL projection onto $\mathcal{P}_\lambda$, that is,

$$p_{t+1}^\star \in \operatorname*{arg\,min}_{p\in\mathcal{P}_\lambda} \mathrm{KL}(p\|\widetilde{p}_{t+1}). \tag{16}$$

Algorithm 1

Require: $T \geq 1, \delta \in (0,1)$

Set $\eta \leftarrow \min\left\{\frac{1}{256d}, \sqrt{\frac{\log(12K/\delta)}{320dT}}\right\}$

Set $\lambda \leftarrow 128\eta d$ and $\varepsilon_{\mathrm{p}} \leftarrow \eta/T$

Initialize $\theta_1 \leftarrow \mathbf{0}$ and $p_1 \leftarrow p_{\theta_1} = U$ ▷ U is the uniform distribution

for $t = 1, \ldots, T$ **do**

  Compute $\mu_t \leftarrow \mu(p_t)$, $M_t \leftarrow M(p_t) = \mathbb{E}_{X\sim p_t}[XX^\top]$

  Compute $c_t \leftarrow c_{p_t}$ using (13)

  Draw $X_t \sim p_t$

  Observe $Y_t = X_t^\top \ell_t$

  Compute $\widehat{\ell}_t \leftarrow M_t^{-1} X_t Y_t$

  Set $z_t \leftarrow \widehat{\ell}_t - \eta c_t$, $\widetilde{\theta}_{t+1} \leftarrow \theta_t - \eta z_t$, and $\widetilde{p}_{t+1} \leftarrow p_{\widetilde{\theta}_{t+1}}$ ▷ MWU update

  Run the projection in Appendix D.1 with input $(\widetilde{\theta}_{t+1}, \eta, T)$ to obtain θ_{t+1}

  Set $p_{t+1} \leftarrow p_{\theta_{t+1}}$

end for

Since $Z_t(x) = z_t^\top x$, where $z_t := \widehat{\ell}_t - \eta c_{p_t}$ and $p_t(x) = p_{\theta_t}(x) \propto e^{\theta_t^\top x}$, the MWU step in (15) preserves the weighted m-set form. Specifically, $\widetilde{p}_{t+1}$ is a weighted m-set distribution described by the vector $\widetilde{\theta}_{t+1} := \theta_t - \eta z_t$.

To implement the KL projection in (16), we use the deterministic procedure described in Appendix D.1. At a high level, the procedure uses the ellipsoid method [19] to approximately solve a suitably defined convex optimization problem. The procedure eventually returns a vector $\theta_{t+1} \in \mathbb{R}^d$ such that the resulting distribution $p_{t+1} = p_{\theta_{t+1}}$ is a weighted m-set distribution that belongs to $\mathcal{P}_{\lambda/2}$ and satisfies, for every $q \in \mathcal{P}_\lambda$, the following inequality

$$\mathrm{KL}(q\|p_{t+1}) + \mathrm{KL}(p_{t+1}\|\widetilde{p}_{t+1}) \leq \mathrm{KL}(q\|\widetilde{p}_{t+1}) + \varepsilon_{\mathrm{p}}, \tag{17}$$

where $\varepsilon_{\mathrm{p}} := \eta/T$. In Theorem 1, we provide the regret guarantees achieved by Algorithm 1.

Theorem 1. *There exists a universal constant $C > 0$ such that the following holds. Let $T \geq 1$ be an integer, let $\delta \in (0,1)$, and assume $1 \leq m \leq d-1$. Run Algorithm 1. Then, with probability at least $1-\delta$,*

$$R_T \leq C\sqrt{dT\left(\log K + \log\frac{1}{\delta}\right)}, \tag{18}$$

where $K = \binom{d}{m}$. One may take $C = 160$.

Proof sketch. For every $x \in \mathcal{X}$, let $q_x := (1-\lambda)\delta_x + \lambda U \in \mathcal{P}_\lambda$, where δ_x is the Dirac distribution on action x. We establish the bounds simultaneously for all x, so we can subsequently choose a best action in hindsight. In the following, we denote $\widehat{y}_t(x) := x^\top \widehat{\ell}_t$, where $\widehat{\ell}_t$ is defined in (6). Since the matrix M_t is symmetric and invertible and satisfies $M_t\mathbf{1} = m\mu_t$, we have

$$\mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)] = \mu_t^\top M_t^{-1} X_t Y_t = \frac{\mathbf{1}^\top X_t}{m} Y_t = Y_t.$$

Thus the learner's average estimated loss equals its realized loss. Since action losses belong to

$[-1, 1]$, it also holds

$$R_T(x) := R_T(\delta_x) = \sum_{t=1}^{T} \big(Y_t - x^\top \ell_t\big) \le \underbrace{\sum_{t=1}^{T} \Big(Y_t - \mathbb{E}_{X\sim q_x}[X^\top \ell_t]\Big)}_{=:(\star)} + 2\lambda T. \tag{19}$$

Adding and subtracting $\mathbb{E}_{X\sim q_x}[\widehat{y}_t(X)]$ gives

$$(\star) = \sum_{t=1}^{T} \left(\mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)] - \mathbb{E}_{X\sim q_x}[\widehat{y}_t(X)]\right) + \sum_{t=1}^{T} \mathbb{E}_{X\sim q_x}[\widehat{y}_t(X) - X^\top \ell_t]. \tag{20}$$

We now bound the two terms in (20).

1. For the first term in (20), we use $\widehat{y}_t(X) = Z_t(X) + \eta c_t^\top X$ and $\mathrm{KL}(q_x \| p_1) \le \log K$. An exponential-weights argument, as in Zimmert and Lattimore [33], combined with the approximate KL projection guarantee in (17) (see Lemma 4), gives

$$\begin{aligned}
&\sum_{t=1}^{T} \left(\mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)] - \mathbb{E}_{X\sim q_x}[\widehat{y}_t(X)]\right) \\
&\quad \le \frac{\log K}{\eta} + 2\eta \sum_{t=1}^{T} \mathbb{E}_{X\sim p_t}[Z_t(X)^2] + \eta \sum_{t=1}^{T} \mathbb{E}_{X\sim p_t}[c_t^\top X] - \eta \sum_{t=1}^{T} \mathbb{E}_{X\sim q_x}[c_t^\top X] + O(1),
\end{aligned}$$

   where the term $O(1)$ accounts for the accumulated projection error ε_{p}. To control the second moment of $Z_t(X)$, we observe that $\mathbb{E}_{X\sim p_t}[Z_t(X)^2] \le 2\mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)^2] + 2\eta^2 \mathbb{E}_{X\sim p_t}[(c_t^\top X)^2]$. Thanks to Lemma 1 and the fact that distributions p_t belong to $\mathcal{P}_{\lambda/2}$ at every round $t \in [T]$ (see Proposition 1), we have $\mathbb{E}_{X\sim p_t}[c_t^\top X] = 4(d+1)$ and $\max_{x\in\mathcal{X}} c_t^\top x = O(d/\lambda)$. Thus, the following holds:

$$\mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)^2] = Y_t^2 X_t^\top M_t^{-1} X_t \le \phi_{p_t}(X_t), \quad \mathbb{E}_{X\sim p_t}[(c_t^\top X)^2] \le O(d^2/\lambda).$$

   Using the above observations and a concentration argument over the quantity $\sum_{t=1}^{T} \phi_{p_t}(X_t)$ (see Lemma 6) gives, with probability at least $1 - \delta/2$, the following:

$$2\eta \sum_{t=1}^{T} \mathbb{E}_{X\sim p_t}[Z_t(X)^2] \le O\left(\eta d T + \frac{\eta d}{\lambda} \log \frac{2}{\delta} + \frac{\eta^3 d^2 T}{\lambda}\right) = O\left(\eta d T + \log \frac{1}{\delta}\right),$$

   where the last step uses $\lambda = \Theta(\eta d)$ and $\eta \le 1$. Moreover, $\log(2/\delta) = \log(1/\delta) + \log 2$, and $\eta d T \ge 1/256$ allows us to absorb the constant $\log 2$ into $O(\eta d T)$. Therefore, we have

$$\sum_{t=1}^{T} \left(\mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)] - \mathbb{E}_{X\sim q_x}[\widehat{y}_t(X)]\right) \le \frac{\log K}{\eta} - \eta \sum_{t=1}^{T} \mathbb{E}_{X\sim q_x}[c_t^\top X] + O\left(\eta d T + \log \frac{1}{\delta}\right). \tag{21}$$

2. For the second term in (20), a concentration bound (see Lemma 5) with failure probability $\delta/(2K)$, followed by a union bound, gives simultaneously for every $x \in \mathcal{X}$, with probability at least $1 - \delta/2$,

$$\sum_{t=1}^{T} \mathbb{E}_{X\sim q_x}[\widehat{y}_t(X) - X^\top \ell_t] \le \frac{\eta}{4} \sum_{t=1}^{T} \mathbb{E}_{X\sim q_x}[c_t^\top X] + \frac{\log(4K/\delta)}{\eta}. \tag{22}$$

Combining (21) and (22) in (20) yields, with probability at least $1-\delta$, simultaneously for all $x \in \mathcal{X}$,

$$(\star) \le O\left(\frac{\log K + \log(1/\delta)}{\eta} + \eta d T + 1\right) + \underbrace{\left(-\eta + \frac{\eta}{4}\right) \sum_{t=1}^{T} \mathbb{E}_{X \sim q_x}[c_t^\top X]}_{\le 0}. \tag{23}$$

Taking x to be a best action in hindsight in (19), applying (23), and substituting the algorithm's parameters gives the claimed rate. When $\eta = 1/(256d)$, we obtain the claimed rate using $R_T \le 2T$. The full proof is provided in Appendix C.1. □

We observe that, since $\log K \le s \log(ed/s)$, the bound in Theorem 1 is $\widetilde{O}(\sqrt{dsT})$, where $s = \min\{m, d-m\}$. In particular, when $m \le d/2$, it is $\widetilde{O}(\sqrt{dmT})$.

5.3 Polynomial-time implementation

We now show that Algorithm 1 can be implemented in polynomial time and space without enumerating $\mathcal{X}$, and that every sampling distribution p_t is a weighted m-set distribution in $\mathcal{P}_{\lambda/2}$.[8]

Proposition 1. *For every $t \in \{2, \dots, T+1\}$, Algorithm 1 maintains a weighted m-set distribution $p_t = p_{\theta_t} \in \mathcal{P}_{\lambda/2}$, represented by $\theta_t \in \mathbb{R}^d$ satisfying Equation (17) at round $t-1$. Moreover, the algorithm can be implemented in $T \cdot \mathrm{poly}(d, m, \log T)$ time and $\mathrm{poly}(d, m)$ space, without enumerating $\mathcal{X}$.*

Proof sketch. We first show by induction that every distribution $p_t = p_{\theta_t}$ chosen by our algorithm is a weighted m-set distribution belonging to $\mathcal{P}_{\lambda/2}$. Initially, $\theta_1 = \mathbf{0}$, so $p_1 = U \in \mathcal{P}_\lambda \subseteq \mathcal{P}_{\lambda/2}$. Suppose the claim holds at round t. Since the surrogate loss is linear, $Z_t(x) = z_t^\top x$, the MWU update preserves the weighted m-set structure. Thus, $\widetilde{p}_{t+1} = p_{\widetilde{\theta}_{t+1}}$, where $\widetilde{\theta}_{t+1} = \theta_t - \eta z_t$.

The challenging step is to show that the approximate KL projection at line 11 of Algorithm 1 also preserves the weighted m-set structure and returns a distribution in $\mathcal{P}_{\lambda/2}$. To achieve this, we formulate a convex optimization problem in d variables. Specifically, in Appendix D.1, we formulate the exact projection onto $\mathcal{P}_\lambda$ through the minimization, with respect to v, of

$$G_{\widetilde{\theta}_{t+1}}(v) = F(\widetilde{\theta}_{t+1} + v) - \underline{\mu}^\top v_+ + \overline{\mu}^\top v_-, \tag{24}$$

where $F(\theta) := \log \sum_{x \in \mathcal{X}} \exp(\theta^\top x)$ is the log-partition function, and $\underline{\mu}, \overline{\mu} \in \mathbb{R}^d$ have coordinates $\underline{\mu}_i = \lambda r$ and $\overline{\mu}_i = 1 - \lambda(1-r)$, with $r = m/d$. Moreover, $(v_+)_i = \max\{v_i, 0\}$ and $(v_-)_i = \max\{-v_i, 0\}$. Using the ellipsoid method, we show how to approximately minimize (24) over a suitably bounded domain and obtain a vector v. Then the procedure returns $\theta_{t+1} = \widetilde{\theta}_{t+1} + v$.

As established in Appendix D, the resulting weighted m-set distribution $p_{t+1} = p_{\theta_{t+1}}$ belongs to $\mathcal{P}_{\lambda/2}$ and satisfies (17) with error $\varepsilon_{\mathrm{p}} = \eta/T$. This closes the induction for all t.

It remains to show that our algorithm can be implemented efficiently. The methods of Chen and Liu [14, Sections 2 and 4], recalled in Appendix D.2, compute the second-moment matrix $M(p_t)$, the marginal vector $\mu(p_t)$, and sample from p_{θ_t} in $\mathrm{poly}(d, m)$ time and space, without enumerating $\mathcal{X}$. Computing the loss estimate $\widehat{\ell}_t$ requires solving a d-dimensional linear system, while the MWU update requires only updating the d parameters. For the projection step, which amounts to minimizing the objective in (24) over a suitable domain, we observe that this objective

[8] In this paper, we focus on the arithmetic model, in which each arithmetic operation on real numbers and each evaluation of exp or log has unit cost, independently of the precision needed to represent the numbers. We also allow exact comparisons of real numbers and independent samples from $\mathrm{Unif}[0,1]$ at unit cost.

is convex and that its value and a subgradient[9] can likewise be evaluated in $\mathrm{poly}(d, m)$ time. The ellipsoid method, as discussed in Appendix D.1, therefore computes the required approximate solution in $\mathrm{poly}(d, m, \log T)$ arithmetic operations and $\mathrm{poly}(d, m)$ space. By storing the required quantities only for the current round and reusing the workspace, we can prove that our algorithm requires time $T \cdot \mathrm{poly}(d, m, \log T)$ and space $\mathrm{poly}(d, m)$. □

5.4 Matching high-probability lower bound

We complement the upper bound with a matching high-probability lower bound.

Theorem 2. *There exist universal constants $c > 0$ and $\delta_0 \in (0, 1)$ and a threshold $\mathfrak{T}(d, m, \delta)$ bounded by a polynomial in d, m, and $\log(1/\delta)$ such that the following holds. Let $d \geq 2$, $1 \leq m \leq d - 1$, $\delta \in (0, \delta_0]$, and $T \geq \mathfrak{T}(d, m, \delta)$. For every randomized algorithm on $\mathcal{X}_{d,m}$, there exists an adaptive non-anticipating adversary whose action losses belong to $[0, 1]$ and such that*

$$\mathbb{P}\left(R_T \geq c\sqrt{dT\left(\log\binom{d}{m} + \log\frac{1}{\delta}\right)}\right) \geq \delta.$$

Proof sketch. To develop our lower bound, we work on $\mathcal{X}_{d,m}$ with $m \leq d/2$. We do this without loss of generality since, by Lemma 11, we can transfer the lower bound obtained for $m \leq d/2$ to $\mathcal{X}_{d,m}$ with $m > d/2$, with the same regret and probability guarantees.

In addition, we temporarily allow randomized non-anticipating adversaries. Given $\mathcal{F}_{t-1}$, the adversary may use its own randomness to choose ℓ_t without observing the learner's current action X_t. This auxiliary randomness is sampled independently of the learner's private randomness. This is without loss of generality, as Lemma 10 allows us to transfer any high-probability lower bound obtained against a randomized adversary to a deterministic adversary of the type introduced in Section 2, while preserving the same guarantee.

We divide the analysis into three cases. The first two establish the $\log K$ contribution, according to whether $r \leq r_\star$ or $r > r_\star$, where $r_\star := 2^{-32}$, and the third establishes the $\log(1/\delta)$ contribution. In all three cases, we show that the scalar feedback Y_t observed by the algorithm, once the instance, the past history, and the current action X_t are fixed, is of the form $Y_t = a_t + b_t\psi_{u_t}(Z_t)$, where $Z_t \sim \mathrm{Unif}[-1, 1]$ is fresh independent noise, and

$$g(z) := 1 - z^2, \qquad \psi_u(z) := z/2 + ug(z), \qquad z \in [-1, 1].$$

Here, u_t is a fixed parameter under this conditioning, as are a_t and $b_t > 0$. This allows us to compare the feedback distributions in two instances by comparing their values of u_t. Writing p_u for the density of $\psi_u(Z_t)$, Lemma 12 gives $\mathrm{KL}(p_u \| p_v) \leq 80(u - v)^2$ for $|u|, |v| \leq 1/16$.

Since the common shift a_t and scaling $b_t > 0$ are fixed conditional on the past history and current action X_t, they do not change KL divergence. Hence, the same bound holds for the feedback distributions conditional on the same quantities.

- **The $\Omega(\sqrt{dT \log K})$ lower bound.** We distinguish two cases, with $r_\star := 2^{-32}$.
  - **The case $r \leq r_\star$.** To build the instances, we independently draw indicator variables $U_i \sim \mathrm{Ber}(\rho)$ and noise variables $Z_t \sim \mathrm{Unif}[-1, 1]$, independently of the algorithm's randomness, with $\rho := 1/256$ and $N_0 := \lceil 8Tr \rceil$. Let $A_{t,i} := \mathbf{1}\{\sum_{\tau<t} X_{\tau,i} < N_0\}$. We say that item i is

[9] For a convex function f, a subgradient at v is a vector ζ such that $f(u) \geq f(v) + \zeta^\top(u - v)$ for every u.

active at round t if it has been selected fewer than N_0 times before round t, equivalently, if $A_{t,i} = 1$. Choose $\varepsilon := \Theta(m\sqrt{\log(1/r)/N_0})$, in a way that $\varepsilon \le 1/16$, and define

$$\ell_{t,i} := \frac{1 + Z_t}{2m} - \frac{\varepsilon}{m} g(Z_t) A_{t,i} U_i. \tag{25}$$

We observe that item i receives a loss discount when $U_i = 1$ and it is still active. Thus, when $U_i = 1$ we say that item i is favorable. Here, the adversary is adaptive because it removes this discount once the algorithm has selected the item N_0 times. We also introduce the following scores

$$S_i := U_i \sum_{t=1}^{T} A_{t,i} g(Z_t), \qquad S_{\mathrm{L}} := \sum_{t=1}^{T} g(Z_t) \sum_{i=1}^{d} A_{t,i} X_{t,i} U_i. \tag{26}$$

Since $(1 + Z_t)/(2m)$ is common to all items, $(\varepsilon/m)S_i$ represents the item i's total discount, while $(\varepsilon/m)S_{\mathrm{L}}$ is the discount collected by the algorithm. The best fixed action selects the m largest scores S_i, whose sum we denote by $\mathrm{Top}_m(S)$, giving

$$R_T = \frac{\varepsilon}{m}\big(\mathrm{Top}_m(S) - S_{\mathrm{L}}\big). \tag{27}$$

Each item contributes to S_{L} at most N_0 times, since, by the definition of $A_{t,i}$, it becomes inactive after its N_0-th selection. Since $g \le 1$, we have $S_{\mathrm{L}} \le N_0 \sum_i U_i$, hence $\mathbb{E}[S_{\mathrm{L}}] \le (N_0 d)/256 \le (9mT)/256$, where the last inequality uses $N_0 \le 9Tr$.

We now want to lower bound $\mathbb{E}[\mathrm{Top}_m(S)]$. A favorable item i (that is, one with $U_i = 1$) that is still active at the start of round T must have been active throughout all T rounds, and therefore has score $S_i = \sum_{t=1}^{T} g(Z_t)$. We will show that at least m favorable items remain active with probability at least $15/16$.

Fix the algorithm's private randomness and all the U_j with $j \neq i$, and let P_i^0, P_i^1 be the history laws for $U_i = 0, 1$.[10] At each round t, given the same past history, the action X_t and the active items are the same under both laws. The feedback in both instances is

$$Y_t = \frac{1}{2} + \psi_{u_t}(Z_t) \quad \text{and} \quad u_t = -\frac{\varepsilon}{m} \sum_j A_{t,j} X_{t,j} U_j,$$

with $U_i = 0$ under P_i^0 and $U_i = 1$ under P_i^1. In addition, in both cases, $|u_t| \le \varepsilon \le 1/16$, so ψ_{u_t} maps $[-1, 1]$ onto $[-1/2, 1/2]$ and the action losses lie in $[0, 1]$. For the same past history and current action X_t, the two values of u_t are uniquely determined and differ in absolute value by ε/m when item i is selected while active, that is, when $X_{t,i} = A_{t,i} = 1$, and coincide otherwise. Thus, Lemma 12 bounds each such round's KL contribution by $80\varepsilon^2/m^2$.

In addition, there are at most N_0 such rounds, so the KL chain rule of Lattimore and Szepesvári [28, Exercise 14.12, Eq. (14.17)] gives

$$\mathrm{KL}(P_i^0 \| P_i^1) \le \frac{80\varepsilon^2 N_0}{m^2} = O(\log(1/r)). \tag{28}$$

Let $N_{\mathrm{act}} := \sum_i U_i A_{T,i}$ be the number of favorable active items at the start of round T. Then $\mathbb{P}(N_{\mathrm{act}} < m)$ is the probability that fewer than m such items remain. For each item i, we introduce the event $H_i := \{A_{T,i} = 1,\ \sum_{j \neq i} U_j A_{T,j} < m\}$, and we write

$$\bar{h}_0 := \frac{1}{d} \sum_i \mathbb{P}(H_i \mid U_i = 0), \qquad \bar{h}_1 := \frac{1}{d} \sum_i \mathbb{P}(H_i \mid U_i = 1).$$

[10] As formalized in Appendix E, we denote by $\mathcal{H}_t = (X_1, Y_1, \ldots, X_t, Y_t)$ the history of actions and observed feedback up to round t, where $Y_t = X_t^\top \ell_t$.

An item becomes inactive only after being selected $N_0 = \lceil 8Tm/d \rceil$ times. Before round T, the algorithm has made $m(T-1)$ item selections in total. Thus, the number of inactive items is at most $m(T-1)/N_0 \le d/8$. Therefore, at least $d - d/8 = 7d/8$ items remain active. This holds for every realization of the randomness of the algorithm. For an unfavorable item i, we have $\{U_i = 0\} \cap H_i = \{U_i = 0,\ A_{T,i} = 1,\ N_{\mathrm{act}} < m\}$. Recalling the definitions of H_i and N_{act}, we obtain $(1-U_i)\mathbf{1}_{H_i} = (1-U_i)A_{T,i}\mathbf{1}_{\{N_{\mathrm{act}}<m\}}$. Summing over items gives

$$\sum_{i=1}^{d}(1-U_i)\mathbf{1}_{H_i} = \left(\sum_{i=1}^{d} A_{T,i} - N_{\mathrm{act}}\right)\mathbf{1}_{\{N_{\mathrm{act}}<m\}} \ge (7d/8 - m)\mathbf{1}_{\{N_{\mathrm{act}}<m\}},$$

where the inequality uses the preceding bound on the number of active items and $N_{\mathrm{act}} < m$ on the indicated event. Taking expectations and using $\mathbb{P}(U_i = 0) = 1 - \rho$ gives

$$(1-\rho)d\bar{h}_0 = \sum_{i=1}^{d}\mathbb{P}(H_i, U_i = 0) = \mathbb{E}\left[\sum_{i=1}^{d}(1-U_i)\mathbf{1}_{H_i}\right] \ge (7d/8 - m)\mathbb{P}(N_{\mathrm{act}} < m).$$

For a favorable active item i, with a similar argument, we obtain

$$\rho d\bar{h}_1 = \sum_{i=1}^{d}\mathbb{P}(H_i, U_i = 1) = \mathbb{E}\left[\sum_{i=1}^{d} U_i\mathbf{1}_{H_i}\right] = \mathbb{E}\left[N_{\mathrm{act}}\mathbf{1}_{\{N_{\mathrm{act}}\le m\}}\right] \le m. \tag{29}$$

Hence $\bar{h}_0 \ge 3\mathbb{P}(N_{\mathrm{act}} < m)/4$ and $\bar{h}_1 \le 256r$. With the algorithm's private randomness and all U_j for $j \ne i$ fixed, H_i is determined by the observed history. Applying data processing to this event gives

$$\mathrm{KL}\big(\mathrm{Ber}(P_i^0(H_i)) \,\big\|\, \mathrm{Ber}(P_i^1(H_i))\big) \le \mathrm{KL}(P_i^0 \| P_i^1).$$

Using (28), averaging over items, the algorithm's randomness, and the remaining variables U_j with $j \ne i$, and using the joint convexity of KL divergence, we obtain $\mathrm{KL}\big(\mathrm{Ber}(\bar{h}_0) \,\big\|\, \mathrm{Ber}(\bar{h}_1)\big) \le 80\varepsilon^2 N_0/m^2$. If $\bar{h}_1 = 0$, the claim follows immediately. Otherwise, the binary-KL inequality gives

$$\mathrm{KL}\big(\mathrm{Ber}(\bar{h}_0) \,\big\|\, \mathrm{Ber}(\bar{h}_1)\big) \ge \bar{h}_0 \log(1/\bar{h}_1) - \log 2.$$

Combining these inequalities with the bounds on $\bar{h}_0$ and $\bar{h}_1$ and $r \le r_\star$, and choosing ε sufficiently small, we obtain $\mathbb{P}(N_{\mathrm{act}} < m) \le 1/16$.

On $\{N_{\mathrm{act}} \ge m\}$, the comparator can select m favorable items that stay active throughout all T rounds, so $\mathrm{Top}_m(S) = m\sum_t g(Z_t)$. In addition, we have

$$\begin{aligned}\mathbb{E}[\mathrm{Top}_m(S)] &\ge m\mathbb{E}\left[\left(\sum_t g(Z_t)\right)\mathbf{1}_{\{N_{\mathrm{act}}\ge m\}}\right] \\ &= m\mathbb{E}\left[\sum_t g(Z_t)\right] - m\mathbb{E}\left[\left(\sum_t g(Z_t)\right)\mathbf{1}_{\{N_{\mathrm{act}}<m\}}\right] \ge \left(\frac{2}{3} - \frac{1}{16}\right)mT.\end{aligned}$$

This uses nonnegative scores, $\sum_t g(Z_t) \le T$, and the bound $\mathbb{P}(N_{\mathrm{act}} < m) \le 1/16$; it does not require independence between this event and the noise. The full argument appears in Appendix E.2.1. Together, the two score bounds give $\mathbb{E}[R_T] \ge \varepsilon T/4$. Since $R_T \le \varepsilon T$, Lemma 13 gives $\mathbb{P}(R_T \ge \varepsilon T/8) \ge 1/8$. Our choice of ε gives $\varepsilon T = \Theta(\sqrt{dmT\log(d/m)}) = \Theta(\sqrt{dT\log K})$.

– **The case $r > r_\star$.** We consider the same losses as in Equation (25), with the only difference being that we now set $A_{t,i} = 1$ for every $t \in [T]$ and take the sign variables U_i to be uniformly distributed over $\{-1, 1\}$ for $i \in [2m]$, with $U_i = 0$ otherwise. These signs are independent of one another, the noises, and the algorithm's private randomness. We also set $\varepsilon := \Theta(m/\sqrt{T})$, in a way that $\varepsilon \le 1/16$ and the KL bound in (30) holds. The scores in (26) have the same interpretation as in the case $r \le r_\star$, and the regret identity (27) remains valid. Notice that, since the losses no longer depend on the algorithm's past actions, the adversary is oblivious.

To lower bound $\mathbb{E}[\mathrm{Top}_m(S)]$ we arrange the first $2m$ items into pairs $(1,2), \dots, (2m-1, 2m)$, and select the item i with the larger score S_i from each pair. This is a valid fixed action with m items, so it can be used to lower bound $\mathrm{Top}_m(S)$. Then, we have

$$\mathbb{E}[\mathrm{Top}_m(S)] \ge \mathbb{E}\left[\sum_{j=1}^{m} \max\{S_{2j-1}, S_{2j}\}\right] = \mathbb{E}\left[\sum_{t=1}^{T} g(Z_t)\right] \sum_{j=1}^{m} \mathbb{E}[\max\{U_{2j-1}, U_{2j}\}] = \frac{mT}{3},$$

where the first equality uses $g(Z_t) \ge 0$ and independence. The second uses $\mathbb{E}[g(Z_t)] = 2/3$ for every t and $\mathbb{E}[\max\{U_{2j-1}, U_{2j}\}] = 1/2$.

To derive an upper bound on the expected score of the algorithm, we fix the randomness of the algorithm and all signs except U_i, and let P_i^+, P_i^- be the history laws for $U_i = 1, -1$, respectively. As in the previous case, we compare the feedback laws conditional on the same past history, with the algorithm's private randomness and all signs other than U_i fixed. Changing U_i from -1 to 1 changes the parameter of ψ_u by $2\varepsilon/m$ in absolute value when item i is selected, and leaves it unchanged otherwise. Thus, Lemma 12 bounds the KL contribution of each round by $80(2\varepsilon/m)^2 X_{t,i}$. Summing these contributions using the KL chain rule gives

$$\mathrm{KL}(P_i^+ \| P_i^-) \le 80 \left(\frac{2\varepsilon}{m}\right)^2 \mathbb{E}_{P_i^+}\left[\sum_{t=1}^{T} X_{t,i}\right] \le \frac{1}{32}. \tag{30}$$

Furthermore, using Pinsker's inequality and averaging over the algorithm's randomness and the signs U_i, we obtain $\mathbb{E}[U_i \sum_{t=1}^{T} X_{t,i}] \le T/16$. Moreover, the fresh noise Z_t is independent of X_t and the signs, and $\mathbb{E}[g(Z_t)] = 2/3$. Consequently, using the definition of S_{L} and the previous bound, we obtain

$$\mathbb{E}[S_{\mathrm{L}}] = \frac{2}{3} \sum_{i=1}^{2m} \mathbb{E}\left[U_i \sum_{t=1}^{T} X_{t,i}\right] \le \frac{2}{3} \cdot 2m \cdot \frac{T}{16} = \frac{mT}{12}.$$

Using the lower bound on $\mathbb{E}[\mathrm{Top}_m(S)]$, the upper bound on $\mathbb{E}[S_{\mathrm{L}}]$, and Equation (27), we obtain $\mathbb{E}[R_T] \ge \varepsilon T/4$. Since both scores lie in $[-mT, mT]$, we have $R_T \le 2\varepsilon T$. Lemma 13 then gives $\mathbb{P}(R_T \ge \varepsilon T/8) \ge 1/16$. To conclude the proof, we observe that $\varepsilon T = \Theta(m\sqrt{T}) = \Omega(\sqrt{dT \log K})$, because $m > r_\star d$ and $\log K \le d \log 2$.

- **The $\Omega(\sqrt{dT \log(1/\delta)})$ lower bound.** We use here a different construction. Specifically, we fix $m - 1$ core items C and let $V := [d] \setminus C$. We sample independent noises $Z_t \sim \mathrm{Unif}[-1, 1]$ and set $\Delta := \Theta(\sqrt{|V| \log(1/\delta)/T})$, with a sufficiently small numerical constant. The horizon assumption then ensures $m\Delta < 1/8$ and $\Delta \le 1/64$. We define a reference instance 0 with losses $\ell_t^{(0)}$ and, for each item $i \in V$ outside the core, an instance i with losses $\ell_t^{(i)}$ as follows:

$$\ell_{t,j}^{(0)} := \frac{1/4 + Z_t/8}{m} + \frac{\mathbf{1}\{j \in V\}}{8m}, \qquad \ell_{t,j}^{(i)} := \ell_{t,j}^{(0)} - \Delta g(Z_t) \mathbf{1}\{j = i\}.$$

Since $m\Delta < 1/8$, in every instance and at every round each core item has a lower loss than every item in V. In instance $i \in V$, item i has its loss reduced by $\Delta g(Z_t)$, making it no larger than that of any other item outside the core. Therefore, since $|C| = m - 1$, the action selecting $C \cup \{i\}$ is optimal in instance i. We also introduce the following quantities

$$D_t := \frac{1}{8m} \sum_{\tau=1}^{t} \left(\sum_{j \in V} X_{\tau,j} - 1 \right), \qquad R_T^{(i)} = D_T + \Delta \sum_{t=1}^{T} g(Z_t)(1 - X_{t,i}).$$

In instance i, the term D_t accounts for the regret accumulated up to round t by selecting more than one item outside the core per round. The term $R_T^{(i)}$ represents the cumulative regret relative to always playing $C \cup \{i\}$ and is therefore the sum of D_T and the cumulative additional loss incurred by not selecting item i.

We construct an auxiliary algorithm that follows the original until its next proposed action would make $D_t \geq \Delta T/8$. It then selects the core items and one fixed item of V, chosen identically in every instance, for all remaining rounds. Run both algorithms with the same internal randomness and noises Z_t. If the two algorithms first recommend different actions at some round t, the original algorithm's action makes $D_t \geq \Delta T/8$, and hence $R_T^{(i)} \geq \Delta T/8$. If their behavior coincides through round T, they incur the same regret. Thus, any lower bound on the probability that the auxiliary algorithm incurs regret at least $\Delta T/8$ also holds for the original algorithm.

For ease of presentation, in the rest of the proof, X_t and D_t refer to the auxiliary algorithm. We now fix its internal randomness and let P, Q_i denote the laws of its observed feedback history in the reference instance and instance i, respectively. In instance i, the feedback is

$$Y_t = \frac{1}{4} + \frac{1}{8m} \sum_{j \in V} X_{t,j} + \frac{1}{4} \psi_{-4\Delta X_{t,i}}(Z_t),$$

while the reference instance has the same expression with $\psi_0(Z_t)$ instead of $\psi_{-4\Delta X_{t,i}}(Z_t)$. For a fixed history and action, subtracting the first two terms and multiplying by four gives a common invertible transformation. Thus, by Lemma 12, the KL contribution is at most $80(4\Delta)^2$ when item i is selected and zero otherwise.

Applying the KL chain rule and averaging over $i \in V$ gives

$$\frac{1}{|V|} \sum_{i \in V} \mathrm{KL}(P \| Q_i) \leq \frac{80(4\Delta)^2}{|V|} \mathbb{E}_P \left[\sum_{t=1}^{T} \sum_{i \in V} X_{t,i} \right] \leq O\left(\frac{\Delta^2 T}{|V|} \right),$$

where the second inequality uses $D_T < \Delta T/8$. After replacing the proposed action, the auxiliary algorithm selects exactly one item outside the core, so D_t no longer increases. Indeed, the definition of D_T gives $\sum_{t=1}^{T} \sum_{j \in V} X_{t,j} = T + 8mD_T < T(1 + m\Delta) < 2T$.

For each item $i \in V$, we consider the event $E_i := \left\{ D_T + \Delta \sum_{t=1}^{T} g(Z_t)(1 - X_{t,i}) \geq \Delta T/8 \right\}$. As shown in Appendix E.3, E_i is a measurable event of the observed history, with the same definition under both P and Q_i.

Furthermore, $Q_i(E_i) \geq \delta$ means that, in instance i, the auxiliary algorithm incurs regret at least $\Delta T/8$ with probability at least δ. Under P, using the definition of D_T, $|V| \geq 2$, $0 \leq g \leq 1$, and $\Delta/|V| \leq 1/(8m)$, we obtain

$$\frac{1}{|V|} \sum_{i \in V} \left(D_T + \Delta \sum_{t=1}^{T} g(Z_t)(1 - X_{t,i}) \right) \geq \frac{\Delta}{2} \sum_{t=1}^{T} g(Z_t).$$

We observe that $D_T + \Delta \sum_{t=1}^{T} g(Z_t)(1 - X_{t,i})$ is at most $9\Delta T/8$ on E_i and below $\Delta T/8$ on its complement E_i^c. Taking expectations, using $\mathbb{E}_P[g(Z_t)] = 2/3$, and splitting each expectation over E_i and E_i^c, we obtain, after rearranging, $\sum_{i \in V} P(E_i) \geq |V|/16$.

Applying data processing to these events, namely $\mathrm{KL}(\mathrm{Ber}(P(E_i)) \,\|\, \mathrm{Ber}(Q_i(E_i))) \leq \mathrm{KL}(P \| Q_i)$, and using joint convexity of KL divergence together with the binary-KL inequality, we obtain

$$\frac{1}{16} \log \frac{1}{|V|^{-1} \sum_{i \in V} Q_i(E_i)} \leq \frac{1}{|V|} \sum_{i \in V} \mathrm{KL}(P \| Q_i) + \log 2 \leq O\left(\frac{\Delta^2 T}{|V|}\right) + \log 2,$$

where the first inequality uses $|V|^{-1} \sum_i P(E_i) \geq 1/16$ and the last uses the preceding bound on the average KL divergence. By suitably choosing the numerical constants in the definition of Δ and taking δ sufficiently small, the preceding inequality implies $\frac{1}{|V|} \sum_{i \in V} Q_i(E_i) \geq \delta$. Averaging over the algorithm's private randomness before choosing the item, we find a fixed $i^\star \in V$, independent of that randomness, such that, in instance $i^\star$, the auxiliary algorithm incurs regret at least $\Delta T/8$ with probability at least δ. By the comparison between the two algorithms, the same guarantee holds for the original algorithm. Finally, our choice of Δ and $|V| \geq d/2$ give $\Delta T/8 = \Omega(\sqrt{dT \log(1/\delta)})$.

Finally, we choose the construction corresponding to the larger of $\log K$ and $\log(1/\delta)$. Since $\max\{\log K, \log(1/\delta)\} \geq \big(\log K + \log(1/\delta)\big)/2$, this gives the claimed lower bound up to a universal constant. The complete proof is in Appendix E. □

Finally, we remark that, when $m = 1$ and $d = K$, our problem reduces to the standard adversarial K-armed bandit problem. Thus, the following corollary holds.

Corollary 1. *There exist universal constants $c > 0$ and $\delta_0 \in (0, 1)$ and a threshold $\mathfrak{T}_{\mathrm{MAB}}(K, \delta)$ bounded by a polynomial in K and $\log(1/\delta)$ such that the following holds. Let $K \geq 2$, $\delta \in (0, \delta_0]$, and $T \geq \mathfrak{T}_{\mathrm{MAB}}(K, \delta)$. For every randomized algorithm, there exists an adaptive non-anticipating adversary with losses $\ell_t \in [0, 1]^K$ such that*

$$\mathbb{P}\left(R_T \geq c\sqrt{KT\left(\log K + \log \frac{1}{\delta}\right)}\right) \geq \delta.$$

Proof. Apply Theorem 2 with $d = K$ and $m = 1$, and set $\mathfrak{T}_{\mathrm{MAB}}(K, \delta) := \mathfrak{T}(K, 1, \delta)$. □

Appendix

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A Related work

Optimal design and bandit linear optimization. Kiefer [24] introduced the general framework of approximate optimal design, and Kiefer and Wolfowitz [25] proved the equivalence between D- and G-optimality. For a finite action set $\mathcal{A} \subset \mathbb{R}^d$, Bubeck et al. [10, Theorem 4] combined exponential weights with John's exploration and obtained $O(\sqrt{dT \log |\mathcal{A}|})$ expected regret against oblivious adversaries under action-normalized losses. For general convex action sets, Abernethy et al. [2, Theorem 1] used self-concordant barriers to obtain an efficient expected-regret bound of $\widetilde{O}(d^{3/2}\sqrt{T})$ against oblivious adversaries.

Early general high-probability guarantees with $\sqrt{T}$ dependence against adaptive adversaries had a larger polynomial dependence on the dimension. Bartlett et al. [7, Theorem 1] obtained $\widetilde{O}(d^{3/2}\sqrt{T})$ regret. Abernethy and Rakhlin [1, Theorem 4 and Section 5.4] then gave a general high-probability reduction whose efficient instantiation requires problem-specific geometric subroutines. Later, Lee et al. [29, Theorem 3.1] gave the first general polynomial-time high-probability algorithm, with worst-case regret $\widetilde{O}(d^{7/2}\sqrt{T})$. Zimmert and Lattimore [33] improved this rate to $\widetilde{O}(d^2\sqrt{T})$. The efficiency guarantees for methods based on self-concordant barriers assume access to the barrier and its derivatives, as specified in the respective sources.

In the finite-action setting, Zimmert and Lattimore [33, Theorem 6 and Algorithm 3] combined EXP3 with Kiefer-Wolfowitz exploration and obtained $O(\sqrt{dT \log(|\mathcal{A}|/\delta)})$ regret with high probability. When applied to $\mathcal{X} = \mathcal{X}_{d,m}$, however, its direct implementation maintains one weight for each of the $K = \binom{d}{m}$ actions, and may therefore require exponential space.

High-probability adversarial multi-armed bandits. For oblivious K-armed bandits with losses in $[0,1]$, setting $G_0 = T/\log(1/\delta)$ in the bound of Audibert and Bubeck [3, Theorem 18] gives an upper bound of $O(\sqrt{KT\log(1/\delta)})$ for $T \geq 81K\log(1/\delta)$, matched by the lower bound of Gerchinovitz and Lattimore [18, Theorem 1]. For adaptive non-anticipating adversaries, the upper bound is $O(\sqrt{KT\log(K/\delta)})$ [11, Theorem 6.10]. The $m = 1$ specialization of our lower bound matches this rate, closing the $\sqrt{\log K}$ gap at fixed confidence (see Corollary 1). These comparisons hold for sufficiently small δ and sufficiently large T.

Expectation-based regret in combinatorial bandits. Early adversarial bandit algorithms for structured action sets were given by McMahan and Blum [31] and Awerbuch and Kleinberg [5]. Cesa-Bianchi and Lugosi [12] subsequently introduced ComBand, a general framework for adversarial combinatorial bandits.

The expectation-based statistical picture was further clarified by Audibert et al. [4, Section 4, after Theorem 5], who considered full-information, semi-bandit, and full-bandit feedback and established several lower bounds. Their results use coordinatewise-bounded item losses, $\ell_t \in [0,1]^d$, so an action selecting m items may incur instantaneous loss as large as m. Under this normalization, they conjectured that the minimax bandit regret in expectation should scale as $m\sqrt{dT}$.

Under the same coordinatewise normalization, Cohen et al. [16, Theorems 1 and 5] disproved this conjecture. Using strongly correlated losses, they proved an expected-regret lower bound of $\Omega(\sqrt{dm^3T/\log T})$ against oblivious adversaries for a multitask action class in which every action selects m items. After rescaling the losses by $1/m$ to match our action-level normalization, this lower bound becomes $\Omega(\sqrt{dmT/\log T})$.

Ito et al. [22, Theorem 2] later removed the $1/\sqrt{\log T}$ factor using i.i.d. binary correlated losses. Crucially, their multiple-play action class is exactly $\{x \in \{0,1\}^d : \|x\|_1 = m\}$, the m-set action class considered here. In the regime $m \leq d/2$ and $T = \Omega(dm^{3/2})$, their expected-regret lower

bound is $\Omega(\sqrt{dm^3T})$ against oblivious adversaries under coordinatewise normalization, and hence $\Omega(\sqrt{dmT})$ after rescaling to our action-level normalization.

These lower bounds concern expected regret. Our results establish matching high-probability upper and lower bounds against adaptive non-anticipating adversaries.

Efficient algorithms for structured action sets. A separate line of work studies how to avoid explicitly enumerating exponentially large action sets. Combes et al. [17, Theorems 6 and 7] introduced CombEXP, which retains the regret scaling of earlier combinatorial-bandit algorithms while using approximate projections and decompositions. Their implementation is polynomial-time when the action polytope admits a polynomial-size description by linear equalities and inequalities. Thus, for $m \le d/2$ and item losses in $[0, 1/m]$, their expected-regret bound against oblivious adversaries becomes $O(\sqrt{dmT\log(d/m)})$ when $T \ge d\log(d/m)$. This efficient guarantee concerns expected regret against oblivious adversaries; our algorithm provides a high-probability guarantee against adaptive non-anticipating adversaries.

For general bandit linear optimization under expectation-based regret, Hazan and Karnin [21, Corollary 35] use volumetric spanners, while Ito et al. [23, Theorem 1] use a linear-optimization oracle. Under their respective access models, both approaches achieve $\widetilde{O}(\sqrt{T})$ expected regret with polynomial dependence on the dimension.

Earlier efficient high-probability guarantees with $T^{2/3}$ dependence were obtained by Braun and Pokutta [9, Theorem 3.1], using a linear-optimization oracle over the action polytope, and by Sakaue et al. [32, Theorem 1], using a zero-suppressed binary decision diagram. These approaches are efficient when the required oracle or compact representation is available, but their high-probability guarantees do not attain $\sqrt{T}$ dependence.

Kontogiannis et al. [26, Theorem 3.2 and Appendix G.5] give an efficient kernelized algorithm for m-set bandits with coordinatewise-bounded losses. Their high-probability regret bound against adaptive adversaries scales as $T^{2/3}$ in the horizon, up to logarithmic factors, compared with our $\sqrt{T}$ dependence.

The work most closely related to ours is Maiti et al. [30, Appendix E.6]. Under action-normalized losses, they represent m-sets as paths in a DAG and obtain a polynomial-time high-probability algorithm against adaptive adversaries with regret $\widetilde{O}(d\sqrt{mT})$. Moreover, applying their general DAG guarantee [30, Theorem 7] to the m-set DAG of [30, Appendix E.6] yields $\widetilde{O}(m\sqrt{dT})$ for $m \le d/2$.

Weighted m-set distributions. The weighted m-set distribution introduced in Section 2 is also known as a conditional Bernoulli or rejective-sampling distribution [14, 15, 20]. Classical methods based on elementary symmetric polynomials allow us to compute its normalizing constant and marginal probabilities, and to sample from it efficiently [14, 27].

Beyond these computational properties, the covariance structure of weighted m-set distributions is central to our analysis. We use the pseudoinverse estimate of Cesari and Colomboni [13, Corollary 1.2] to control leverage scores through an affine majorant.

B Proofs from Section 5.1

We first establish the identity relating the inverse second-moment matrix of a weighted m-set distribution to the pseudoinverse of its covariance matrix. We then combine this identity with the pseudoinverse estimate of Cesari and Colomboni [13, Corollary 1.2] to prove Lemma 1.

Lemma 2. *Let $p \in \Delta(\mathcal{X})$ be a weighted m-set distribution, where $1 \le m \le d-1$. Then its second-moment matrix M is positive definite and, for every $y \in H_0$,*

$$y^\top M^{-1} y = y^\top \Sigma^+ y.$$

Consequently, for every $x \in \mathcal{X}$,

$$x^\top M^{-1} x = 1 + (x-\mu)^\top \Sigma^+ (x-\mu),$$

where $\mu = \mu(p)$ is the marginal vector of p and $\Sigma = \Sigma(p)$ is the covariance matrix of p.

Proof. Let $\Sigma := \Sigma(p)$ and $\mu := \mu(p)$. We first show that M is positive definite. Suppose that $a^\top M a = 0$. Since

$$a^\top M a = \mathbb{E}_{X\sim p}\big[(a^\top X)^2\big],$$

we have $a^\top x = 0$ for every $x \in \mathcal{X}$, because p has full support.

Fix $i \neq j$. Since $1 \le m \le d-1$, we can choose two actions that differ only by exchanging item i with item j. Comparing their inner products with a gives $a_i = a_j$. Thus, $a = c\mathbf{1}$ for some $c \in \mathbb{R}$. Since every $x \in \mathcal{X}$ has m nonzero coordinates, $0 = a^\top x = cm$. Therefore, $c = 0$, and hence $a = \mathbf{0}$. This proves that $M \succ 0$.

By [13, Corollary 1.2], $\ker(\Sigma) = \operatorname{span}\{\mathbf{1}\}$. Since Σ is symmetric, its range satisfies $\operatorname{range}(\Sigma) = \ker(\Sigma)^\perp = H_0$. Every $X \in \mathcal{X}$ satisfies $X^\top \mathbf{1} = m$. Therefore,

$$M\mathbf{1} = \mathbb{E}_{X\sim p}\big[X(X^\top \mathbf{1})\big] = m\mu.$$

Since M is invertible,

$$M^{-1}\mu = \frac{1}{m}\mathbf{1}. \tag{31}$$

Now fix $y \in H_0$ and let $z := M^{-1}y$. By (31),

$$\mu^\top z = (M^{-1}\mu)^\top y = \frac{1}{m}\mathbf{1}^\top y = 0.$$

It follows that $\Sigma z = (M - \mu\mu^\top)z = y$. Moreover, $\Sigma^+ y$ also solves the equation $\Sigma u = y$. Therefore, $z - \Sigma^+ y \in \ker(\Sigma) = \operatorname{span}\{\mathbf{1}\}$. Since $y \in H_0$, it is orthogonal to this difference. Hence,

$$y^\top M^{-1} y = y^\top \Sigma^+ y. \tag{32}$$

Finally, fix $x \in \mathcal{X}$ and let $y := x - \mu$. Since $\mathbf{1}^\top x = \mathbf{1}^\top \mu = m$, we have $y \in H_0$. Expanding $x = \mu + y$ gives

$$x^\top M^{-1} x = \mu^\top M^{-1}\mu + 2y^\top M^{-1}\mu + y^\top M^{-1} y = 1 + y^\top \Sigma^+ y,$$

where the second equality follows from (31) and (32). Substituting $y = x - \mu$ proves that

$$x^\top M^{-1} x = 1 + (x-\mu)^\top \Sigma^+ (x-\mu).$$

This concludes the proof. □

We now combine this lemma with [13, Corollary 1.2] to prove the affine leverage bound stated in Lemma 1.[11]

[11] In an earlier preprint of this work [6, Lemmas 2 and 4], we used the majorant $\phi_p^{\mathrm{old}}(x) := 1 + 2\sum_{i=1}^d (x_i - \mu_i)^2/[\mu_i(1-\mu_i)]$, which satisfies $\mathbb{E}_{X\sim p}[\phi_p^{\mathrm{old}}(X)] = 2d+1$ and $0 \le \phi_p^{\mathrm{old}}(x) \le 1 + 4d/\lambda$. Here we use the improved pseudoinverse estimate of Cesari and Colomboni [13, Corollary 1.2], which reduces the coefficient from 2 to 1, to obtain $\mathbb{E}_{X\sim p}[\phi_p(X)] = d+1$ and $0 \le \phi_p(x) \le 1 + 2d/\lambda$ for every $x \in \mathcal{X}$, as stated in Lemma 1.

Lemma 1. *Let $p \in \mathcal{P}_\lambda$ be a weighted m-set distribution, where $1 \le m \le d-1$, and let μ, M, and Σ be its marginal vector, second-moment matrix, and covariance matrix, respectively. For every $x \in \mathcal{X}$, define*

$$\phi_p(x) := 1 + \sum_{i=1}^{d} \frac{(x_i - \mu_i)^2}{\mu_i(1-\mu_i)}.$$

Then, for every $x \in \mathcal{X}$, $x^\top M^{-1} x \le \phi_p(x)$. Moreover, $4\phi_p(x) = c_p^\top x$, where

$$c_{p,i} := 4\left(\frac{1 + \sum_{j=1}^{d} \frac{\mu_j}{1-\mu_j}}{m} + \frac{1-2\mu_i}{\mu_i(1-\mu_i)}\right) \qquad \text{for every } i \in [d]. \tag{13}$$

Finally, $\mathbb{E}_{X\sim p}[\phi_p(X)] = d+1$, and, for every $x \in \mathcal{X}$, $0 \le \phi_p(x) \le 1 + 2d/\lambda$.

Proof. For $x \in \mathcal{X}$, let $y = x - \mu \in H_0$. By Lemma 2,

$$x^\top M^{-1} x = 1 + y^\top \Sigma^+ y.$$

The pseudoinverse estimate in [13, Corollary 1.2] gives

$$y^\top \Sigma^+ y \le \sum_{i\in[d]} \frac{y_i^2}{\mu_i(1-\mu_i)},$$

and hence $x^\top M^{-1} x \le \phi_p(x)$. Taking expectations and using $\mathbb{E}[(X_i - \mu_i)^2] = \mu_i(1-\mu_i)$ yields $\mathbb{E}_{X\sim p}[\phi_p(X)] = d+1$. In addition, we write $r = m/d$. If $x_i = 1$, then

$$\frac{(x_i-\mu_i)^2}{\mu_i(1-\mu_i)} = \frac{1-\mu_i}{\mu_i} \le \frac{1}{\lambda r};$$

if $x_i = 0$, then

$$\frac{(x_i-\mu_i)^2}{\mu_i(1-\mu_i)} = \frac{\mu_i}{1-\mu_i} \le \frac{1}{\lambda(1-r)}.$$

Since x has m ones and $d-m$ zeros,

$$\phi_p(x) \le 1 + \frac{m}{\lambda r} + \frac{d-m}{\lambda(1-r)} = 1 + \frac{2d}{\lambda}.$$

Finally, because $x_i \in \{0,1\}$,

$$\frac{(x_i-\mu_i)^2}{\mu_i(1-\mu_i)} = \frac{\mu_i}{1-\mu_i} + x_i \frac{1-2\mu_i}{\mu_i(1-\mu_i)}.$$

Thus

$$\phi_p(x) = 1 + \sum_{i\in[d]} \frac{\mu_i}{1-\mu_i} + \sum_{i\in[d]} x_i \frac{1-2\mu_i}{\mu_i(1-\mu_i)}.$$

Distribute the constant term evenly over the selected coordinates using $\sum_i x_i = m$, and multiply the resulting affine representation by four, to obtain $4\phi_p(x) = c_p^\top x$. □

C Proofs from Section 5.2

Throughout this appendix, $T \geq 1$ is an integer, $\delta \in (0,1)$, and the parameter choices in Algorithm 1 give

$$0 < \eta \leq \frac{1}{64}, \qquad \lambda = 128\eta d \leq \frac{1}{2}.$$

The loss sequence $(\ell_t)_{t=1}^T$ is chosen by a non-anticipating adaptive adversary, that is, ℓ_t is $\mathcal{F}_{t-1}$-measurable for every $t \in [T]$, and satisfies the action-normalized condition in Section 2. Recall from Algorithm 1 that

$$\phi_t := \phi_{p_t}, \qquad c_t := c_{p_t}, \qquad c_t^\top x = 4\phi_t(x), \qquad Z_t(x) := \widehat{y}_t(x) - \eta c_t^\top x.$$

Furthermore, $\widehat{y}_t(x) = x^\top \widehat{\ell}_t$, where $\widehat{\ell}_t = M_t^{-1} X_t Y_t$ is the loss estimator computed in Algorithm 1.

Lemma 3. *Fix $t \in [T]$. Suppose that $0 < \eta \leq 1/64$, $\lambda = 128\eta d \leq 1/2$, the loss at round t satisfies the action-normalized condition, and p_t is a weighted m-set distribution in $\mathcal{P}_{\lambda/2}$. Then, for every $x \in \mathcal{X}$, $\eta|Z_t(x)| \leq 1/4$.*

Proof. Since $|Y_t| \leq 1$, the Cauchy-Schwarz inequality in the norm induced by M_t^{-1} gives

$$|\widehat{y}_t(x)| = |x^\top M_t^{-1} X_t Y_t| \leq \sqrt{x^\top M_t^{-1} x}\sqrt{X_t^\top M_t^{-1} X_t}.$$

Applying Lemma 1 to $p_t \in \mathcal{P}_{\lambda/2}$, both leverage scores in the last inequality are at most $\phi_t(x)$ and $\phi_t(X_t)$, respectively, and both ϕ_t-values are at most $1 + 4d/\lambda$. Thus $|\widehat{y}_t(x)| \leq (1 + 4d/\lambda)$. Also $0 \leq c_t^\top x \leq 4(1 + 4d/\lambda)$. Hence

$$|Z_t(x)| = |\widehat{y}_t(x) - \eta c_t^\top x| \leq (1 + 4\eta)\left(1 + \frac{4d}{\lambda}\right).$$

The parameter choice implies

$$\eta\left(1 + \frac{4d}{\lambda}\right) = \eta + \frac{4\eta d}{\lambda} = \eta + \frac{1}{32} \leq \frac{3}{64}.$$

Therefore

$$\eta|Z_t(x)| \leq (1 + 4\eta)\eta\left(1 + \frac{4d}{\lambda}\right) \leq \left(1 + \frac{1}{16}\right)\frac{3}{64} = \frac{51}{1024} < \frac{1}{4},$$

where we used $\eta \leq 1/64$. This proves the claim. □

As shown in the proof of Proposition 1, at every round t, p_t is a weighted m-set distribution and $p_t \in \mathcal{P}_{\lambda/2}$. Hence, Lemma 1 applies at every round with smoothing parameter $\lambda/2$. Applying Lemma 1 with $\lambda/2$, for every $x \in \mathcal{X}$, we have

$$0 \leq c_t^\top x \leq 4\left(1 + \frac{4d}{\lambda}\right), \qquad \mathbb{E}_{X \sim p_t}[c_t^\top X] = 4(d+1), \qquad x^\top M_t^{-1} x \leq \frac{c_t^\top x}{4}.$$

Lemma 4. *When running Algorithm 1, for every distribution $q \in \mathcal{P}_\lambda$,*

$$\sum_{t=1}^T \left(\mathbb{E}_{X \sim p_t}[Z_t(X)] - \mathbb{E}_{X \sim q}[Z_t(X)]\right) \leq \frac{\mathrm{KL}(q\|p_1)}{\eta} + 1 + 2\eta \sum_{t=1}^T \mathbb{E}_{X \sim p_t}[Z_t(X)^2].$$

Proof. Fix $t \in [T]$ and define the unprojected multiplicative-weights distribution

$$\widetilde{p}_{t+1}(x) := \frac{p_t(x)\exp(-\eta Z_t(x))}{\mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))]}, \qquad x \in \mathcal{X}.$$

Since $Z_t(x) = z_t^\top x$ and $p_t = p_{\theta_t}$, this is precisely $p_{\widetilde{\theta}_{t+1}}$ from Algorithm 1. By Proposition 1, the update p_{t+1} satisfies (17). Hence, for every $q \in \mathcal{P}_\lambda$,

$$\begin{aligned}
\mathrm{KL}(q\|p_{t+1}) &\le \mathrm{KL}(q\|\widetilde{p}_{t+1}) - \mathrm{KL}(p_{t+1}\|\widetilde{p}_{t+1}) + \varepsilon_{\mathrm{p}} \\
&\le \mathrm{KL}(q\|\widetilde{p}_{t+1}) + \varepsilon_{\mathrm{p}} \\
&= \mathrm{KL}(q\|p_t) + \eta\mathbb{E}_{X\sim q}[Z_t(X)] + \log \mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))] + \varepsilon_{\mathrm{p}}. \qquad (33)
\end{aligned}$$

Here, the first inequality is (17), the second uses nonnegativity of KL divergence, and the equality follows from the definition of $\widetilde{p}_{t+1}$.

By Lemma 3, $|\eta Z_t(x)| \le 1/4$ for every $x \in \mathcal{X}$. Therefore,

$$\begin{aligned}
\log \mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))] &\le \log\bigl(1 - \eta\mathbb{E}_{X\sim p_t}[Z_t(X)] + 2\eta^2\mathbb{E}_{X\sim p_t}[Z_t(X)^2]\bigr) \\
&\le -\eta\mathbb{E}_{X\sim p_t}[Z_t(X)] + 2\eta^2\mathbb{E}_{X\sim p_t}[Z_t(X)^2]. \qquad (34)
\end{aligned}$$

The first inequality uses $e^{-u} \le 1 - u + 2u^2$ for $|u| \le 1/4$ and monotonicity of the logarithm. The second uses $\log(1+s) \le s$ with $s = -\eta\mathbb{E}_{X\sim p_t}[Z_t(X)] + 2\eta^2\mathbb{E}_{X\sim p_t}[Z_t(X)^2]$. Combining (33) with (34) and rearranging gives

$$\begin{aligned}
&\eta\left(\mathbb{E}_{X\sim p_t}[Z_t(X)] - \mathbb{E}_{X\sim q}[Z_t(X)]\right) \\
&\quad\le \mathrm{KL}(q\|p_t) - \mathrm{KL}(q\|p_{t+1}) + \eta\mathbb{E}_{X\sim p_t}[Z_t(X)] + \log \mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))] + \varepsilon_{\mathrm{p}} \\
&\quad\le \mathrm{KL}(q\|p_t) - \mathrm{KL}(q\|p_{t+1}) + 2\eta^2\mathbb{E}_{X\sim p_t}[Z_t(X)^2] + \varepsilon_{\mathrm{p}}. \qquad (35)
\end{aligned}$$

Summing (35) over t and dividing by η yields

$$\begin{aligned}
\sum_{t=1}^{T}\left(\mathbb{E}_{X\sim p_t}[Z_t(X)] - \mathbb{E}_{X\sim q}[Z_t(X)]\right) &\le \frac{\mathrm{KL}(q\|p_1) - \mathrm{KL}(q\|p_{T+1})}{\eta} + 2\eta\sum_{t=1}^{T}\mathbb{E}_{X\sim p_t}[Z_t(X)^2] + \frac{T\varepsilon_{\mathrm{p}}}{\eta} \\
&\le \frac{\mathrm{KL}(q\|p_1)}{\eta} + 2\eta\sum_{t=1}^{T}\mathbb{E}_{X\sim p_t}[Z_t(X)^2] + 1.
\end{aligned}$$

The first inequality uses telescoping of the KL terms. The second uses $\mathrm{KL}(q\|p_{T+1}) \ge 0$ and $\varepsilon_{\mathrm{p}} = \eta/T$, which proves the claim. □

Lemma 5. *When running Algorithm 1, let $\delta_{\mathrm{aux}} \in (0,1)$, and let $(q_t)_{t=1}^T$ be a sequence of distributions in $\Delta(\mathcal{X})$ such that q_t is $\mathcal{F}_{t-1}$-measurable for every $t \in [T]$. For every $t \in [T]$, define*

$$L_t(q_t) := \mathbb{E}_{X\sim q_t}[X^\top \ell_t], \qquad \widehat{L}_t(q_t) := \mathbb{E}_{X\sim q_t}[\widehat{y}_t(X)].$$

Then, with probability at least $1 - \delta_{\mathrm{aux}}$,

$$\left|\sum_{t=1}^{T}\left(\widehat{L}_t(q_t) - L_t(q_t)\right)\right| \le \frac{\eta}{4}\sum_{t=1}^{T}\mathbb{E}_{X\sim q_t}[c_t^\top X] + \frac{\log(2/\delta_{\mathrm{aux}})}{\eta}.$$

Proof. For every $t \in [T]$, define $\Delta_t := \widehat{L}_t(q_t) - L_t(q_t)$. Recall that $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}_{t-1}]$. Since p_t, q_t, and ℓ_t are $\mathcal{F}_{t-1}$-measurable, they are fixed conditionally on $\mathcal{F}_{t-1}$, and the only randomness at round t comes from $X_t \sim p_t$.

We first show that Δ_t has conditional mean zero. By the definition of the loss estimate,

$$\mathbb{E}_t[\widehat{\ell}_t] = M_t^{-1}\mathbb{E}_t[X_t X_t^\top]\ell_t = M_t^{-1}M_t\ell_t = \ell_t. \tag{36}$$

Since q_t is $\mathcal{F}_{t-1}$-measurable, (36) gives

$$\mathbb{E}_t[\widehat{L}_t(q_t)] = \mathbb{E}_{X\sim q_t}[X^\top \mathbb{E}_t[\widehat{\ell}_t]] = \mathbb{E}_{X\sim q_t}[X^\top \ell_t] = L_t(q_t). \tag{37}$$

Therefore, by the definition of Δ_t and (37),

$$\mathbb{E}_t[\Delta_t] = \mathbb{E}_t[\widehat{L}_t(q_t)] - L_t(q_t) = 0. \tag{38}$$

Define $\bar{x}_t := \mathbb{E}_{X\sim q_t}[X]$, so that $\widehat{L}_t(q_t) = \bar{x}_t^\top M_t^{-1} X_t Y_t$. Then

$$\begin{aligned}
\mathbb{E}_t[\widehat{L}_t(q_t)^2] &\le \bar{x}_t^\top M_t^{-1}\mathbb{E}_t[X_t X_t^\top]M_t^{-1}\bar{x}_t \\
&= \bar{x}_t^\top M_t^{-1}\bar{x}_t \\
&\le \mathbb{E}_{X\sim q_t}[X^\top M_t^{-1} X] \\
&\le \frac{1}{4}\mathbb{E}_{X\sim q_t}[c_t^\top X].
\end{aligned} \tag{39}$$

The first inequality uses $|Y_t| \le 1$, and the equality uses $\mathbb{E}_t[X_t X_t^\top] = M_t$. The second inequality is Jensen's inequality for the convex function $x \mapsto x^\top M_t^{-1} x$, since M_t^{-1} is positive semidefinite. The third follows from Lemma 1 and $c_t^\top X = 4\phi_t(X)$, which give $X^\top M_t^{-1} X \le c_t^\top X/4$. Combining (37) with (39) yields

$$\mathbb{E}_t[\Delta_t^2] = \mathbb{E}_t[\widehat{L}_t(q_t)^2] - L_t(q_t)^2 \le \mathbb{E}_t[\widehat{L}_t(q_t)^2] \le \frac{1}{4}\mathbb{E}_{X\sim q_t}[c_t^\top X], \tag{40}$$

where the equality uses (37), the first inequality drops the nonnegative square, and the second applies (39). We next bound Δ_t uniformly. By (39) and $c_t^\top x \le 4\,(1 + 4d/\lambda)$ for every $x \in \mathcal{X}$,

$$\bar{x}_t^\top M_t^{-1}\bar{x}_t \le \frac{1}{4}\mathbb{E}_{X\sim q_t}[c_t^\top X] \le \left(1 + \frac{4d}{\lambda}\right). \tag{41}$$

Here, the first inequality is contained in (39), and the second uses $c_t^\top x \le 4\,(1 + 4d/\lambda)$. For the sampled action, we similarly have

$$X_t^\top M_t^{-1} X_t \le \frac{1}{4} c_t^\top X_t \le \left(1 + \frac{4d}{\lambda}\right), \tag{42}$$

where the first inequality is due to Lemma 1, and the second uses $c_t^\top X_t \le 4\,(1 + 4d/\lambda)$. Consequently,

$$|\widehat{L}_t(q_t)| = |\bar{x}_t^\top M_t^{-1} X_t Y_t| \le \sqrt{\bar{x}_t^\top M_t^{-1}\bar{x}_t}\sqrt{X_t^\top M_t^{-1} X_t} \le 1 + \frac{4d}{\lambda}. \tag{43}$$

By (43), the action-normalized condition $|L_t(q_t)| \le 1$, and $(1 + 4d/\lambda) \ge 1$, we have

$$|\Delta_t| \le |\widehat{L}_t(q_t)| + |L_t(q_t)| \le 2 + \frac{4d}{\lambda} \le 2\left(1 + \frac{4d}{\lambda}\right). \tag{44}$$

Moreover, the parameter choices give

$$\eta\left(1+\frac{4d}{\lambda}\right)=\eta+\frac{4\eta d}{\lambda}=\eta+\frac{1}{32}\le\frac{3}{64}, \tag{45}$$

where the second equality uses $\lambda = 128\eta d$ and the inequality uses $\eta \le 1/64$, both ensured by Algorithm 1. Combining (44) with (45) yields

$$|\eta\Delta_t|\le 2\eta\left(1+\frac{4d}{\lambda}\right)\le\frac{3}{32}<1. \tag{46}$$

At the same time, combining (46), (38), and (40) yields

$$\begin{aligned}\mathbb{E}_t[\exp(\eta\Delta_t)] &\le 1+\eta\mathbb{E}_t[\Delta_t]+\eta^2\mathbb{E}_t[\Delta_t^2]\\ &= 1+\eta^2\mathbb{E}_t[\Delta_t^2]\\ &\le \exp\left(\eta^2\mathbb{E}_t[\Delta_t^2]\right)\\ &\le \exp\left(\frac{\eta^2}{4}\mathbb{E}_{X\sim q_t}[c_t^\top X]\right).\end{aligned} \tag{47}$$

The first inequality uses $e^u \le 1+u+u^2$ for $|u|\le 1$, which applies by (46). The equality uses (38); the second inequality uses $1+v\le e^v$; and the third uses (40).

For $s \in \{0,\dots,T\}$, define

$$\mathcal{M}_s := \exp\left(\eta\sum_{t=1}^{s}\Delta_t-\frac{\eta^2}{4}\sum_{t=1}^{s}\mathbb{E}_{X\sim q_t}[c_t^\top X]\right). \tag{48}$$

Since $\mathbb{E}_{X\sim q_t}[c_t^\top X]$ is $\mathcal{F}_{t-1}$-measurable, (47) and (48) imply

$$\mathbb{E}_t[\mathcal{M}_t]=\mathcal{M}_{t-1}\exp\left(-\frac{\eta^2}{4}\mathbb{E}_{X\sim q_t}[c_t^\top X]\right)\mathbb{E}_t[\exp(\eta\Delta_t)]\le\mathcal{M}_{t-1}.$$

Thus $(\mathcal{M}_s)_{s=0}^T$ is a nonnegative supermartingale with $\mathcal{M}_0 = 1$, and $\mathbb{E}[\mathcal{M}_T] \le 1$. By Markov's inequality, $\mathbb{P}(\mathcal{M}_T \ge 2/\delta_{\text{aux}}) \le (\delta_{\text{aux}}/2)\mathbb{E}[\mathcal{M}_T] \le \delta_{\text{aux}}/2$. Using (48), we conclude that, with probability at least $1-\delta_{\text{aux}}/2$,

$$\sum_{t=1}^{T}\Delta_t\le\frac{\eta}{4}\sum_{t=1}^{T}\mathbb{E}_{X\sim q_t}[c_t^\top X]+\frac{\log(2/\delta_{\text{aux}})}{\eta}. \tag{49}$$

The same argument applies to $-\Delta_t$: the conditional mean remains zero by (38), and the bounds (40) and (46) are unchanged. Hence, with probability at least $1-\delta_{\text{aux}}/2$,

$$-\sum_{t=1}^{T}\Delta_t\le\frac{\eta}{4}\sum_{t=1}^{T}\mathbb{E}_{X\sim q_t}[c_t^\top X]+\frac{\log(2/\delta_{\text{aux}})}{\eta}. \tag{50}$$

A union bound on the events in (49) and (50) gives, with probability at least $1-\delta_{\text{aux}}$,

$$\left|\sum_{t=1}^{T}\Delta_t\right|\le\frac{\eta}{4}\sum_{t=1}^{T}\mathbb{E}_{X\sim q_t}[c_t^\top X]+\frac{\log(2/\delta_{\text{aux}})}{\eta}.$$

Substituting $\Delta_t = \widehat{L}_t(q_t) - L_t(q_t)$ proves the claim. □

Lemma 6. *When running Algorithm 1, for every $\delta_{\mathrm{aux}} \in (0,1)$, with probability at least $1-\delta_{\mathrm{aux}}$,*

$$\sum_{t=1}^{T} \phi_t(X_t) \le 2(d+1)T + 2\left(1+\frac{4d}{\lambda}\right)\log\frac{1}{\delta_{\mathrm{aux}}}.$$

Proof. Conditionally on $\mathcal{F}_{t-1}$, the distribution p_t and the function ϕ_t are fixed, while $X_t \sim p_t$. Therefore, Lemma 1 gives

$$\mathbb{E}_t[\phi_t(X_t)] = \mathbb{E}_{X\sim p_t}[\phi_t(X)] = d+1, \qquad 0 \le \phi_t(X_t) \le \left(1+\frac{4d}{\lambda}\right).$$

Since $\phi_t(X_t)/(1+4d/\lambda) \in [0,1]$, convexity of the exponential gives $e^u \le 1+(e-1)u$, $u\in[0,1]$. It follows that

$$\mathbb{E}_t\left[\exp\left(\frac{\phi_t(X_t)}{(1+4d/\lambda)}\right)\right] \le 1 + \frac{(e-1)(d+1)}{(1+4d/\lambda)} \le \exp\left(\frac{(e-1)(d+1)}{(1+4d/\lambda)}\right).$$

For $s\in\{0,\dots,T\}$, define

$$\mathcal{M}_s := \exp\left(\frac{1}{(1+4d/\lambda)}\sum_{t=1}^{s}\phi_t(X_t) - \frac{(e-1)(d+1)s}{(1+4d/\lambda)}\right).$$

The previous conditional moment bound shows that $(\mathcal{M}_s)_{s=0}^{T}$ is a nonnegative supermartingale with $\mathcal{M}_0 = 1$. Hence, $\mathbb{E}[\mathcal{M}_T] \le 1$. By Markov's inequality, with probability at least $1-\delta_{\mathrm{aux}}$,

$$\sum_{t=1}^{T}\phi_t(X_t) \le (e-1)(d+1)T + \left(1+\frac{4d}{\lambda}\right)\log\frac{1}{\delta_{\mathrm{aux}}}.$$

Using $e-1\le 2$ proves the result. □

C.1 Proof of the upper bound

Theorem 1. *There exists a universal constant $C>0$ such that the following holds. Let $T\ge 1$ be an integer, let $\delta\in(0,1)$, and assume $1\le m\le d-1$. Run Algorithm 1. Then, with probability at least $1-\delta$,*

$$R_T \le C\sqrt{dT\left(\log K + \log\frac{1}{\delta}\right)}, \tag{18}$$

where $K=\binom{d}{m}$. One may take $C=160$.

Proof. For every $x\in\mathcal{X}$, define the smoothed comparator

$$q_x := (1-\lambda)\delta_x + \lambda U,$$

where δ_x assigns probability one to x, and U is the uniform distribution over $\mathcal{X}$.[12] If $x_i=1$, the i-th marginal of q_x is $1-\lambda(1-r)$, while, if $x_i=0$, it is λr. Therefore, $q_x\in\mathcal{P}_\lambda$. Moreover, since $p_1 = U$,

$$\mathrm{KL}(q_x\|p_1) = \mathrm{KL}(q_x\|U) = \log K - H(q_x) \le \log K.$$

[12] For $p\in\Delta(\mathcal{X})$, its entropy is $H(p) := -\sum_{x\in\mathcal{X}} p(x)\log p(x)$, with the convention $0\log 0 = 0$.

Before comparing losses, observe that the learner's average estimated loss equals its realized loss at every round. Indeed, $\mathbf{1}^\top X = m$ for every $X \in \mathcal{X}$, so $M_t \mathbf{1} = \mathbb{E}_{X\sim p_t}[X(X^\top \mathbf{1})] = m\mu_t$. Since M_t is symmetric and invertible, this gives

$$\mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)] = \mu_t^\top M_t^{-1} X_t Y_t = \frac{\mathbf{1}^\top X_t}{m} Y_t = Y_t. \tag{51}$$

Applying Lemma 4 with $q = q_x$ gives

$$\sum_{t=1}^{T} (\mathbb{E}_{X\sim p_t}[Z_t(X)] - \mathbb{E}_{X\sim q_x}[Z_t(X)]) \le \frac{\log K}{\eta} + 1 + 2\eta \sum_{t=1}^{T} \mathbb{E}_{X\sim p_t}[Z_t(X)^2]. \tag{52}$$

We first bound the second-order term. Since $Z_t(X) = \widehat{y}_t(X) - \eta c_t^\top X$, we have

$$Z_t(X)^2 \le 2\widehat{y}_t(X)^2 + 2\eta^2 (c_t^\top X)^2.$$

Furthermore, we have

$$\begin{aligned}
\mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)^2] &= Y_t^2 X_t^\top M_t^{-1} \mathbb{E}_{X\sim p_t}[XX^\top] M_t^{-1} X_t \\
&= Y_t^2 X_t^\top M_t^{-1} X_t \\
&\le \phi_t(X_t),
\end{aligned}$$

where the last inequality follows from $|Y_t| \le 1$ and Lemma 1. Recall that $0 \le c_t^\top X \le 4(1 + 4d/\lambda)$ and $\mathbb{E}_{X\sim p_t}[c_t^\top X] = 4(d+1)$. Hence,

$$\mathbb{E}_{X\sim p_t}[(c_t^\top X)^2] \le 4\left(1 + \frac{4d}{\lambda}\right) \mathbb{E}_{X\sim p_t}[c_t^\top X] = 16\left(1 + \frac{4d}{\lambda}\right)(d+1).$$

It follows that

$$\begin{aligned}
2\eta \mathbb{E}_{X\sim p_t}[Z_t(X)^2] &\le 4\eta\phi_t(X_t) + 64\eta^3 \left(1 + \frac{4d}{\lambda}\right)(d+1) \\
&= 4\eta\phi_t(X_t) + 2\eta^2(1 + 32\eta)(d+1) \\
&\le 4\eta\phi_t(X_t) + 3\eta^2(d+1).
\end{aligned} \tag{53}$$

The equality uses $\lambda = 128\eta d$, and the last inequality uses $\eta \le 1/64$. Using $Z_t(X) = \widehat{y}_t(X) - \eta c_t^\top X$ in (52), and then applying (53) and (51), gives

$$\begin{aligned}
&\sum_{t=1}^{T} (Y_t - \mathbb{E}_{X\sim q_x}[\widehat{y}_t(X)]) \\
&\quad \le \frac{\log K}{\eta} + 1 + 4\eta \sum_{t=1}^{T} \phi_t(X_t) + 3\eta^2(d+1)T + 4\eta(d+1)T - \eta \sum_{t=1}^{T} \mathbb{E}_{X\sim q_x}[c_t^\top X].
\end{aligned} \tag{54}$$

We now apply Lemma 6 with $\delta_{\text{aux}} = \delta/2$. With probability at least $1 - \delta/2$,

$$4\eta \sum_{t=1}^{T} \phi_t(X_t) \le 8\eta(d+1)T + \left(8\eta + \frac{1}{4}\right) \log \frac{2}{\delta} \le 8\eta(d+1)T + \log \frac{2}{\delta}. \tag{55}$$

Here we used $\lambda = 128\eta d$ and $8\eta + 1/4 \le 3/8 \le 1$. For each fixed $x \in \mathcal{X}$, apply Lemma 5 to the constant deterministic sequence $q_t = q_x$, with $\delta_{\mathrm{aux}} = \delta/(2K)$. A union bound over the K actions gives, with probability at least $1 - \delta/2$, simultaneously for every $x \in \mathcal{X}$,

$$\sum_{t=1}^{T} \mathbb{E}_{X\sim q_x}[\widehat{y}_t(X) - X^\top \ell_t] \le \frac{\eta}{4} \sum_{t=1}^{T} \mathbb{E}_{X\sim q_x}[c_t^\top X] + \frac{\log(4K/\delta)}{\eta}. \tag{56}$$

We work on the intersection of these two events, whose probability is at least $1 - \delta$. Combining (54), (55), and (56) gives, for every $x \in \mathcal{X}$,

$$\begin{aligned}\sum_{t=1}^{T} \Big(Y_t - \mathbb{E}_{X\sim q_x}[X^\top \ell_t]\Big) \le \frac{\log K + \log(4K/\delta)}{\eta} + 1 + 12\eta(d+1)T \\ + 3\eta^2(d+1)T + \log\frac{2}{\delta} - \frac{3\eta}{4}\sum_{t=1}^{T} \mathbb{E}_{X\sim q_x}[c_t^\top X].\end{aligned} \tag{57}$$

The last term is nonpositive because $c_t^\top X \ge 0$. The negative correction $-\eta$ in (54) absorbs the $\eta/4$ term in (56). Furthermore, $3\eta^2 \le 3\eta$ since $\eta \le 1$. Thus, simultaneously for every $x \in \mathcal{X}$,

$$\sum_{t=1}^{T} \Big(Y_t - \mathbb{E}_{X\sim q_x}[X^\top \ell_t]\Big) \le \frac{\log K + \log(4K/\delta)}{\eta} + 1 + 15\eta(d+1)T + \log\frac{2}{\delta}. \tag{58}$$

We may now choose a best action in hindsight $x^\star$, even though it can depend on the realized history. Since $q_{x^\star} = (1-\lambda)\delta_{x^\star} + \lambda U$ and every action loss belongs to $[-1, 1]$,

$$\mathbb{E}_{X\sim q_{x^\star}}[X^\top \ell_t] - (x^\star)^\top \ell_t = \lambda\Big(\mathbb{E}_{X\sim U}[X^\top \ell_t] - (x^\star)^\top \ell_t\Big) \le 2\lambda.$$

Applying (58) with $x = x^\star$, using $d + 1 \le 2d$ and $\lambda = 128\eta d$, therefore yields

$$R_T \le \frac{\log K + \log(4K/\delta)}{\eta} + 1 + 286\eta dT + \log\frac{2}{\delta}.$$

Since $\log K + \log(4K/\delta) \le 2\log(12K/\delta)$, $1 + \log(2/\delta) \le \log(12K/\delta)$, and $\eta \le 1$, we obtain

$$R_T \le \frac{2\log(12K/\delta)}{\eta} + 286\eta dT + \log(12K/\delta) \le \frac{3\log(12K/\delta)}{\eta} + 320\eta dT. \tag{59}$$

Moreover, $K = \binom{d}{m} \ge 2$, so $\log(K/\delta) \ge \log 2$ and $\log(12K/\delta) \le 5\log(K/\delta)$. We now substitute the learning rate from Algorithm 1. If $\eta = \sqrt{\log(12K/\delta)/(320dT)}$, then

$$R_T \le 4\sqrt{320dT\log(12K/\delta)} \le 160\sqrt{dT\log(K/\delta)}.$$

Otherwise, $\eta = 1/(256d)$, which implies $T \le (256^2/320)d\log(12K/\delta)$. The deterministic bound $R_T \le 2T$ then gives

$$R_T \le 2T \le \frac{512}{\sqrt{320}}\sqrt{dT\log(12K/\delta)} \le 64\sqrt{dT\log(K/\delta)}.$$

Thus in both cases $R_T \le 160\sqrt{dT(\log K + \log(1/\delta))}$, as claimed. □

D Polynomial-time implementation

This appendix proves Proposition 1 by giving a polynomial-time implementation of Algorithm 1 that does not enumerate $\mathcal{X}$. We show how to compute the required moments and draw samples while maintaining each p_t as a weighted m-set distribution. We also implement an approximate KL-projection procedure that returns a weighted m-set distribution in $\mathcal{P}_{\lambda/2}$ satisfying (17).

The appendix is organized as follows. We first introduce additional notation. Appendix D.1 presents the approximate KL-projection procedure and establishes its guarantees. Appendix D.2 describes the moment and sampling routines. Finally, Appendix D.3 combines these results to complete the proof of Proposition 1.

We introduce the following additional notation. For a vector $z = (z_1, \dots, z_n)$ and $k \in [n]$, we define its k-th elementary symmetric polynomial as

$$e_k(z) := \sum_{S \subseteq [n]:\, |S|=k} \prod_{i \in S} z_i, \qquad e_0(z) := 1, \qquad e_k(z) := 0 \quad \text{for } k < 0 \text{ or } k > n.$$

Let $\theta \in \mathbb{R}^d$. For the weights $w_i = e^{\theta_i}$, the normalizing constant of p_θ is $e_m(w)$. The log-partition function associated with p_θ is

$$F(\theta) := \log \sum_{A \subseteq [d]:\, |A|=m} \exp\left(\sum_{i \in A} \theta_i\right) = \log e_m(e^{\theta_1}, \dots, e^{\theta_d}).$$

In particular, if $S \sim p_\theta$ and $X := \mathbf{1}_S \in \mathcal{X}$ is its indicator vector, then

$$\nabla F(\theta) = \mathbb{E}_{X \sim p_\theta}[X] = \mu(p_\theta), \qquad \nabla^2 F(\theta) = \mathrm{Cov}_{X \sim p_\theta}(X) = \Sigma(p_\theta).$$

Furthermore, given $v \in \mathbb{R}^d$, we write $(v_+)_i := \max\{v_i, 0\}$ and $(v_-)_i := \max\{-v_i, 0\}$ for every $i \in [d]$. Moreover, we recall that, for a convex function $f : \mathbb{R}^d \to \mathbb{R}$, a vector $\zeta \in \mathbb{R}^d$ is a subgradient of f at $v \in \mathbb{R}^d$ if

$$f(u) \geq f(v) + \zeta^\top (u - v) \qquad \text{for every } u \in \mathbb{R}^d.$$

D.1 Approximate KL projection

This subsection describes the procedure used in line 11 of Algorithm 1. Its input is a parameter vector $\widetilde{\theta} \in \mathbb{R}^d$ representing the weighted m-set distribution $p_{\widetilde{\theta}}$, together with $T \geq 1$ and $0 < \eta \leq 1/(256d)$. Let λ and ε_{p} be defined from η and T as in Algorithm 1, and define

$$B_{\mathrm{p}} := 4dT, \qquad \mathcal{D}_{\mathrm{p}} := \{v \in \mathbb{R}^d : \|v\|_1 \leq B_{\mathrm{p}}\}, \qquad \tau := \frac{\varepsilon_{\mathrm{p}}^2}{2B_{\mathrm{p}}^2} = \frac{\eta^2}{32d^2T^4}. \tag{60}$$

Our procedure consists of applying the ellipsoid method [19, Sections 2.1 and 3.1], initialized with the Euclidean ball $B_2(\mathbf{0}, B_{\mathrm{p}}) := \{u \in \mathbb{R}^d : \|u\|_2 \leq B_{\mathrm{p}}\}$, to the convex problem

$$\min_{v \in \mathcal{D}_{\mathrm{p}}} G_{\widetilde{\theta}}(v), \tag{61}$$

where

$$G_{\widetilde{\theta}}(v) := F(\widetilde{\theta} + v) - \underline{\mu}^\top v_+ + \overline{\mu}^\top v_-, \tag{62}$$

with $\underline{\mu}_i := \lambda r$, $\overline{\mu}_i := 1 - \lambda(1 - r)$ for every $i \in [d]$, and $r = m/d$.

Under the assumption $\mathrm{KL}(U\|p_{\widetilde{\theta}}) \le T$, Lemma 8 shows that the ellipsoid method computes a vector $v \in \mathcal{D}_{\mathrm{p}}$ satisfying

$$G_{\widetilde{\theta}}(v) \le \min_{u\in\mathcal{D}_{\mathrm{p}}} G_{\widetilde{\theta}}(u) + \tau$$

using $\mathrm{poly}(d, m, \log B_{\mathrm{p}}, \log(1/\tau))$ arithmetic operations and $\mathrm{poly}(d, m)$ space. At round t, the input vector is $\widetilde{\theta} = \widetilde{\theta}_{t+1}$. Setting $\theta_{t+1} := \widetilde{\theta}_{t+1} + v$ then gives the next sampling distribution $p_{t+1} := p_{\theta_{t+1}}$. By Lemma 8, this is a weighted m-set distribution in $\mathcal{P}_{\lambda/2}$ satisfying (17) for every $q \in \mathcal{P}_\lambda$.

Lemma 7. *Let $0 < \lambda \le 1/2$ and $\widetilde{\theta} \in \mathbb{R}^d$. The function $G_{\widetilde{\theta}}$ defined in (62) is convex and attains its minimum over $\mathbb{R}^d$. For every minimizer $v^\star \in \mathbb{R}^d$, we have*

$$p_{\widetilde{\theta}+v^\star} \in \arg\min_{p\in\mathcal{P}_\lambda} \mathrm{KL}(p\|p_{\widetilde{\theta}}). \tag{63}$$

Moreover, for every $v \in \mathbb{R}^d$, writing $\mu := \mu(p_{\widetilde{\theta}+v})$, the vector

$$\zeta_i(v) := \begin{cases} \mu_i - \underline{\mu}_i, & v_i > 0, \\ \mu_i - \overline{\mu}_i, & v_i < 0, \\ \mu_i - (\underline{\mu}_i + \overline{\mu}_i)/2, & v_i = 0 \end{cases} \qquad (i \in [d]) \tag{64}$$

is a subgradient of $G_{\widetilde{\theta}}$ at v, with $\|\zeta(v)\|_\infty \le 1$.

Proof. The projection in (63) minimizes $\mathrm{KL}(p\|p_{\widetilde{\theta}})$ over $p \in \Delta(\mathcal{X})$ subject to $\underline{\mu} \le \mu(p) \le \overline{\mu}$. The uniform distribution has positive mass on every action and satisfies $\underline{\mu}_i < r < \overline{\mu}_i$ for every $i \in [d]$. Hence, the strong-duality and Karush-Kuhn-Tucker (KKT) optimality results of Boyd and Vandenberghe [8, Sections 5.2.3 and 5.5.3] apply. For multipliers $\alpha, \beta \ge \mathbf{0}$ associated with the constraints $\underline{\mu} - \mu(p) \le \mathbf{0}$ and $\mu(p) - \overline{\mu} \le \mathbf{0}$, respectively, set $v := \alpha - \beta$. Using $\log p_\theta(x) = \theta^\top x - F(\theta)$, we obtain

$$\begin{aligned}
\mathcal{L}(p, \alpha, \beta) &= \mathrm{KL}(p\|p_{\widetilde{\theta}}) + \alpha^\top(\underline{\mu} - \mu(p)) + \beta^\top(\mu(p) - \overline{\mu}) \\
&= \mathrm{KL}(p\|p_{\widetilde{\theta}}) - v^\top\mu(p) + \underline{\mu}^\top\alpha - \overline{\mu}^\top\beta \\
&= \sum_{x\in\mathcal{X}} p(x)\Big[\log p(x) - \widetilde{\theta}^\top x + F(\widetilde{\theta}) - v^\top x\Big] + \underline{\mu}^\top\alpha - \overline{\mu}^\top\beta \\
&= \sum_{x\in\mathcal{X}} p(x)\Big[\log p(x) - (\widetilde{\theta}+v)^\top x + F(\widetilde{\theta}+v)\Big] + F(\widetilde{\theta}) - F(\widetilde{\theta}+v) + \underline{\mu}^\top\alpha - \overline{\mu}^\top\beta \\
&= \mathrm{KL}(p\|p_{\widetilde{\theta}+v}) + F(\widetilde{\theta}) - F(\widetilde{\theta}+v) + \underline{\mu}^\top\alpha - \overline{\mu}^\top\beta \\
&= \mathrm{KL}(p\|p_{\widetilde{\theta}+\alpha-\beta}) + F(\widetilde{\theta}) - F(\widetilde{\theta}+\alpha-\beta) + \underline{\mu}^\top\alpha - \overline{\mu}^\top\beta.
\end{aligned}$$

Only the KL term in the last expression depends on p. It is nonnegative and equals zero at $p = p_{\widetilde{\theta}+\alpha-\beta}$. For fixed multipliers, minimizing over $p \in \Delta(\mathcal{X})$ defines the dual function

$$D(\alpha, \beta) := \min_{p\in\Delta(\mathcal{X})} \mathcal{L}(p, \alpha, \beta) = F(\widetilde{\theta}) - F(\widetilde{\theta}+\alpha-\beta) + \underline{\mu}^\top\alpha - \overline{\mu}^\top\beta.$$

The dual problem is then $\max_{\alpha,\beta\ge\mathbf{0}} D(\alpha, \beta)$.

Take any pair $\alpha, \beta \geq \mathbf{0}$. We define $h \in \mathbb{R}^d$ by $h_i := \min\{\alpha_i, \beta_i\}$ for every $i \in [d]$, and set $\alpha' := \alpha - h$ and $\beta' := \beta - h$. We observe that $\alpha', \beta' \geq \mathbf{0}$ and $\alpha' - \beta' = \alpha - \beta = v$. Hence,

$$\begin{aligned} D(\alpha', \beta') &= D(v_+, v_-) \\ &= F(\widetilde{\theta}) - F(\widetilde{\theta} + v) + \underline{\mu}^\top(\alpha - h) - \overline{\mu}^\top(\beta - h) \\ &= D(\alpha, \beta) + (\overline{\mu} - \underline{\mu})^\top h \\ &\geq D(\alpha, \beta), \end{aligned}$$

where the first equality follows from $\alpha'_i = \max\{\alpha_i - \beta_i, 0\} = (v_+)_i$ and $\beta'_i = \max\{\beta_i - \alpha_i, 0\} = (v_-)_i$ for every $i \in [d]$, the second uses $\alpha' - \beta' = v$, and the inequality follows from $\overline{\mu}_i \geq \underline{\mu}_i$ for every i and $h \geq \mathbf{0}$. This shows that every feasible pair can be replaced by (v_+, v_-) without decreasing the dual objective. We may therefore restrict to $\alpha = v_+$ and $\beta = v_-$, for which $D(v_+, v_-) = F(\widetilde{\theta}) - G_{\widetilde{\theta}}(v)$. Since $F(\widetilde{\theta})$ is independent of v, maximizing the dual is equivalent to minimizing (62). By strong duality and the KKT conditions, a minimizer $v^\star$ yields the KL projection $p_{\widetilde{\theta}+v^\star}$.

The function $G_{\widetilde{\theta}}$ is convex because F is convex and each coordinate term $-\underline{\mu}_i(v_+)_i + \overline{\mu}_i(v_-)_i$ has slope $-\overline{\mu}_i$ for $v_i < 0$ and $-\underline{\mu}_i$ for $v_i > 0$, with $-\overline{\mu}_i \leq -\underline{\mu}_i$. Since $\nabla F(\widetilde{\theta} + v) = \mu$, the corresponding coordinate derivatives of $G_{\widetilde{\theta}}$ are $\mu_i - \overline{\mu}_i$ for $v_i < 0$ and $\mu_i - \underline{\mu}_i$ for $v_i > 0$. At $v_i = 0$, any value in $[\mu_i - \overline{\mu}_i, \mu_i - \underline{\mu}_i]$ is valid; choosing the midpoint gives (64). Consequently, for every $u, v \in \mathbb{R}^d$,

$$G_{\widetilde{\theta}}(u) \geq G_{\widetilde{\theta}}(v) + \zeta(v)^\top(u - v).$$

Finally, $\underline{\mu}_i, \overline{\mu}_i, \mu_i \in [0, 1]$ implies $|\zeta_i(v)| \leq 1$ for every $i \in [d]$, and hence $\|\zeta(v)\|_\infty \leq 1$. □

Lemma 8. *Let $T \geq 1$, $0 < \eta \leq 1/(256d)$, and $\widetilde{\theta} \in \mathbb{R}^d$ satisfy $\mathrm{KL}(U \| p_{\widetilde{\theta}}) \leq T$. Let λ and ε_{p} be defined from (η, T) as in Algorithm 1, and let B_{p}, $\mathcal{D}_{\mathrm{p}}$, and τ be as in (60). The ellipsoid method applied to (61), initialized with $B_2(\mathbf{0}, B_{\mathrm{p}})$, computes a vector $v \in \mathcal{D}_{\mathrm{p}}$ satisfying*

$$G_{\widetilde{\theta}}(v) \leq \min_{u \in \mathcal{D}_{\mathrm{p}}} G_{\widetilde{\theta}}(u) + \tau \tag{65}$$

using $\mathrm{poly}(d, m, \log B_{\mathrm{p}}, \log(1/\tau))$ arithmetic operations and $\mathrm{poly}(d, m)$ space. Setting $\theta^+ := \widetilde{\theta} + v$ gives a weighted m-set distribution $p_{\theta^+} \in \mathcal{P}_{\lambda/2}$ that satisfies, for every $q \in \mathcal{P}_\lambda$,

$$\mathrm{KL}(q \| p_{\theta^+}) + \mathrm{KL}(p_{\theta^+} \| p_{\widetilde{\theta}}) \leq \mathrm{KL}(q \| p_{\widetilde{\theta}}) + \varepsilon_{\mathrm{p}}. \tag{66}$$

Proof. Choose any global minimizer $v^\star \in \arg\min_{u \in \mathbb{R}^d} G_{\widetilde{\theta}}(u)$, whose existence is guaranteed by Lemma 7. We first bound $\|v^\star\|_1$ to show that this minimizer lies inside $\mathcal{D}_{\mathrm{p}}$. Since $\mathbb{E}_{X \sim U}[X] = r\mathbf{1}$,

Jensen's inequality gives, for every $v \in \mathbb{R}^d$,

$$\begin{aligned}
F(\widetilde{\theta}+v) - F(\widetilde{\theta}) &= \log\left(\frac{\sum_{x\in\mathcal{X}} e^{(\widetilde{\theta}+v)^\top x}}{\sum_{z\in\mathcal{X}} e^{\widetilde{\theta}^\top z}}\right) \\
&= \log\left(\sum_{x\in\mathcal{X}} \frac{e^{\widetilde{\theta}^\top x}}{\sum_{z\in\mathcal{X}} e^{\widetilde{\theta}^\top z}} e^{v^\top x}\right) \\
&= \log\left(\sum_{x\in\mathcal{X}} p_{\widetilde{\theta}}(x) e^{v^\top x}\right) \\
&= \log \mathbb{E}_{X\sim U}\left[\frac{p_{\widetilde{\theta}}(X)}{U(X)} e^{v^\top X}\right] \\
&\geq \mathbb{E}_{X\sim U}\left[\log\frac{p_{\widetilde{\theta}}(X)}{U(X)} + v^\top X\right] \\
&= -\mathrm{KL}(U\|p_{\widetilde{\theta}}) + r\mathbf{1}^\top v.
\end{aligned}$$

By the definition of $G_{\widetilde{\theta}}$ in (62), the preceding bound, and $\mathrm{KL}(U\|p_{\widetilde{\theta}}) \leq T$, we obtain

$$G_{\widetilde{\theta}}(v) \geq F(\widetilde{\theta}) - T + (1-\lambda)\big(r\|v_+\|_1 + (1-r)\|v_-\|_1\big).$$

Because $v^\star$ minimizes $G_{\widetilde{\theta}}$ over all of $\mathbb{R}^d$, comparison with $v = \mathbf{0}$ gives $G_{\widetilde{\theta}}(v^\star) \leq G_{\widetilde{\theta}}(\mathbf{0}) = F(\widetilde{\theta})$. Applying the preceding bound at $v^\star$ therefore yields

$$(1-\lambda)\min\{r, 1-r\}\|v^\star\|_1 \leq (1-\lambda)\big(r\|(v^\star)_+\|_1 + (1-r)\|(v^\star)_-\|_1\big) \leq T,$$

where we used $\|(v^\star)_+\|_1 + \|(v^\star)_-\|_1 = \|v^\star\|_1$. Since $\lambda \leq 1/2$ and $\min\{r, 1-r\} = \min\{m, d-m\}/d \geq 1/d$, it follows that

$$\|v^\star\|_1 \leq \frac{T}{(1-\lambda)\min\{r, 1-r\}} \leq 2dT = B_{\mathrm{p}}/2. \tag{67}$$

In particular, $v^\star \in \mathcal{D}_{\mathrm{p}}$. Thus, $\mathcal{D}_{\mathrm{p}}$ contains a global minimizer of $G_{\widetilde{\theta}}$. Together with $\mathcal{D}_{\mathrm{p}} \subseteq \mathbb{R}^d$, this gives

$$\min_{u\in\mathbb{R}^d} G_{\widetilde{\theta}}(u) \leq \min_{u\in\mathcal{D}_{\mathrm{p}}} G_{\widetilde{\theta}}(u) \leq G_{\widetilde{\theta}}(v^\star) = \min_{u\in\mathbb{R}^d} G_{\widetilde{\theta}}(u).$$

Hence the minimum over $\mathcal{D}_{\mathrm{p}}$ in (65) equals the minimum over all of $\mathbb{R}^d$.

Let $v \in \mathcal{D}_{\mathrm{p}}$ satisfy (65). We first show that $p_{\widetilde{\theta}+v} \in \mathcal{P}_{\lambda/2}$ and that (66) holds for every $q \in \mathcal{P}_\lambda$. We then show that the ellipsoid method computes such a vector v. Since $v^\star \in \mathcal{D}_{\mathrm{p}}$ is a global minimizer, (65) gives

$$0 \leq G_{\widetilde{\theta}}(v) - G_{\widetilde{\theta}}(v^\star) \leq \tau. \tag{68}$$

By Lemma 7, $p_{\widetilde{\theta}+v^\star}$ minimizes $\mathrm{KL}(q\|p_{\widetilde{\theta}})$ over $q \in \mathcal{P}_\lambda$. In particular, $p_{\widetilde{\theta}+v^\star} \in \mathcal{P}_\lambda$, therefore $\underline{\mu} \leq \mu(p_{\widetilde{\theta}+v^\star}) \leq \overline{\mu}$ by (3) and the definitions of $\underline{\mu}_i = \lambda r$, $\overline{\mu}_i = 1 - \lambda(1-r)$ for every $i \in [d]$.

We also show two equalities that follow directly from the definition of $v^\star$. For every $i \in [d]$ with $v_i^\star > 0$, differentiating (62) in coordinate i at $v^\star$ gives $0 = \mu_i(p_{\widetilde{\theta}+v^\star}) - \underline{\mu}_i$, using $\nabla F(\widetilde{\theta}+v^\star) = \mu(p_{\widetilde{\theta}+v^\star})$. If $v_i^\star < 0$, the same argument gives $0 = \mu_i(p_{\widetilde{\theta}+v^\star}) - \overline{\mu}_i$. If $v_i^\star = 0$, both $((v^\star)_+)_i$ and $((v^\star)_-)_i$ are zero. Multiplying by the corresponding coordinates of $(v^\star)_+$ and $(v^\star)_-$ and summing, we obtain

$$(v^\star)_+^\top(\mu(p_{\widetilde{\theta}+v^\star}) - \underline{\mu}) = (v^\star)_-^\top(\overline{\mu} - \mu(p_{\widetilde{\theta}+v^\star})) = 0. \tag{69}$$

The definitions of the weighted m-set distribution p_θ and of F give

$$\log p_\theta(x) = \theta^\top x - F(\theta) \qquad \text{for every } x \in \mathcal{X}. \tag{70}$$

Substituting (70) into the definition of KL divergence yields

$$\mathrm{KL}(p_{\widetilde{\theta}+v^\star} \| p_{\widetilde{\theta}+v}) = F(\widetilde{\theta}+v) - F(\widetilde{\theta}+v^\star) - \mu(p_{\widetilde{\theta}+v^\star})^\top (v - v^\star). \tag{71}$$

Combining (69) with (71) and using (62), we obtain

$$G_{\widetilde{\theta}}(v) - G_{\widetilde{\theta}}(v^\star) = \mathrm{KL}(p_{\widetilde{\theta}+v^\star} \| p_{\widetilde{\theta}+v}) + v_+^\top \big(\mu(p_{\widetilde{\theta}+v^\star}) - \underline{\mu}\big) + v_-^\top \big(\overline{\mu} - \mu(p_{\widetilde{\theta}+v^\star})\big). \tag{72}$$

All terms on the right-hand side of (72) are nonnegative. Combining (72) with (68) therefore gives $\mathrm{KL}(p_{\widetilde{\theta}+v^\star} \| p_{\widetilde{\theta}+v}) \le \tau$. Pinsker's inequality now implies

$$\|\mu(p_{\widetilde{\theta}+v}) - \mu(p_{\widetilde{\theta}+v^\star})\|_\infty \le \mathrm{TV}(p_{\widetilde{\theta}+v}, p_{\widetilde{\theta}+v^\star}) \le \sqrt{\tau/2} = \frac{\eta}{8dT^2} \le \frac{\lambda}{2d}. \tag{73}$$

Each marginal of $p_{\widetilde{\theta}+v^\star}$, namely $\mu_i(p_{\widetilde{\theta}+v^\star})$ for $i \in [d]$, belongs to $[\lambda r, 1-\lambda(1-r)]$. The lower and upper boundaries in the definition of $\mathcal{P}_{\lambda/2}$ are $\lambda r/2$ and $1-\lambda(1-r)/2$, respectively. Their distances from the corresponding endpoints of the preceding interval are $\lambda r/2$ and $\lambda(1-r)/2$, both at least $\lambda \min\{r, 1-r\}/2 \ge \lambda/(2d)$. Thus (73) ensures that every $\mu_i(p_{\widetilde{\theta}+v})$ lies between these boundaries. By (3), this proves $p_{\widetilde{\theta}+v} \in \mathcal{P}_{\lambda/2}$.
We next prove (66). Since $\mathrm{KL}(p_{\widetilde{\theta}+v^\star} \| p_{\widetilde{\theta}+v}) \ge 0$, combining (72) with (68) also gives

$$v_+^\top (\mu(p_{\widetilde{\theta}+v^\star}) - \underline{\mu}) + v_-^\top (\overline{\mu} - \mu(p_{\widetilde{\theta}+v^\star})) \le \tau. \tag{74}$$

For every $q \in \mathcal{P}_\lambda$, we have

$$\begin{aligned}
&\mathrm{KL}(q \| p_{\widetilde{\theta}+v}) + \mathrm{KL}(p_{\widetilde{\theta}+v} \| p_{\widetilde{\theta}}) - \mathrm{KL}(q \| p_{\widetilde{\theta}}) \\
&\quad = v^\top \big(\mu(p_{\widetilde{\theta}+v}) - \mu(q)\big) \\
&\quad \le v^\top \big(\mu(p_{\widetilde{\theta}+v}) - \mu(p_{\widetilde{\theta}+v^\star})\big) + v_+^\top \big(\mu(p_{\widetilde{\theta}+v^\star}) - \underline{\mu}\big) + v_-^\top \big(\overline{\mu} - \mu(p_{\widetilde{\theta}+v^\star})\big) \\
&\quad \le \|v\|_1 \|\mu(p_{\widetilde{\theta}+v}) - \mu(p_{\widetilde{\theta}+v^\star})\|_\infty + \tau \\
&\quad \le B_{\mathrm{p}} \sqrt{\tau/2} + \tau = \frac{\varepsilon_{\mathrm{p}}}{2} + \frac{\varepsilon_{\mathrm{p}}^2}{2B_{\mathrm{p}}^2} \le \varepsilon_{\mathrm{p}}.
\end{aligned}$$

The first equality follows by substituting (70) into the three KL divergences and canceling the terms involving F. The first inequality uses $\underline{\mu} \le \mu(q) \le \overline{\mu}$ and $v = v_+ - v_-$. The second uses Hölder's inequality and (74); the third uses $v \in \mathcal{D}_{\mathrm{p}}$ and (73). The last equality follows from (60), and the last inequality uses $\varepsilon_{\mathrm{p}} \le 1 \le B_{\mathrm{p}}^2$. This proves (66).

It remains to show that the ellipsoid method computes a point satisfying (65), within the claimed computational bounds. By Lemma 7, $G_{\widetilde{\theta}}$ is convex, and the subgradient in (64) satisfies $\|\zeta(u)\|_\infty \le 1$. Hence $G_{\widetilde{\theta}}$ is $\sqrt{d}$-Lipschitz in the Euclidean norm , meaning that $|G_{\widetilde{\theta}}(u) - G_{\widetilde{\theta}}(v)| \le \sqrt{d}\,\|u - v\|_2$ for all $u, v \in \mathbb{R}^d$. Its value and a subgradient can be computed using $\mathrm{poly}(d, m)$ operations by Lemma 9. Moreover, $\mathcal{D}_{\mathrm{p}} \subseteq B_2(\mathbf{0}, B_{\mathrm{p}})$ admits an efficient separation oracle for the constraint $\|v\|_1 \le B_{\mathrm{p}}$. This procedure checks whether a point satisfies the constraint and, if not, returns a hyperplane separating that point from the feasible set. At each iteration, we use a separation cut at an infeasible center and a subgradient cut at a feasible center. We retain the feasible center with the smallest objective value.

Set $r_{\mathrm{ball}} := \tau/(2\sqrt{d})$, where $B_2(w, r_{\mathrm{ball}})$ denotes the Euclidean ball of radius r_{ball} centered at w. By (67) and $\tau \le B_{\mathrm{p}}$, $B_2(v^\star, r_{\mathrm{ball}}) \subseteq \mathcal{D}_{\mathrm{p}}$. For every $u \in B_2(v^\star, r_{\mathrm{ball}})$, we have

$$G_{\widetilde{\theta}}(u) \le G_{\widetilde{\theta}}(v^\star) + \sqrt{d}\,\|u - v^\star\|_2 \le G_{\widetilde{\theta}}(v^\star) + \tau/2. \tag{75}$$

If none of the feasible centers visited so far satisfies (65), (75) ensures that the ellipsoid at each iteration contains the ball $B_2(v^\star, r_{\mathrm{ball}})$. Since the ellipsoid volume decreases by a fixed factor at each iteration, such a point must eventually be found. The standard ellipsoid bound [19, Sections 2.1 and 3.1] therefore yields a vector $v \in \mathcal{D}_{\mathrm{p}}$ satisfying (65) in $O(d^2 \log(B_{\mathrm{p}}/r_{\mathrm{ball}}))$ oracle calls. We run the method for a predetermined number of iterations given by this bound, with a sufficiently large constant, and return the retained feasible center. For this vector, setting $\theta^+ := \widetilde{\theta} + v$ gives $p_{\theta^+} \in \mathcal{P}_{\lambda/2}$ and (66), as proved above. Since $\log(B_{\mathrm{p}}/r_{\mathrm{ball}}) = O(\log d + \log B_{\mathrm{p}} + \log(1/\tau))$, the procedure uses $\mathrm{poly}(d, m, \log B_{\mathrm{p}}, \log(1/\tau))$ operations. The moment tables, ellipsoid matrix, and retained best feasible center require $\mathrm{poly}(d, m)$ space and can be reused at each iteration. □

D.2 Moment and sampling routines

We use the standard conditional-Bernoulli routines of Chen and Liu [14, Sections 2 and 4] to compute $F(\theta)$, $\mu(p_\theta)$, and $M(p_\theta)$, and to draw exact samples from p_θ without enumerating $\mathcal{X}$.

Lemma 9. *Given $\theta \in \mathbb{R}^d$, the log-partition function $F(\theta)$, all marginals $\mu_i(p_\theta)$, and all pairwise marginals*

$$\pi_{ij}(p_\theta) := \mathbb{P}_{X \sim p_\theta}(X_i = X_j = 1)$$

can be computed, and an exact sample from p_θ can be drawn, in polynomial time, without enumerating $\mathcal{X}$.

Proof. By its definition in Section 2, p_θ is the distribution of independent Bernoulli variables with success probabilities $w_i/(1+w_i)$, conditioned on their sum being m. Thus the conditional-Bernoulli routines of Chen and Liu [14, Sections 2 and 4] apply.

We first compute the normalizing constant. Splitting the sum defining e_k according to whether item j is selected gives, for $j \in [d]$ and $1 \le k \le m$,

$$e_k(w_1, \dots, w_j) = e_k(w_1, \dots, w_{j-1}) + w_j e_{k-1}(w_1, \dots, w_{j-1}). \tag{76}$$

This is Method 2 of Chen and Liu [14, Section 2]. With the conventions for e_0 and e_k given above, (76) computes the table for $0 \le k \le m$ and $0 \le j \le d$ in $O(dm)$ operations and space. In particular, it computes $e_m(w)$ and hence $F(\theta) = \log e_m(w)$.

For every $i \in [d]$, the definitions of $\mu_i(p_\theta)$ and p_θ give

$$\mu_i(p_\theta) = \sum_{\substack{S \subseteq [d]:\, |S| = m \\ i \in S}} p_\theta(S) = \frac{w_i}{e_m(w)} \sum_{\substack{A \subseteq [d] \setminus \{i\} \\ |A| = m-1}} \prod_{\ell \in A} w_\ell = \frac{w_i e_{m-1}((w_\ell)_{\ell \in [d] \setminus \{i\}})}{e_m(w)}. \tag{77}$$

Similarly, for distinct $i, j \in [d]$,

$$\pi_{ij}(p_\theta) = \sum_{\substack{S \subseteq [d]:\, |S| = m \\ i,j \in S}} p_\theta(S) = \frac{w_i w_j}{e_m(w)} \sum_{\substack{A \subseteq [d] \setminus \{i,j\} \\ |A| = m-2}} \prod_{\ell \in A} w_\ell = \frac{w_i w_j e_{m-2}((w_\ell)_{\ell \in [d] \setminus \{i,j\}})}{e_m(w)}. \tag{78}$$

The last expression is zero when $m = 1$, by the convention $e_{-1} = 0$. On the diagonal, $\pi_{ii}(p_\theta) = \mu_i(p_\theta)$. Since $X_i X_j$ is the indicator of $X_i = X_j = 1$, these probabilities also give every entry of

$M(p_\theta)$. Applying (76) to the $O(d^2)$ lists obtained by deleting one or two weights therefore computes (77) and (78) in $O(d^3 m)$ operations. Processing one list at a time and storing the resulting moments requires $O(dm + d^2)$ space.

Finally, Procedure 3 of Chen and Liu [14, Section 4] examines items $1, \dots, d$ in order. If k items remain to be selected from $\{i, \dots, d\}$, it selects item i with probability

$$\frac{w_i e_{k-1}(w_{i+1}, \dots, w_d)}{e_k(w_i, \dots, w_d)}. \tag{79}$$

By the same calculation as in (77), (79) is the conditional probability of selecting item i under p_θ, given all preceding choices. Thus this procedure draws an exact sample from p_θ. Applying (76) in reverse order computes the required values $e_k(w_i, \dots, w_d)$ in $O(dm)$ operations and space. After this preprocessing, sampling uses $O(d)$ operations and at most d independent uniform random draws. All these computations are polynomial in d and m and do not enumerate $\mathcal{X}$. □

D.3 Proof of Proposition 1

Proposition 1. *For every $t \in \{2, \dots, T+1\}$, Algorithm 1 maintains a weighted m-set distribution $p_t = p_{\theta_t} \in \mathcal{P}_{\lambda/2}$, represented by $\theta_t \in \mathbb{R}^d$ satisfying Equation (17) at round $t-1$. Moreover, the algorithm can be implemented in $T \cdot \mathrm{poly}(d, m, \log T)$ time and $\mathrm{poly}(d, m)$ space, without enumerating $\mathcal{X}$.*

Proof. We first prove by induction that, for every $t \in [T+1]$, $p_t = p_{\theta_t}$ is a weighted m-set distribution in $\mathcal{P}_{\lambda/2}$ and

$$\mathrm{KL}(U \| p_t) \le t - 1. \tag{80}$$

The additional KL bound in (80) will allow us to apply Lemma 8 at each round. The base case holds because $\theta_1 = \mathbf{0}$ and $p_1 = p_{\theta_1} = U \in \mathcal{P}_\lambda \subseteq \mathcal{P}_{\lambda/2}$.

Suppose that the claims hold at round $t \in [T]$. By Lemma 2, M_t is positive definite. In addition, c_t and $\widehat{\ell}_t = M_t^{-1} X_t Y_t$ are well defined. Since $Z_t(x) = z_t^\top x$, with $z_t = \widehat{\ell}_t - \eta c_t$, the MWU step in (15) gives $\widetilde{p}_{t+1} = p_{\widetilde{\theta}_{t+1}}$, where $\widetilde{\theta}_{t+1} = \theta_t - \eta z_t$. Thus, the unprojected distribution remains a weighted m-set distribution.

We next verify the input condition of Lemma 8. The induction hypothesis ensures that p_t is a weighted m-set distribution in $\mathcal{P}_{\lambda/2}$. Together with the algorithm's parameter choices, this verifies the assumptions of Lemma 3 at round t, giving $|\eta Z_t(x)| \le 1/4$ for every $x \in \mathcal{X}$. Hence $|\log \mathbb{E}_{X \sim p_t}[e^{-\eta Z_t(X)}]| \le 1/4$, and

$$\left| \log \frac{\widetilde{p}_{t+1}(x)}{p_t(x)} \right| = \left| -\eta Z_t(x) - \log \mathbb{E}_{X \sim p_t}[e^{-\eta Z_t(X)}] \right| \le \frac{1}{2}.$$

By the definition of KL divergence, this implies

$$\mathrm{KL}(U \| \widetilde{p}_{t+1}) \le \mathrm{KL}(U \| p_t) + \frac{1}{2} \le t - 1 + \frac{1}{2} \le T.$$

The projection procedure described in Appendix D.1 with input $(\widetilde{\theta}_{t+1}, \eta, T)$ therefore satisfies all the assumptions of Lemma 8. Its output θ_{t+1} defines a weighted m-set distribution $p_{t+1} = p_{\theta_{t+1}} \in \mathcal{P}_{\lambda/2}$ satisfying (17). Taking $q = U \in \mathcal{P}_\lambda$ in that inequality and dropping the nonnegative term $\mathrm{KL}(p_{t+1} \| \widetilde{p}_{t+1})$ gives

$$\mathrm{KL}(U \| p_{t+1}) \le \mathrm{KL}(U \| \widetilde{p}_{t+1}) + \varepsilon_{\mathrm{p}} \le t - 1 + \frac{1}{2} + \varepsilon_{\mathrm{p}} \le t,$$

where $\varepsilon_{\mathrm{p}} = \eta/T \le 1/64$. This closes the induction.

We now bound the computational cost. Given θ_t, Lemma 9 computes the marginals and M_t in $O(d^3 m)$ operations and draws X_t in polynomial time without enumerating $\mathcal{X}$. Computing c_t takes $O(d)$ operations, solving $M_t \widehat{\ell}_t = X_t Y_t$ takes $O(d^3)$, and updating z_t and $\widetilde{\theta}_{t+1}$ takes $O(d)$. For the projection step, since $K \ge 2$ and $\delta < 1$, the learning rate in Algorithm 1 satisfies

$$\eta \ge \frac{1}{256 d\sqrt{T}}.$$

Together with $B_{\mathrm{p}} = 4dT$ and $\tau = \eta^2/(32 d^2 T^4)$, this gives $\log B_{\mathrm{p}} + \log(1/\tau) = O(\log(dT))$. Thus, Lemma 8 bounds the projection cost by $\mathrm{poly}(d, m, \log T)$ operations per round. The moment and sampling routines and the projection use $\mathrm{poly}(d, m)$ space, which can be reused across rounds. Including the polynomial-time initialization, the total cost is $T \cdot \mathrm{poly}(d, m, \log T)$ time and $\mathrm{poly}(d, m)$ space, without enumerating $\mathcal{X}$. □

E Proof of the matching lower bound

We prove the following fully explicit version of Theorem 2.

Theorem 3. *Let $d \ge 2$, $1 \le m \le d-1$, and $\delta \in (0, 2^{-36}]$, and assume that*

$$T \ge \left\lceil 64 d^2 + 4d \min\{m, d-m\}^2 \log(1/\delta) \right\rceil.$$

Then, for every randomized algorithm on $\mathcal{X}_{d,m}$, there exists a deterministic adaptive non-anticipating loss process whose action losses belong to $[0,1]$ and such that

$$\mathbb{P}\left(R_T \ge 2^{-43} \sqrt{dT \left(\log \binom{d}{m} + \log(1/\delta) \right)} \right) \ge \delta.$$

Thus, Theorem 3 implies Theorem 2 by taking $c = 2^{-43}$ and $\delta_0 = 2^{-36}$. The horizon threshold is at most $1 + 64d^2 + 4d \min\{m, d-m\}^2 \log(1/\delta)$, and is therefore bounded by a polynomial in d, m, and $\log(1/\delta)$ since $\min\{m, d-m\} \le m$. The rest of this appendix proves Theorem 3.

Our strategy is to prove two separate lower bounds of order $\sqrt{dT \log K}$ (see Propositions 2 and 3) and $\sqrt{dT \log(1/\delta)}$ (see Proposition 4) on the action space $\mathcal{X}_{d,m}$ with $m \le d/2$. We then combine the two bounds and use Lemma 11 to extend the result to $m > d/2$. In these lower bounds, we temporarily allow randomized adaptive non-anticipating adversaries. Thus, conditionally on $\mathcal{F}_{t-1}$, the adversary may randomize the choice of ℓ_t before the learner draws $X_t \sim p_t$. Notice that such auxiliary randomness used by the adversary is independent of the learner's private randomness. By Lemma 10, we can then transfer the regret guarantees obtained against a randomized adversary to a deterministic adaptive non-anticipating adversary, as required by our model.

Before presenting Lemmas 10 and 11, we introduce some additional notation. For $t \in [T]$, we denote by $\mathcal{H}_t := (X_1, Y_1, \dots, X_t, Y_t)$ the history of actions and observed feedback up to round t, where $Y_t := X_t^\top \ell_t$. We set $\mathcal{H}_0 := \varnothing$. The filtration $(\mathcal{F}_t)_{t=0}^T$ introduced in Section 2 satisfies $\mathcal{F}_t = \sigma(\mathcal{H}_t)$ for every $t \in \{0, \dots, T\}$. Thus, conditioning on $\mathcal{H}_{t-1}$ is equivalent to conditioning on $\mathcal{F}_{t-1}$. We use the KL divergence and total variation distance defined in Section 2. For probability laws Q and P on the same measurable space, we also define the χ^2 divergence by

$$\chi^2(Q\|P) := \int \left(\frac{dQ}{dP} - 1 \right)^2 dP$$

when $Q \ll P$, and set it to $+\infty$ otherwise.

To formalize the randomized adversaries introduced above, let Υ collect the algorithm's private randomness and let Ξ collect the adversary's auxiliary randomness. We assume, without loss of generality, that these two collections are sampled independently before the interaction begins. This is equivalent to generating fresh randomness at each round since sampling it in advance does not change the law of the interaction, and decisions still depend on the history observed so far. We can then write the algorithm's chosen action at time $t \in [T]$ as $X_t = h_t(\mathcal{H}_{t-1}, \Upsilon)$ and the adversary's chosen loss vector as $\ell_t = f_t(\mathcal{H}_{t-1}, \Xi)$, where h_t and f_t are deterministic functions. Fixing Ξ therefore yields a deterministic adaptive non-anticipating adversary, as required by our model.

We can prove that the following lemma holds.

Lemma 10. *Fix an algorithm, and let Υ and Ξ denote its private randomness and the adversary's independent randomness, respectively. For any $b \geq 0$ and $p \in [0,1]$, if $\mathbb{P}(R_T \geq b) \geq p$ over Υ and Ξ, then there exists a realization $\bar{\xi}$ of Ξ such that the deterministic adversary obtained by fixing $\Xi = \bar{\xi}$ satisfies $\mathbb{P}(R_T \geq b) \geq p$ over Υ alone.*

Proof. For each realization ξ of Ξ, let $q(\xi)$ be the probability of $R_T \geq b$ against the deterministic adversary obtained by fixing $\Xi = \xi$, where the probability is taken over Υ alone. By independence and the law of total expectation,

$$\mathbb{P}(R_T \geq b) = \mathbb{E}_{\Xi}[q(\Xi)] \geq p,$$

where the probability on the left is taken over both Υ and Ξ. If no such realization $\bar{\xi}$ existed, then $q(\xi) < p$ for every realization ξ of Ξ, and hence $\mathbb{E}_{\Xi}[q(\Xi)] < p$. This contradicts the above inequality and proves the claim. □

Lemma 10 allows us to use a randomized adaptive non-anticipating adversary in our lower-bound construction. This is because, for every fixed algorithm, we can transfer the lower bound on regret to a deterministic adaptive non-anticipating adversary, preserving the same probability guarantee. The following lemma transfers a regret lower bound from m-set to $(d-m)$-set actions at the same time horizon. It therefore suffices to construct the lower-bound instances for $m \leq d/2$.

Lemma 11. *Fix integers $d \geq 2$, $1 \leq m \leq d-1$, and $T \geq 1$, and constants $c_0 \geq 0$ and $\delta_0 \in [0,1]$. Suppose that, for every randomized algorithm on $\mathcal{X}_{d,m}$, there exists a deterministic adaptive non-anticipating adversary whose action losses belong to $[0,1]$ at every round and such that the regret is at least c_0 with probability at least δ_0. Then the same statement holds on $\mathcal{X}_{d,d-m}$, with the same horizon T and parameters c_0 and δ_0.*

Proof. Fix any randomized algorithm on $\mathcal{X}_{d,d-m}$, and call it algorithm 1. Construct algorithm 2 on $\mathcal{X}_{d,m}$ that runs algorithm 1 internally, plays $y_t = \mathbf{1} - x_t$ whenever algorithm 1 selects x_t, and returns the observed scalar feedback to algorithm 1. By assumption, there exists a deterministic adaptive non-anticipating adversary 2 for algorithm 2, with loss vectors g_t satisfying $y^\top g_t \in [0,1]$ for every $y \in \mathcal{X}_{d,m}$, such that algorithm 2's regret is at least c_0 with probability at least δ_0. We construct adversary 1 for algorithm 1 as follows. At each round, replace every past action x_u by $\mathbf{1} - x_u$ and keep its observed feedback to obtain the corresponding m-set history. Given this history, adversary 2 returns the loss vector g_t. We then set

$$\widetilde{\ell}_t := -g_t + \frac{\mathbf{1}^\top g_t}{d-m}\mathbf{1}. \tag{81}$$

This defines adversary 1, a deterministic adaptive non-anticipating adversary for algorithm 1, since it uses only past actions and feedback. For every $x \in \mathcal{X}_{d,d-m}$ and its complement $y = \mathbf{1} - x \in \mathcal{X}_{d,m}$, we have $x^\top \mathbf{1} = d - m$ and hence

$$x^\top \widetilde{\ell}_t = -x^\top g_t + \mathbf{1}^\top g_t = y^\top g_t.$$

In particular, all $(d-m)$-set action losses lie in $[0,1]$.

Run algorithm 1 against adversary 1 and algorithm 2 against adversary 2 with the same private randomness. The above identity implies, by induction over the rounds, that they select complementary actions and receive the same scalar feedback. Let $R_T^{(d-m)}$ and $R_T^{(m)}$ denote their respective regrets. The same identity pairs every fixed comparator in $\mathcal{X}_{d,d-m}$ with its complement in $\mathcal{X}_{d,m}$, so both the accumulated losses of the algorithms and the minimum accumulated comparator losses are equal. Thus $R_T^{(d-m)} = R_T^{(m)}$ for every realization of their common private randomness, and consequently

$$\mathbb{P}\big(R_T^{(d-m)} \geq c_0\big) = \mathbb{P}\big(R_T^{(m)} \geq c_0\big) \geq \delta_0.$$

Since algorithm 1 was arbitrary, this proves the claim. □

E.1 Preliminary lemmas

In the following we denote

$$g(z) := 1 - z^2,\ \psi_u(z) := \frac{z}{2} + ug(z) \quad \text{with} \quad z \in [-1,1].$$

The next lemma bounds the KL divergence between the distributions that we use to generate the feedback in our lower-bound instances.

Lemma 12. *For every $|u| \leq 1/16$, the map ψ_u is an increasing diffeomorphism from $[-1,1]$ onto $[-1/2,1/2]$, meaning a differentiable bijection with a differentiable inverse. Let p_u be the density of $\psi_u(Z)$ when $Z \sim \mathrm{Unif}[-1,1]$. For every $|u|, |v| \leq 1/16$,*

$$\mathrm{KL}(p_u \| p_v) \leq 80(u-v)^2. \tag{82}$$

Proof. For $|u| \leq 1/16$ and $z \in [-1,1]$,

$$\frac{3}{8} \leq \psi_u'(z) = \frac{1}{2} - 2uz \leq \frac{5}{8}. \tag{83}$$

Together with $\psi_u(-1) = -1/2$ and $\psi_u(1) = 1/2$, this proves the diffeomorphism claim. Fix $y \in [-1/2, 1/2]$, and let $z := \psi_u^{-1}(y)$. Since $y = \psi_u(z) = z/2 + u(1-z^2)$, we have

$$\begin{aligned}
1 - 16uy + 16u^2 &= 1 - 16u\left(\frac{z}{2} + u(1-z^2)\right) + 16u^2 \\
&= 1 - 8uz - 16u^2 + 16u^2 z^2 + 16u^2 \\
&= 1 - 8uz + 16u^2 z^2 \\
&= (1 - 4uz)^2 \\
&= 4\left(\frac{1}{2} - 2uz\right)^2 \\
&= 4\psi_u'(z)^2.
\end{aligned}$$

The change-of-variables formula therefore gives

$$p_u(y) = \frac{1}{2\psi_u'(z)} = \frac{1}{\sqrt{1 - 16uy + 16u^2}},$$

with $9/16 \le 1 - 16uy + 16u^2 \le 25/16$. Moreover, since $|y| \le 1/2$,

$$(32u - 16y)^2 = 64(-16uy + 16u^2) + 256y^2 \le 64(1 - 16uy + 16u^2).$$

It follows that

$$|\partial_u p_u(y)| = \frac{|32u - 16y|}{2(1 - 16uy + 16u^2)^{3/2}} \le \frac{4}{1 - 16uy + 16u^2} \le \frac{64}{9} < 8.$$

Moreover, since $1 - 16vy + 16v^2 \le 25/16$, we have

$$p_v(y) = \frac{1}{\sqrt{1 - 16vy + 16v^2}} \ge \frac{1}{\sqrt{25/16}} = \frac{4}{5}.$$

The mean-value theorem, on the common support of length one, yields

$$\chi^2(p_u \| p_v) = \int_{-1/2}^{1/2} \frac{(p_u(y) - p_v(y))^2}{p_v(y)}\, dy \le 80(u - v)^2.$$

Since $\mathrm{KL} \le \chi^2$, this proves (82). □

The next lemma turns a lower bound on the expectation into a probability bound.

Lemma 13. *Let X be a real-valued random variable, and let $0 < \mu \le M$. Suppose that $X \le M$ almost surely and $\mathbb{E}[X] \ge \mu$. Then*

$$\mathbb{P}\left(X \ge \frac{\mu}{2}\right) \ge \frac{\mu}{2M}.$$

Proof. Since

$$\mu \le \mathbb{E}[X] \le \mathbb{E}\left[X\mathbf{1}\left\{X < \frac{\mu}{2}\right\}\right] + \mathbb{E}\left[X\mathbf{1}\left\{X \ge \frac{\mu}{2}\right\}\right] \le \frac{\mu}{2} + M\mathbb{P}(X \ge \mu/2),$$

rearranging proves the claim. □

E.2 An $\Omega(\sqrt{dT \log K})$ lower bound

We prove the lower bound separately for $r = m/d \le r_\star$ and $r = m/d > r_\star$, with $m \le d/2$ and $r_\star := 2^{-32}$.

E.2.1 The case $r \le r_\star$

Proposition 2. *Let $1 \le m \le d/2$, with $m/d \le r_\star$, and assume $T \ge 64d^2$. For every randomized algorithm on $\mathcal{X}_{d,m}$, there exists an adaptive non-anticipating adversary choosing action losses in $[0,1]$ at every round such that*

$$\mathbb{P}\left(R_T \ge 2^{-24}\sqrt{dT \log \binom{d}{m}}\right) \ge \frac{1}{8}. \tag{84}$$

Proof. **Instances and regret.** First, we set

$$N_0 := \lceil 8Tr \rceil \quad \text{and} \quad \varepsilon := m\sqrt{2^{-36}\log(1/r)/N_0}.$$

Independently draw indicators $U_i \sim \text{Bernoulli}(\rho)$, with $\rho := 2^{-8}$, for every item $i \in [d]$ and uniform noises $Z_t \sim \text{Unif}[-1,1]$ for every round $t \in [T]$, and collect these random variables in $\Xi = (U_1, \dots, U_d, Z_1, \dots, Z_T)$, independently of the algorithm's private randomness Υ.

At round $t \in [T]$, let $A_{t,i} := \mathbf{1}\{\sum_{\tau<t} X_{\tau,i} < N_0\}$ for each item i, and define its loss by

$$\ell_{t,i} := \frac{1+Z_t}{2m} - \frac{\varepsilon}{m} g(Z_t) A_{t,i} U_i. \tag{85}$$

Intuitively, $A_{t,i} = 1$ exactly when the algorithm has selected item i fewer than N_0 times before round t, and $A_{t,i} = 0$ otherwise. Thus, when $U_i = 1$, we subtract a positive quantity from the loss of item i until that item has been selected N_0 times. To verify that action losses lie in $[0,1]$, note that $N_0 \geq 8Tr$ and $T \geq 64d^2$ give

$$\varepsilon^2 = \frac{m^2\log(1/r)}{2^{36}N_0} \leq \frac{m^2\log(1/r)}{2^{39}Tr} \leq \frac{m^2\log(1/r)}{2^{45}d^2r} = 2^{-45} r\log(1/r) \leq 2^{-45},$$

so $\varepsilon < 1/16$.

Since $m\ell_{t,i} = 1/2 + \psi_{-\varepsilon A_{t,i}U_i}(Z_t)$, Lemma 12 therefore gives $\ell_{t,i} \in [0, 1/m]$, and hence every action x is such that $x^\top \ell_t \in [0,1]$ for every t. Moreover, once Ξ is fixed, the losses in (85) are deterministic functions of past actions, as required for an adaptive non-anticipating adversary.

Finally, to express the regret, we define the following scores

$$S_i := U_i \sum_{t=1}^{T} A_{t,i} g(Z_t), \qquad S_{\mathrm{L}} := \sum_{t=1}^{T} g(Z_t) \sum_{i=1}^{d} A_{t,i} X_{t,i} U_i.$$

The quantity $(\varepsilon/m)S_i$ is the amount subtracted from the common term $\sum_{t=1}^{T}(1+Z_t)/(2m)$ in the cumulative loss of item i. Similarly, $(\varepsilon/m)S_{\mathrm{L}}$ is the amount subtracted from the common term $\sum_{t=1}^{T}(1+Z_t)/2$ in the algorithm's cumulative loss. Thus, a larger S_i means a smaller cumulative item loss, so the best fixed action in hindsight selects the m items with the largest S_i. We denote their sum by $\mathrm{Top}_m(S)$. Since the common loss term cancels, the regret can be written as follows:

$$R_T = \frac{\varepsilon}{m}\Gamma, \qquad \Gamma := \mathrm{Top}_m(S) - S_{\mathrm{L}} \leq mT, \tag{86}$$

since $S_i \leq T$ for every $i \in [d]$.

Two deterministic bounds. For each item i, we have $A_{t,i}X_{t,i} = 1$ at most N_0 times, because of the definition of $A_{t,i}$. Since $0 \leq g(x) \leq 1$, for every $x \in [-1,1]$, we have:

$$S_{\mathrm{L}} \leq N_0 \sum_{i=1}^{d} U_i.$$

Furthermore, $Tr \geq 1$, so $N_0 \leq 8Tr + 1 \leq 9Tr$. Consequently,

$$\mathbb{E}[S_{\mathrm{L}}] \leq \frac{N_0 d}{256} \leq \frac{9}{256} mT, \tag{87}$$

since $\mathbb{E}[U_i] = 1/256$ and $r = m/d$ by definition. Before round T, there have been exactly $m(T-1)$ item selections in total, and each item with $A_{T,i} = 0$ accounts for at least N_0 of them. Thus

$$\begin{aligned}
\sum_{i=1}^{d} A_{T,i} &= d - \sum_{i=1}^{d} (1 - A_{T,i}) \\
&= d - \sum_{i=1}^{d} \mathbf{1}\left\{ \sum_{\tau=1}^{T-1} X_{\tau,i} \geq N_0 \right\} \\
&\geq d - \frac{1}{N_0} \sum_{i=1}^{d} \sum_{\tau=1}^{T-1} X_{\tau,i} \\
&= d - \frac{m(T-1)}{N_0} \\
&\geq d - \frac{d(T-1)}{8T} \geq \frac{7}{8} d.
\end{aligned} \tag{88}$$

The first inequality uses $N_0 \, \mathbf{1}\{x \geq N_0\} \leq x$ for $x \geq 0$, and the third equality uses $\sum_{i=1}^{d} X_{\tau,i} = m$. The remaining inequality follows from $N_0 \geq 8Tm/d$. Moreover, $A_{T,i} = 1$ implies $A_{t,i} = 1$ for every $t \in [T]$. Since $S_i = U_i \sum_{t=1}^{T} A_{t,i} g(Z_t)$ and $g \geq 0$, we have $0 \leq S_i \leq \sum_{t=1}^{T} g(Z_t)$ for every item i, with $S_i = \sum_{t=1}^{T} g(Z_t)$ whenever $U_i A_{T,i} = 1$. Consequently, on the event $\{\sum_{i=1}^{d} U_i A_{T,i} \geq m\}$, at least m items attain this upper bound, and therefore

$$\mathrm{Top}_m(S) = m \sum_{t=1}^{T} g(Z_t).$$

We next show that this event has large probability.

Comparing the feedback distributions. Fix an item $i \in [d]$, a realization of the algorithm's private randomness Υ, and realizations of all indicators U_j with $j \neq i$. Let P_i^0 and P_i^1 denote the resulting laws of $\mathcal{H}_T$ when $U_i = 0$ and $U_i = 1$, respectively, with only $Z_1, \ldots, Z_T$ remaining random. Let $U_{-i} := (U_j)_{j \neq i}$. The dependence of these laws on the fixed values of Υ and U_{-i} is left implicit. We write $\mathbb{E}_{P_i^0}$ and $\mathbb{E}_{P_i^1}$ for the corresponding expectations. Summing the item losses in (85) over the m selected items gives

$$Y_t - \frac{1}{2} = \psi_{u_t}(Z_t), \qquad u_t = -\frac{\varepsilon}{m} \sum_{j=1}^{d} A_{t,j} X_{t,j} U_j,$$

with $|u_t| \leq \varepsilon < 1/16$. Given the fixed values of Υ and U_{-i}, the same preceding history $\mathcal{H}_{t-1}$ determines the same action X_t and indicators $A_{t,j}$ under the two laws. Since Z_t remains independent and uniform, the corresponding parameters satisfy

$$|u_t^0 - u_t^1| = \frac{\varepsilon}{m} A_{t,i} X_{t,i}.$$

The chain rule for KL divergence, Lemma 12, and the bound $\sum_{t \in [T]} A_{t,i} X_{t,i} \leq N_0$ give

$$\mathrm{KL}(P_i^0 \| P_i^1) \leq \frac{128\varepsilon^2}{m^2} \mathbb{E}_{P_i^0}\left[\sum_{t=1}^{T} A_{t,i} X_{t,i} \right] \leq \frac{128\varepsilon^2 N_0}{m^2} = 2^{-29} \log(1/r). \tag{89}$$

This bound holds for every fixed realization of Υ and U_{-i}.

Counting items that remain active. For each $i \in [d]$, define the event

$$H_i := \left\{ A_{T,i} = 1,\ \sum_{j \neq i} U_j A_{T,j} < m \right\}.$$

For fixed Υ, U_{-i}, the indicators $A_{T,j}$ are determined by $\mathcal{H}_T$, so H_i is the same measurable event of the history under P_i^0 and P_i^1. We now average over Υ and U_{-i} and define

$$h_{0,i} := \mathbb{P}(H_i \mid U_i = 0) \qquad \text{and} \qquad h_{1,i} := \mathbb{P}(H_i \mid U_i = 1),$$

where the probabilities are taken over Υ, U_{-i}, and $Z_1, \ldots, Z_T$, with U_i fixed to 0 and 1, respectively. Let

$$\bar{h}_0 := \frac{1}{d} \sum_{i=1}^{d} h_{0,i}, \qquad \bar{h}_1 := \frac{1}{d} \sum_{i=1}^{d} h_{1,i}.$$

First, we bound $(1-\rho) d \bar{h}_0$ as follows

$$\begin{aligned}
(1-\rho) d \bar{h}_0 &= (1-\rho) \sum_{i=1}^{d} h_{0,i} \\
&= (1-\rho) \sum_{i=1}^{d} \mathbb{P}(H_i \mid U_i = 0) \\
&= \sum_{i=1}^{d} \mathbb{P}(U_i = 0, H_i) \\
&= \sum_{i=1}^{d} \mathbb{E}[\mathbf{1}\{U_i = 0, H_i\}] \\
&= \sum_{i=1}^{d} \mathbb{E}\left[\mathbf{1}\left\{ U_i = 0, A_{T,i} = 1,\ \sum_{j \neq i} U_j A_{T,j} < m \right\} \right] \\
&= \sum_{i=1}^{d} \mathbb{E}\left[\mathbf{1}\left\{ U_i = 0, A_{T,i} = 1,\ \sum_{j=1}^{d} U_j A_{T,j} < m \right\} \right] \\
&= \sum_{i=1}^{d} \mathbb{E}\left[(1-U_i) A_{T,i} \mathbf{1}\left\{ \sum_{j=1}^{d} U_j A_{T,j} < m \right\} \right] \\
&= \mathbb{E}\left[\left(\sum_{i=1}^{d} A_{T,i} - \sum_{i=1}^{d} U_i A_{T,i} \right) \mathbf{1}\left\{ \sum_{j=1}^{d} U_j A_{T,j} < m \right\} \right] \\
&\geq \mathbb{E}\left[\left(\frac{7}{8} d - \sum_{i=1}^{d} U_i A_{T,i} \right) \mathbf{1}\left\{ \sum_{j=1}^{d} U_j A_{T,j} < m \right\} \right] \\
&\geq \left(\frac{7}{8} d - m \right) \mathbb{E}\left[\mathbf{1}\left\{ \sum_{j=1}^{d} U_j A_{T,j} < m \right\} \right] \\
&= \left(\frac{7}{8} d - m \right) \mathbb{P}\left(\sum_{i=1}^{d} U_i A_{T,i} < m \right), \qquad (90)
\end{aligned}$$

where the probabilities are taken over Υ, $U_1, \dots, U_d$, and $Z_1, \dots, Z_T$. The first inequality uses $\sum_{i=1}^d A_{T,i} \geq 7d/8$ from (88). Writing $N_{\mathrm{act}} := \sum_{i=1}^d U_i A_{T,i}$, we also bound $\rho d \bar{h}_1$ as follows:

$$\rho d \bar{h}_1 = \sum_{i=1}^d \mathbb{E}\left[U_i A_{T,i} \mathbf{1}\left\{\sum_{j \neq i} U_j A_{T,j} < m\right\}\right] = \mathbb{E}[N_{\mathrm{act}} \mathbf{1}\{N_{\mathrm{act}} \leq m\}] \leq m. \tag{91}$$

The second equality holds pointwise: if $U_i A_{T,i} = 1$, then $\sum_{j \neq i} U_j A_{T,j} = N_{\mathrm{act}} - 1$, and $N_{\mathrm{act}} - 1 < m$ is equivalent to $N_{\mathrm{act}} \leq m$ because N_{act} is integer-valued. If $U_i A_{T,i} = 0$, the summand vanishes. Thus, this identity requires no independence assumption. In particular, (91) gives $\bar{h}_1 \leq m/(\rho d) = 256r$. Recall that $\mathrm{kl}(a, b)$ denotes the KL divergence between Bernoulli laws with parameters a and b. Then

$$\begin{aligned}
\mathrm{kl}(\bar{h}_0, \bar{h}_1) &\leq \frac{1}{d} \sum_{i=1}^d \mathrm{kl}(h_{0,i}, h_{1,i}) \\
&\leq \frac{1}{d} \sum_{i=1}^d \mathbb{E}_{\Upsilon, U_{-i}}\left[\mathrm{kl}\big(P_i^0(H_i), P_i^1(H_i)\big)\right] \\
&\leq \frac{1}{d} \sum_{i=1}^d \mathbb{E}_{\Upsilon, U_{-i}}\left[\mathrm{KL}(P_i^0 \| P_i^1)\right] \\
&\leq 2^{-29} \log(1/r).
\end{aligned} \tag{92}$$

The first two inequalities use joint convexity of binary KL divergence, first over the items and then over Υ, U_{-i}. The third applies data processing to H_i for each fixed Υ, U_{-i}, and the last uses the uniform bound in (89).

We notice that if $\bar{h}_1 = 0$, we must also have $\bar{h}_0 = 0$, since otherwise $\mathrm{kl}(\bar{h}_0, \bar{h}_1) = +\infty$, contradicting the upper bound in (92). Then (90) gives $\mathbb{P}(\sum_{i=1}^d U_i A_{T,i} < m) = 0$. We may therefore assume that $\bar{h}_1 > 0$. Using $\mathrm{kl}(a, b) \geq a \log(1/b) - \log 2$ and $\bar{h}_1 \leq 256r$ according to (91), we obtain

$$\bar{h}_0 \leq \frac{2^{-29} \log(1/r) + \log 2}{\log(1/r) - 8 \log 2} \leq \frac{4}{3}\left(2^{-29} + \frac{1}{32}\right) \leq \frac{3}{64},$$

where we used $r \leq r_\star = 2^{-32}$. Since $r \leq 1/8$, (90) now gives

$$\mathbb{P}\left(\sum_{i=1}^d U_i A_{T,i} < m\right) \leq \frac{1 - \rho}{7/8 - r} \bar{h}_0 \leq \frac{4}{3} \bar{h}_0 \leq \frac{1}{16}. \tag{93}$$

Since $\mathrm{Top}_m(S) = m \sum_{t=1}^T g(Z_t)$ on the event $\{\sum_{i=1}^d U_i A_{T,i} \geq m\}$, we have

$$\begin{aligned}
\mathbb{E}[\mathrm{Top}_m(S)] &\geq m \mathbb{E}\left[\left(\sum_{t=1}^T g(Z_t)\right) \mathbf{1}\left\{\sum_{i=1}^d U_i A_{T,i} \geq m\right\}\right] \\
&\geq m \mathbb{E}\left[\sum_{t=1}^T g(Z_t)\right] - mT \mathbb{P}\left(\sum_{i=1}^d U_i A_{T,i} < m\right) \\
&\geq \left(\frac{2}{3} - \frac{1}{16}\right) mT = \frac{29}{48} mT,
\end{aligned} \tag{94}$$

where the last inequality holds by (93). Notice that this bound does not require the sums $\sum_{t=1}^T g(Z_t)$ and $\sum_{i=1}^d U_i A_{T,i}$ to be independent since we only use $\sum_{t=1}^T g(Z_t) \leq T$ and $\mathbb{E}[\sum_{t=1}^T g(Z_t)] = 2T/3$.

Putting it all together. Combining (87) and (94) gives

$$\mathbb{E}[\Gamma] \geq \left(\frac{29}{48} - \frac{9}{256}\right) mT = \frac{437}{768} mT \geq \frac{1}{4} mT.$$

Since $\mathbb{E}[\Gamma] \geq mT/4$ and $\Gamma \leq mT$ by (86), Lemma 13 gives

$$\mathbb{P}(\Gamma \geq mT/8) \geq \frac{1}{8}. \tag{95}$$

Here the probability is taken over the algorithm's private randomness Υ and the adversary's randomness $\Xi = (U_1, \ldots, U_d, Z_1, \ldots, Z_T)$. Notice that this argument does not require Γ to be nonnegative. We next bound $\varepsilon T/8$. Using $N_0 \leq 9Tr$, we have

$$(\varepsilon T)^2 = \frac{m^2 \log(1/r) T^2}{2^{36} N_0} \geq \frac{dmT \log(1/r)}{9 \cdot 2^{36}}.$$

Moreover, $\binom{d}{m} \leq (ed/m)^m$ and $\log(1/r) \geq 1$ imply $\log \binom{d}{m} \leq 2m \log(1/r)$. Therefore,

$$\frac{\varepsilon T}{8} \geq \frac{2^{-21}}{\sqrt{18}} \sqrt{dT \log \binom{d}{m}} \geq 2^{-24} \sqrt{dT \log \binom{d}{m}}.$$

Using $R_T = (\varepsilon/m)\Gamma$ from (86) and (95), we conclude that

$$\mathbb{P}\left(R_T \geq 2^{-24} \sqrt{dT \log \binom{d}{m}}\right) \geq \mathbb{P}(\Gamma \geq mT/8) \geq \frac{1}{8}. \tag{96}$$

Since Ξ is independent of Υ, Lemma 10, with $b = 2^{-24}\sqrt{dT \log \binom{d}{m}}$ and $p = 1/8$, guarantees the existence of a realization $\bar{\xi}$ of Ξ for which this regret bound holds with probability at least $1/8$ over Υ alone. This proves (84). □

E.2.2 The case $r > r_\star$

Proposition 3. *Let $1 \leq m \leq d/2$, with $m/d > r_\star$, and assume $T \geq 64d^2$. For every randomized algorithm on $\mathcal{X}_{d,m}$, there exists a deterministic oblivious loss sequence with action losses in $[0,1]$ at every round such that*

$$\mathbb{P}\left(R_T \geq 2^{-42} \sqrt{dT \log \binom{d}{m}}\right) \geq 2^{-14}. \tag{97}$$

Proof. **Instances and regret.** We use the first $2m$ items, arranged into the pairs $(1,2), \ldots, (2m-1, 2m)$, and set $\varepsilon := m/(128\sqrt{T})$. Independently draw a sign $U_i \in \{-1, 1\}$ with equal probabilities for every $i \in [2m]$ and a uniform noise $Z_t \sim \mathrm{Unif}[-1,1]$ for every $t \in [T]$. Set $U_i := 0$ for $i > 2m$. Collect the adversary's randomness in $\Xi = (U_1, \ldots, U_{2m}, Z_1, \ldots, Z_T)$, independently of the algorithm's private randomness Υ. At every round, define the item losses by

$$\ell_{t,i} := \frac{1 + Z_t}{2m} - \frac{\varepsilon}{m} g(Z_t) U_i, \tag{98}$$

for every $i \in [d]$. This construction is oblivious as fixing Ξ determines the entire loss sequence before the interaction starts. We observe that $U_i = 1$ reduces the loss of item i, while $U_i = -1$ increases

it. Since $T \geq d^2$ and $m \leq d/2$, we have $\varepsilon \leq 1/256 < 1/16$. The identity $m\ell_{t,i} = 1/2 + \psi_{-\varepsilon U_i}(Z_t)$ and Lemma 12 give $\ell_{t,i} \in [0, 1/m]$. Hence $x^\top \ell_t \in [0, 1]$ for every t and $x \in \mathcal{X}$. In particular, the observed loss satisfies

$$Y_t = \frac{1}{2} + \psi_{u_t}(Z_t), \qquad u_t := -\frac{\varepsilon}{m} \sum_{i=1}^{d} X_{t,i} U_i \in [-\varepsilon, \varepsilon]. \tag{99}$$

As in the proof of Proposition 2, define the item and algorithm scores by

$$S_i := U_i \sum_{t=1}^{T} g(Z_t), \qquad S_{\mathrm{L}} := \sum_{t=1}^{T} g(Z_t) \sum_{i=1}^{d} X_{t,i} U_i.$$

As before, a larger S_i means a smaller cumulative item loss. To build a comparator, we choose the item with the larger S_i in each pair $(1, 2), \ldots, (2m-1, 2m)$. This gives a fixed comparator selecting exactly m items, with total score $\sum_{j=1}^{m} \max\{S_{2j-1}, S_{2j}\}$. As in the proof of Proposition 2, the common loss terms cancel, so the regret compared with this fixed action gives

$$R_T \geq \frac{\varepsilon}{m} \Gamma, \qquad \Gamma := \sum_{j=1}^{m} \max\{S_{2j-1}, S_{2j}\} - S_{\mathrm{L}}. \tag{100}$$

The comparator's score. The expected comparator score satisfies

$$\begin{aligned} \mathbb{E}\left[\sum_{j=1}^{m} \max\{S_{2j-1}, S_{2j}\}\right] &= \mathbb{E}\left[\sum_{t=1}^{T} g(Z_t)\right] \sum_{j=1}^{m} \mathbb{E}\left[\max\{U_{2j-1}, U_{2j}\}\right] \\ &= \frac{2T}{3} \cdot \frac{m}{2} = \frac{mT}{3}, \end{aligned} \tag{101}$$

where the expectation is taken over $U_1, \ldots, U_{2m}$ and the noises $Z_1, \ldots, Z_T$. The first equality uses $g(Z_t) \geq 0$ and the independence of the signs and noises. The second uses $\mathbb{E}[g(Z_t)] = 2/3$ and $\mathbb{E}[\max\{U_{2j-1}, U_{2j}\}] = 1/2$.

The algorithm's score. Fix an item $i \in [2m]$, a realization of the algorithm's private randomness Υ, and realizations of all signs U_j with $j \neq i$. Let P_i^+ and P_i^- denote the resulting laws of $\mathcal{H}_T$ when $U_i = 1$ and $U_i = -1$, respectively, with only $Z_1, \ldots, Z_T$ remaining random. Write $\mathbb{E}_{P_i^+}$ and $\mathbb{E}_{P_i^-}$ for the corresponding expectations. Given the algorithm's private randomness Υ and the history up to round $t-1$, namely $\mathcal{H}_{t-1}$, the action X_t played by the algorithm is fixed. Therefore, the two parameters in (99),

$$u_t^+ := -\frac{\varepsilon}{m} \left(\sum_{j \neq i} X_{t,j} U_j + X_{t,i} \right), \qquad u_t^- := -\frac{\varepsilon}{m} \left(\sum_{j \neq i} X_{t,j} U_j - X_{t,i} \right),$$

are also fixed, and their absolute difference is $2\varepsilon X_{t,i}/m$. By (99), the KL divergence between the conditional distributions of Y_t in the two instances, given the same history up to round $t-1$ and the same realization of the private randomness Υ, is equal to that between the distributions of $\psi_{u_t^+}(Z)$ and $\psi_{u_t^-}(Z)$, where $Z \sim \mathrm{Unif}[-1, 1]$, since adding the common constant $1/2$ does not change the

KL divergence. Writing p_u for the density of $\psi_u(Z)$, the KL chain rule and (82) give

$$\begin{aligned}\mathrm{KL}(P_i^+\|P_i^-) &= \sum_{t=1}^{T}\mathbb{E}_{P_i^+}\left[\mathrm{KL}(p_{u_t^+}\|p_{u_t^-})\right]\\ &\le 128\sum_{t=1}^{T}\mathbb{E}_{P_i^+}\left[(u_t^+-u_t^-)^2\right]\\ &= 128\left(\frac{2\varepsilon}{m}\right)^2\mathbb{E}_{P_i^+}\left[\sum_{t=1}^{T}X_{t,i}\right]\le\frac{1}{32}.\end{aligned}$$

The second equality uses $X_{t,i}^2 = X_{t,i}$, since $X_{t,i}\in\{0,1\}$. The last inequality follows from $\sum_{t=1}^{T}X_{t,i}\le T$ and $\varepsilon = m/(128\sqrt{T})$. Pinsker's inequality then yields

$$\mathrm{TV}(P_i^+,P_i^-)\le\sqrt{\frac{1}{2}\mathrm{KL}(P_i^+\|P_i^-)}\le\frac{1}{8}.$$

The number of times item i is selected depends only on $\mathcal{H}_T$ and lies in $[0,T]$. Since the two signs of U_i, corresponding to instances P_i^+ and P_i^-, are equally likely, it follows that

$$\begin{aligned}\left|\mathbb{E}\left[U_i\sum_{t=1}^{T}X_{t,i}\,\middle|\,\Upsilon,(U_j)_{j\neq i}\right]\right| &= \frac{1}{2}\left|\mathbb{E}_{P_i^+}\left[\sum_{t=1}^{T}X_{t,i}\right]-\mathbb{E}_{P_i^-}\left[\sum_{t=1}^{T}X_{t,i}\right]\right|\\ &\le\frac{T}{2}\mathrm{TV}(P_i^+,P_i^-)\le\frac{T}{16}.\end{aligned}$$

Conditionally on Υ, $U_1,\dots,U_{2m}$, and $Z_1,\dots,Z_{t-1}$, the action X_t is fixed, while Z_t remains uniformly distributed on $[-1,1]$. Using the above inequality, the tower property, and the fact that $\mathbb{E}[g(Z_t)] = 2/3$ for every t, we obtain

$$\mathbb{E}[S_{\mathrm{L}}] = \frac{2}{3}\sum_{i=1}^{2m}\mathbb{E}\left[U_i\sum_{t=1}^{T}X_{t,i}\right]\le\frac{2}{3}\cdot 2m\cdot\frac{T}{16} = \frac{mT}{12},$$

where the expectations are taken over $U_1,\dots,U_{2m}$, the noises $Z_1,\dots,Z_T$, and the algorithm's private randomness Υ. The equality uses $\mathbb{E}[g(Z_t)] = 2/3$ and $U_i = 0$ for $i > 2m$. The inequality follows by averaging the preceding conditional bound over the private randomness and the remaining signs.

Putting it all together. Taking expectations over the signs $U_1,\dots,U_{2m}$, the noises $Z_1,\dots,Z_T$, and the algorithm's private randomness Υ, and using (101), we obtain

$$\mathbb{E}[\Gamma]\ge\frac{mT}{3}-\frac{mT}{12} = \frac{mT}{4}. \tag{102}$$

Since $0\le g\le 1$ and $|U_i|\le 1$, we have $S_i\in[-T,T]$ for every $i\in[d]$. Also, each round contributes at least $-m$ to S_{L}, so $S_{\mathrm{L}}\ge -mT$. Consequently,

$$\Gamma = \sum_{j=1}^{m}\max\{S_{2j-1},S_{2j}\}-S_{\mathrm{L}}\le mT-(-mT) = 2mT.$$

Using (100) and Lemma 13, we obtain

$$\mathbb{P}\left(R_T\ge\frac{m\sqrt{T}}{1024}\right)\ge\mathbb{P}\left(\Gamma\ge\frac{mT}{8}\right)\ge\frac{1}{16}, \tag{103}$$

where the probability is taken over both the adversary's randomness and the algorithm's private randomness. The first inequality uses $R_T \geq (\varepsilon/m)\Gamma$ and $\varepsilon = m/(128\sqrt{T})$. The second applies Lemma 13 using (102) and $\Gamma \leq 2mT$. Since $r = m/d > 2^{-32}$, we have $m > 2^{-32}d$. Moreover, there are 2^d subsets of $[d]$, so $\binom{d}{m} \leq 2^d$ and $\log\binom{d}{m} \leq d\log 2 < d$. Consequently,

$$\frac{m\sqrt{T}}{1024} > 2^{-42}d\sqrt{T} > 2^{-42}\sqrt{dT\log\binom{d}{m}}.$$

Applying Lemma 10 with $b = m\sqrt{T}/1024$ and $p = 1/16$, we can fix a realization of Ξ for which the same guarantee holds over Υ alone. The losses in (98) are then fixed in advance and remain in $[0, 1/m]$ at every round. Since $1/16 \geq 2^{-14}$, this proves (97). □

E.3 An $\Omega(\sqrt{dT\log(1/\delta)})$ lower bound

Proposition 4. *Let $1 \leq m \leq d/2$, let $0 < \delta \leq 2^{-36}$, and suppose that $T \geq 4dm^2\log(1/\delta)$. For every randomized algorithm on $\mathcal{X}_{d,m}$, there exists a deterministic oblivious loss sequence with every action loss in $[0,1]$ such that*

$$\mathbb{P}\left(R_T \geq 2^{-16}\sqrt{dT\log\frac{1}{\delta}}\right) \geq \delta. \tag{104}$$

Proof. **Instances and regret.** Fix a core set of items $C \subset [d]$ with $|C| = m-1$, and let $V := [d]\setminus C$ be the set of remaining items. Define $\Delta := 2^{-12}\sqrt{|V|\log(1/\delta)/T}$.

We consider $|V|+1$ instances, indexed by $V \cup \{0\}$. Instance 0 is the reference instance, while each instance $i \in V$ is designed to favor item i. Specifically, for each $i \in V \cup \{0\}$ and $t \in [T]$, we define the loss vector $\ell_t^{(i)}$ as follows:

$$\begin{aligned}\ell_{t,j}^{(0)} &:= \frac{1/4 + Z_t/8}{m} + \frac{\mathbf{1}\{j \in V\}}{8m}, \qquad j \in [d],\\ \ell_{t,j}^{(i)} &:= \ell_{t,j}^{(0)} - \Delta g(Z_t)\mathbf{1}\{j = i\},\end{aligned} \tag{105}$$

where $Z_1, \ldots, Z_T$ are i.i.d. random variables with distribution $\mathrm{Unif}[-1,1]$, independent of the algorithm's private randomness Υ. In each instance $i \in V \cup \{0\}$, fixing $Z_1, \ldots, Z_T$ determines all loss vectors before the interaction starts, independently of the algorithm's actions; hence the adversary is oblivious. Notice that every core item has a lower loss than every item in V since $\Delta < 1/(8m)$ and this is true in all the instances $i \in V \cup \{0\}$. Moreover, the loss of an item $i \in V$ is reduced only in its associated instance.

In addition, for every action $x \in \mathcal{X}_{d,m}$, its loss in instance 0 satisfies $1/8 + 1/(8m) \leq x^\top \ell_t^{(0)} \leq 1/2$. In each instance $i \in V$, this loss is reduced by at most Δ. Since $\Delta < 1/(8m)$, every action has loss in $[0,1]$ in every instance.

Furthermore, we have

$$m\Delta = 2^{-12}\sqrt{\frac{m^2|V|\log(1/\delta)}{T}} \leq 2^{-13}\sqrt{\frac{|V|}{d}} \leq 2^{-13}, \tag{106}$$

where the first inequality uses $T \geq 4dm^2\log(1/\delta)$ and the second uses $|V| \leq d$.

As observed above, selecting more than one item outside the core incurs an additional loss. To track the cumulative cost of these additional selections, we define

$$D_t := \frac{1}{8m}\sum_{\tau=1}^{t}\left(\sum_{j\in V} X_{\tau,j} - 1\right), \qquad t = 0, \ldots, T. \tag{107}$$

Thus $D_0 = 0$ and D_t is nondecreasing. In each instance $i \in V$, the action selecting $C \cup \{i\}$ is optimal at every round, since every item in C has smaller loss than every item in V, and item i has no larger loss than any other item in V. In instance $i \in V$, subtracting the loss of $C \cup \{i\}$ from the learner's loss and summing over rounds gives the following regret decomposition:

$$R_T^{(i)} = D_T + \Delta \sum_{t=1}^{T} g(Z_t)(1 - X_{t,i}) \geq D_T. \tag{108}$$

The first term accounts for selecting additional items outside the core, while the second accounts for not selecting item i. The inequality follows from $g \geq 0$. Notice that $R_T^{(i)}$ denotes the regret of the algorithm when run in instance i, for every $i \in V$. Indeed, in the reference instance 0, the regret is simply $R_T^{(0)} = D_T$.

An auxiliary algorithm. We aim to force regret at least $\Delta T/8$. Given the original algorithm $\mathcal{A}$, construct an auxiliary algorithm $\overline{\mathcal{A}}$ that follows $\mathcal{A}$ as long as its proposed action would keep the quantity D_t strictly below $\Delta T/8$. At the first action recommendation by $\mathcal{A}$ that would make $D_t \geq \Delta T/8$, it instead selects all core items and a fixed item in V, the same in every instance, and repeats this action thereafter.

We use bars to distinguish the quantities associated with $\overline{\mathcal{A}}$. In particular, $\bar{X}_t$, $\overline{\mathcal{H}}_t$, $\bar{D}_t$, and $\bar{R}_T^{(i)}$ denote its actions, history, cumulative cost, and regret in instance i, respectively. The identities in (107) and (108) hold with the quantities associated to $\overline{\mathcal{A}}$. Thus, we have

$$\bar{D}_T < \Delta T/8 \quad \text{and} \quad \sum_{t=1}^{T} \sum_{j \in V} \bar{X}_{t,j} = T + 8m\bar{D}_T < T + m\Delta T = T(1 + m\Delta). \tag{109}$$

The inequality $\bar{D}_T < \Delta T/8$ follows because the fixed action chosen by $\overline{\mathcal{A}}$ does not increase $\bar{D}_t$. The second inequality follows from (107), and $\bar{D}_T < \Delta T/8$.

To compare the two regrets $R_T^{(i)}$ and $\bar{R}_T^{(i)}$, fix an instance $i \in V$ and a realization of $Z_1, \ldots, Z_T$, and run both algorithms using the same private randomness Υ. Let τ be the first round at which $\overline{\mathcal{A}}$ replaces the action selected by $\mathcal{A}$, with $\tau := T + 1$ if no replacement occurs. Their actions and feedback coincide up to round $\tau - 1$. If $\tau \leq T$, the action selected by $\mathcal{A}$ at round τ makes $D_\tau \geq \Delta T/8$. Since D_t is nondecreasing, (108) gives $R_T^{(i)} \geq D_T \geq \Delta T/8$. If $\tau = T + 1$, their entire trajectories agree, so $R_T^{(i)} = \bar{R}_T^{(i)}$. Consequently, with i and $Z_1, \ldots, Z_T$ fixed and $\mathbb{P}_\Upsilon$ denoting probability over the algorithm's private randomness,

$$\begin{aligned}
\mathbb{P}_\Upsilon\left(R_T^{(i)} \geq \frac{\Delta T}{8}\right) &= \mathbb{P}_\Upsilon\left(R_T^{(i)} \geq \frac{\Delta T}{8}, \tau \leq T\right) + \mathbb{P}_\Upsilon\left(R_T^{(i)} \geq \frac{\Delta T}{8}, \tau = T + 1\right) \\
&= \mathbb{P}_\Upsilon(\tau \leq T) + \mathbb{P}_\Upsilon\left(\bar{R}_T^{(i)} \geq \frac{\Delta T}{8}, \tau = T + 1\right) \\
&\geq \mathbb{P}_\Upsilon\left(\bar{R}_T^{(i)} \geq \frac{\Delta T}{8}, \tau \leq T\right) + \mathbb{P}_\Upsilon\left(\bar{R}_T^{(i)} \geq \frac{\Delta T}{8}, \tau = T + 1\right) \\
&= \mathbb{P}_\Upsilon\left(\bar{R}_T^{(i)} \geq \frac{\Delta T}{8}\right).
\end{aligned} \tag{110}$$

Therefore, it is enough to prove the lower bound for $\overline{\mathcal{A}}$. To simplify notation, from now on we omit the bars from the auxiliary algorithm's quantities. Thus, X_t, Y_t, $\mathcal{H}_t$, and D_t now refer to $\overline{\mathcal{A}}$. We retain the bar in its regret $\bar{R}_T^{(i)}$ to distinguish it from the original algorithm's regret $R_T^{(i)}$.

Fix a realization of the algorithm's private randomness Υ. Let P denote the resulting law of $\mathcal{H}_T$ when the auxiliary algorithm is run in instance 0, and, for each $i \in V$, let Q_i denote its law in instance i, with only $Z_1, \dots, Z_T$ remaining random. The dependence of these laws on the fixed realization of Υ is left implicit. We write $\mathbb{E}_P$ and $\mathbb{E}_{Q_i}$ for expectations under P and Q_i, respectively.

For each $i \in V$, consider the event

$$E_i := \left\{ D_T + \Delta \sum_{t=1}^{T} g(Z_t)(1 - X_{t,i}) \geq \frac{\Delta T}{8} \right\}. \tag{111}$$

Under P, the algorithm is run in instance 0. At every round, we can recover Z_t from the observed action and feedback using

$$Z_t = 8 \left(Y_t - \frac{1}{4} - \frac{1}{8m} \sum_{j \in V} X_{t,j} \right). \tag{112}$$

Also, D_T depends only on the actions $X_1, \dots, X_T$. Therefore, all the quantities defining E_i can be computed from the history $\mathcal{H}_T$ under P. Now fix $i \in V$ and consider Q_i. There are two cases:

- If $X_{t,i} = 0$, Equation (112) recovers Z_t.
- If $X_{t,i} = 1$, the summand $g(Z_t)(1 - X_{t,i})$ is zero, so we do not need to recover Z_t.

Since D_T depends only on the actions, we can thus compute $D_T + \Delta \sum_{t=1}^{T} g(Z_t)(1 - X_{t,i})$ from the history under Q_i as well. Under both P and Q_i, the event E_i can therefore be written as

$$E_i = \left\{ D_T + \Delta \sum_{\substack{t=1 \\ X_{t,i}=0}}^{T} g\left(8 \left(Y_t - \frac{1}{4} - \frac{1}{8m} \sum_{j \in V} X_{t,j} \right) \right) \geq \frac{\Delta T}{8} \right\}.$$

Indeed, when $X_{t,i} = 0$, the argument of g equals Z_t in Equation (111) under both laws. When $X_{t,i} = 1$, the corresponding term in the original definition of E_i is equal to zero. Since D_T depends only on the actions, the above condition involves only the observed history $\mathcal{H}_T$. Thus, for each $i \in V$, the event E_i is entirely determined by the observed history $\mathcal{H}_T$, in the same way under both P and Q_i. Consequently, $P(E_i)$ and $Q_i(E_i)$ are well-defined. Moreover, under Q_i, the left-hand side of the inequality defining E_i equals $\bar{R}_T^{(i)}$, whereas under P the auxiliary algorithm's regret is only D_T. Moreover, under P, (112) holds at every round, including when $X_{t,i} = 1$. Consequently, conditioning on $\mathcal{H}_T$ fixes the actions, D_T, and all the variables Z_t. Averaging over the indices $i \in V$ gives

$$\begin{aligned}
&\frac{1}{|V|} \sum_{i \in V} \mathbb{E}_P \left[D_T + \Delta \sum_{t=1}^{T} g(Z_t)(1 - X_{t,i}) \,\middle|\, \mathcal{H}_T \right] \\
&= D_T + \Delta \sum_{t=1}^{T} g(Z_t) \left(1 - \frac{1}{|V|} \sum_{j \in V} X_{t,j} \right) \\
&= \Delta \left(1 - \frac{1}{|V|} \right) \sum_{t=1}^{T} g(Z_t) + \sum_{t=1}^{T} \left(\sum_{j \in V} X_{t,j} - 1 \right) \left(\frac{1}{8m} - \frac{\Delta g(Z_t)}{|V|} \right) \\
&\geq \frac{\Delta}{2} \sum_{t=1}^{T} g(Z_t).
\end{aligned} \tag{113}$$

The first equality uses the fact that the quantities inside each conditional expectation are fixed by that history. The second equality substitutes (107). The inequality uses $|V| \geq 2$, $0 \leq g \leq 1$, and (106). Taking expectation under P and using the tower property yields

$$\frac{\Delta T}{3} \leq \frac{1}{|V|} \sum_{i \in V} \mathbb{E}_P \left[D_T + \Delta \sum_{t=1}^{T} g(Z_t)(1 - X_{t,i}) \right] \leq \frac{\Delta T}{8} + \frac{\Delta T}{|V|} \sum_{i \in V} P(E_i). \tag{114}$$

The first inequality follows from (113), because $\mathbb{E}_P[g(Z_t)] = 2/3$ for every t. For the second inequality, we split each expectation over E_i and E_i^c. The expression $D_T + \Delta \sum_{t=1}^{T} g(Z_t)(1 - X_{t,i})$ is at most $\Delta T/8$ on E_i^c and at most $9\Delta T/8$ on E_i, by (109) and $0 \leq g(x) \leq 1$ for every $x \in [-1, 1]$. Thus each expectation is at most $(\Delta T/8)P(E_i^c) + (9\Delta T/8)P(E_i) = \Delta T/8 + \Delta T\, P(E_i)$. Subtracting $\Delta T/8$ and dividing by ΔT in (114) gives

$$\frac{1}{|V|} \sum_{i \in V} P(E_i) \geq \frac{1}{3} - \frac{1}{8} = \frac{5}{24} \geq \frac{1}{16}. \tag{115}$$

Thus the events E_i have constant probability on average under P. We next compare P with each Q_i to show that the average of $Q_i(E_i)$ over $i \in V$ is larger than δ, for every fixed realization of Υ.

Comparing P and Q_i. Fix $i \in V$. Since Υ is already fixed, conditioning on the same preceding history $\mathcal{H}_{t-1}$ determines the same action X_t under P and Q_i. In either instance, Z_t remains independent of $\mathcal{H}_{t-1}$ and uniform on $[-1, 1]$. If $X_{t,i} = 0$, the conditional feedback laws coincide. If $X_{t,i} = 1$, subtracting $1/4 + (8m)^{-1} \sum_{j \in V} X_{t,j}$ from the feedback and multiplying by four gives $\psi_0(Z_t)$ under P and $\psi_{-4\Delta}(Z_t)$ under Q_i. This common invertible affine transformation preserves KL divergence. Thus, we have

$$\mathrm{KL}\left(P(\cdot \mid \mathcal{H}_{t-1}) \,\|\, Q_i(\cdot \mid \mathcal{H}_{t-1})\right) \leq 128(4\Delta)^2 X_{t,i}. \tag{116}$$

Here, $P(\cdot \mid \mathcal{H}_{t-1})$ and $Q_i(\cdot \mid \mathcal{H}_{t-1})$ denote the conditional distributions of Y_t under P and Q_i, respectively, given the same preceding history $\mathcal{H}_{t-1}$, with Υ fixed throughout. The bound in (116) is zero when $X_{t,i} = 0$. Otherwise, we can employ Lemma 12 with parameters 0 and -4Δ. Since $4\Delta \leq 2^{-11} < 1/16$, we have (116). Using the KL chain rule and (116), we have

$$\begin{aligned} \mathrm{KL}(P \| Q_i) &= \sum_{t=1}^{T} \mathbb{E}_P[\mathrm{KL}(P(\cdot \mid \mathcal{H}_{t-1}) \,\|\, Q_i(\cdot \mid \mathcal{H}_{t-1}))] \\ &\leq \sum_{t=1}^{T} \mathbb{E}_P\left[128(4\Delta)^2 X_{t,i}\right] \\ &= 128(4\Delta)^2 \mathbb{E}_P\left[\sum_{t=1}^{T} X_{t,i}\right]. \end{aligned} \tag{117}$$

Averaging over $i \in V$ gives

$$\begin{aligned} \frac{1}{|V|} \sum_{i \in V} \mathrm{KL}(P \| Q_i) &\leq \frac{128(4\Delta)^2}{|V|} \mathbb{E}_P\left[\sum_{t=1}^{T} \sum_{i \in V} X_{t,i}\right] \\ &\leq 2^{11} \frac{\Delta^2 T}{|V|}(1 + m\Delta) \\ &\leq 2^{-12} \log \frac{1}{\delta}. \end{aligned} \tag{118}$$

The first inequality averages (117) and uses linearity of expectation under the same law P. The second uses (109). The last uses $1 + m\Delta < 2$ and $\Delta^2 T/|V| = 2^{-24}\log(1/\delta)$.

We now bound the average of $Q_i(E_i)$ with Υ still fixed. In particular, we have

$$\begin{aligned}\left(\frac{1}{|V|}\sum_{i\in V} P(E_i)\right)\log\frac{1}{\frac{1}{|V|}\sum_{i\in V} Q_i(E_i)} &\le \mathrm{kl}\left(\frac{1}{|V|}\sum_{i\in V} P(E_i), \frac{1}{|V|}\sum_{i\in V} Q_i(E_i)\right) + \log 2 \\ &\le \frac{1}{|V|}\sum_{i\in V} \mathrm{kl}\big(P(E_i), Q_i(E_i)\big) + \log 2 \\ &\le \frac{1}{|V|}\sum_{i\in V} \mathrm{KL}(P\|Q_i) + \log 2. \end{aligned} \tag{119}$$

As before, $\mathrm{kl}(a,b)$ denotes the KL divergence between Bernoulli laws with success probabilities a and b. The first inequality uses the binary entropy bound $\mathrm{kl}(a,b) \ge a\log(1/b) - \log 2$. The second follows from joint convexity of KL divergence. The last follows from data processing applied to the indicator of E_i under P and Q_i, for each $i \in V$.

Suppose by contradiction that $|V|^{-1}\sum_{i\in V} Q_i(E_i) \le \delta$. Then

$$\begin{aligned}\frac{1}{16}\log\frac{1}{\delta} &\le \left(\frac{1}{|V|}\sum_{i\in V} P(E_i)\right)\log\frac{1}{\frac{1}{|V|}\sum_{i\in V} Q_i(E_i)} \\ &\le \frac{1}{|V|}\sum_{i\in V}\mathrm{KL}(P\|Q_i) + \log 2 \\ &\le \left(2^{-12} + \frac{1}{36}\right)\log\frac{1}{\delta} < \frac{1}{16}\log\frac{1}{\delta}.\end{aligned}$$

The first inequality uses (115) and the assumed bound on the average of $Q_i(E_i)$. The second uses (119). The third uses (118) and $\log(1/\delta) \ge 36\log 2$. This contradiction proves that $|V|^{-1}\sum_{i\in V} Q_i(E_i) > \delta$ for every fixed realization of Υ.

For each fixed realization of Υ, $Q_i(E_i)$ is the probability of E_i with respect to the randomness of $Z_1, \dots, Z_T$. Averaging over Υ, before choosing i, gives

$$\frac{1}{|V|}\sum_{i\in V}\mathbb{E}_\Upsilon[Q_i(E_i)] = \mathbb{E}_\Upsilon\left[\frac{1}{|V|}\sum_{i\in V} Q_i(E_i)\right] > \delta, \tag{120}$$

where $\mathbb{E}_\Upsilon$ denotes expectation over the algorithm's private randomness Υ. Therefore, there is a fixed index $i^\star \in V$, which does not depend on the realization of Υ, such that $\mathbb{E}_\Upsilon[Q_{i^\star}(E_{i^\star})] > \delta$. By the law of total expectation,

$$\mathbb{E}_\Upsilon\left[Q_{i^\star}(E_{i^\star})\right] = \mathbb{E}_\Upsilon\left[\mathbb{E}_{Q_{i^\star}}\left[\mathbf{1}\left\{\bar{R}_T^{(i^\star)} \ge \frac{\Delta T}{8}\right\}\right]\right] = \mathbb{P}\left(\bar{R}_T^{(i^\star)} \ge \frac{\Delta T}{8}\right),$$

where the probability on the right is taken over both the algorithm's private randomness Υ and $Z = (Z_1, \dots, Z_T)$, in instance $i^\star$. Fix this index $i^\star$. The comparison in (110) holds for every fixed realization of $Z_1, \dots, Z_T$, with probabilities taken over Υ. Averaging it over $Z_1, \dots, Z_T$ therefore yields

$$\mathbb{P}\left(R_T^{(i^\star)} \ge \frac{\Delta T}{8}\right) \ge \mathbb{P}\left(\bar{R}_T^{(i^\star)} \ge \frac{\Delta T}{8}\right) = \mathbb{E}_\Upsilon[Q_{i^\star}(E_{i^\star})] > \delta,$$

where both probabilities are over $Z_1, \dots, Z_T$ and the algorithm's private randomness Υ. Moreover,

$$\frac{\Delta T}{8} = 2^{-15}\sqrt{|V| T \log\frac{1}{\delta}} \geq 2^{-16}\sqrt{dT\log\frac{1}{\delta}}.$$

The equality follows from the definition of Δ, and the inequality uses $|V| = d - m + 1 \geq d/2$. For this fixed $i^\star$, we encode the adversary's randomness in $\Xi = (Z_1, \dots, Z_T)$, independently of Υ. By Lemma 10, there exists a realization $\bar{\xi}$ of Ξ for which the original algorithm incurs regret at least $\Delta T/8$ with probability at least δ over Υ alone. The losses in (105) then form a deterministic oblivious sequence, proving the claim. □

E.4 Proof of Theorem 3

Theorem 3. *Let $d \geq 2$, $1 \leq m \leq d-1$, and $\delta \in (0, 2^{-36}]$, and assume that*

$$T \geq \left\lceil 64d^2 + 4d\min\{m, d-m\}^2 \log(1/\delta)\right\rceil.$$

Then, for every randomized algorithm on $\mathcal{X}_{d,m}$, there exists a deterministic adaptive non-anticipating loss process whose action losses belong to $[0,1]$ and such that

$$\mathbb{P}\left(R_T \geq 2^{-43}\sqrt{dT\left(\log\binom{d}{m} + \log(1/\delta)\right)}\right) \geq \delta.$$

Proof. We first assume that $1 \leq m \leq d/2$. The assumption on T implies both $T \geq 64d^2$ and $T \geq 4dm^2\log(1/\delta)$. Thus the horizon requirements of Propositions 2 to 4 are satisfied. Suppose that $\log\binom{d}{m} \geq \log(1/\delta)$. If $m/d \leq 2^{-32}$, Proposition 2 gives

$$\mathbb{P}\left(R_T \geq 2^{-24}\sqrt{dT\log\binom{d}{m}}\right) \geq \frac{1}{8}.$$

As in the proof of that proposition, Lemma 10 allows us to fix the adversary's auxiliary randomness while preserving this guarantee over the algorithm's private randomness. The resulting adversary is deterministic, adaptive, and non-anticipating. If $m/d > 2^{-32}$, Proposition 3 gives a deterministic oblivious loss sequence satisfying

$$\mathbb{P}\left(R_T \geq 2^{-42}\sqrt{dT\log\binom{d}{m}}\right) \geq 2^{-14}.$$

In both cases, every m-set action has loss in $[0,1]$. Since $2^{-24} \geq 2^{-42}$ and $1/8 \geq 2^{-14} \geq \delta$, we obtain

$$\mathbb{P}\left(R_T \geq 2^{-42}\sqrt{dT\log\binom{d}{m}}\right) \geq \delta.$$

Suppose instead that $\log(1/\delta) > \log\binom{d}{m}$. Since $0 < \delta \leq 2^{-36}$ and $T \geq 4dm^2\log(1/\delta)$, Proposition 4 gives a deterministic oblivious loss sequence with action losses in $[0,1]$ such that

$$\mathbb{P}\left(R_T \geq 2^{-16}\sqrt{dT\log(1/\delta)}\right) \geq \delta.$$

This also implies $\mathbb{P}(R_T \geq 2^{-42}\sqrt{dT\log(1/\delta)}) \geq \delta$.

Consequently, in either case, every randomized algorithm on $\mathcal{X}_{d,m}$ admits a deterministic adaptive non-anticipating adversary with action losses in $[0,1]$ for which

$$\mathbb{P}\left(R_T \geq 2^{-42}\sqrt{dT\max\left\{\log\binom{d}{m}, \log(1/\delta)\right\}}\right) \geq \delta.$$

Since

$$\max\left\{\log\binom{d}{m}, \log(1/\delta)\right\} \geq \frac{1}{2}\left(\log\binom{d}{m} + \log(1/\delta)\right),$$

we obtain

$$\mathbb{P}\left(R_T \geq 2^{-43}\sqrt{dT\left(\log\binom{d}{m} + \log(1/\delta)\right)}\right) \geq \delta.$$

This proves the theorem when $m \leq d/2$.

Now suppose that $m > d/2$. Then $1 \leq d-m \leq d/2$ and $\min\{m, d-m\} = d-m$, so the horizon assumption gives

$$T \geq \left\lceil 64d^2 + 4d(d-m)^2\log(1/\delta)\right\rceil.$$

The case already proved therefore applies to every randomized algorithm on $\mathcal{X}_{d,d-m}$ at the same horizon T and confidence level δ. By Lemma 11, the resulting lower bound transfers to $\mathcal{X}_{d,m}$, preserving the probability guarantee and the action-loss range $[0,1]$. Since $\binom{d}{d-m} = \binom{d}{m}$, this is exactly the claimed bound. □

AI methodology

We used OpenAI's GPT-5.5, GPT-5.6 Sol, and GPT-6 Astra to brainstorm and refine algorithmic ideas, develop preliminary proof arguments, and assist with literature searches and LaTeX formatting. Since the authors are not native English speakers, we also used these models to improve grammar, punctuation, and phrasing throughout the manuscript. All AI-assisted material incorporated into the manuscript was subsequently checked by the authors and revised where appropriate.

OpenAI's GPT-6 Astra was also used to produce the accompanying Lean formalization. Its correspondence with the manuscript was assessed using separate instances of GPT-6 Astra, distinct from those used to produce the formalization. The authors thoroughly reviewed the resulting reports. The Lean sources and instructions for independent verification are available in the companion GitHub repository.

The authors take full responsibility for the paper and its accompanying materials.

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